Simple and Log Returns

What changes when you take the logarithm, and when it matters

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Abstract

Simple returns and log returns carry the same information, and neither approximates the other. Each one is exactly right for a different job. Log returns add across time. Simple returns add across positions. Wherever the two disagree, the gap is about half the variance. This chapter derives these facts and traces their consequences through econometrics, risk models and machine-learning features.

No return measure can add both across time and across positions. Compounding creates a cross-term, and plain addition cannot represent it. Models of a price through time use logs: volatility models, option pricing, simulation and realised variance. Calculations that count money use simple returns: portfolio returns, attribution, profit and loss, costs, leverage and client reporting. A well-built system converts between the two in one place.

The half-variance gap appears in many forms. It is the volatility drag on compound growth and the drift correction in geometric Brownian motion. It is also the Kelly growth rate, the rebalancing premium, and the difference between two versions of the Sharpe ratio. A 10% rise followed by a 10% fall leaves an investor down 1%. A threefold leveraged ETF on an index with 20% annual volatility trails three times the index's growth by about 12 percentage points a year, from arithmetic alone.

In many settings the choice does not matter. On daily stock data the two series correlate at 0.99996. Tree-based models give identical results for both, because a monotone transform preserves the order of a feature. The Sharpe ratio is the main exception. Its two versions differ by about half the annual volatility. The difference is 0.1 for a long-only stock portfolio and 0.3 for a volatile one. Volatility scaling matters about a thousand times more than the choice of measure. The same 3% daily move ranges from half a standard deviation to 30 standard deviations across a realistic universe.

Log returns do not exist for spreads, rates or profit-and-loss series that can be zero or negative. Taking the log of a return instead of one plus the return turns the return into a sign indicator without raising any error. Converting a predicted log return to a simple return gives a median rather than a mean. The resulting bias reorders a ranking. The chapter recommends scaling returns by volatility before choosing a measure. It also recommends adding across positions in simple returns and across time in log returns, with one conversion function between the two.

How to read this document. Sections 1–3 are the foundations. They cover what a return is and the theorem that explains why two conventions exist. That theorem also shows why no third convention can replace them. Section 3 is a reference catalogue of the identities in daily use. Sections 4–6 are the statistical theory. They cover distributions, comparability across assets and across time, and the time-series econometrics that make the log transform compulsory in some settings. Sections 7–10 are practice. They cover where practice forces the choice in each direction, where it provably does not matter, how it goes wrong, and how to build working intuition. Section 11 covers machine learning, with returns as features and targets for gradient-boosted trees and neural networks. It stands alone for readers who need only that. Sections 12–14 cover conventions, references, and the synthesis. Appendix A defines the supporting concepts in dependency order, and Appendix B lists additional works cited.

For the short answer, read §2, the table in §3.6, and §14.

Objectives. After this chapter, you should be able to:

- explain why simple and log returns carry the same information, and why neither one approximates the other.
- state the impossibility theorem, which says that no return measure adds up both across time and across positions.
- recognise the half-variance gap $\sigma^2/2$ in its common forms, and estimate its size.
- choose the convention that each stage of a pipeline requires, and convert between conventions correctly.
- say when the choice cannot matter, including for tree-based models, and when it changes a result.
- diagnose the standard failures: domain errors, `log` in place of `log1p`, regressions in the wrong space, and retransformation bias.

Epistemic tags. The chapters in this collection flag claims by status:

- **[Fact]** — a mathematical identity, or a result replicated across independent datasets or implementations, with broad agreement.

- **[Contested]** — documented, but researchers still disagree about magnitude, robustness, or cause.
- **[Hypothesis]** — a proposed mechanism, not decisively tested.
- **[Practice]** — practitioner convention. The evidence may be private or absent. A [Practice] claim has not been debunked.

Tags appear only where the status changes what a reader should do. A tag governs the sentence or clause it opens.

Notation. The table lists every symbol that recurs in the chapter, with the section that defines or first uses it. Symbols used in a single derivation are defined where they appear.

Symbol	Meaning	Defined in
P_t; $p_t = \ln P_t$	Price of the asset at time t; log price	§1.3, §6.1
D_t	Cash distribution paid over $(t-1, t]$	§1.3
$G_t = 1 + R_t = e^{r_t}$	**Gross return**, the primitive object	§1.1
$R_t = (P_t + D_t)/P_{t-1} - 1$	**Simple return**	§1.1
$r_t = \ln(1 + R_t)$	**Log return**; equal to $p_t - p_{t-1}$ when distributions are ignored, as in most formulas below	§1.1
$R_{1:h}$; $r_{1:h}$	Cumulative simple and log return over periods 1 to h	§2.1
$\mu = \mathbb{E}[R]$; $\sigma^2 = \mathrm{Var}(R)$	Mean and variance of the **simple** return	§2.6
$m = \mathbb{E}[r]$; $s^2 = \mathrm{Var}(r)$	Mean and variance of the **log** return	§2.6
g	Geometric mean return, defined by $1 + g = \exp(\mathbb{E}[r])$	§1.3, §3.4
$\bar{R}$; $\bar{r}$	Sample means of the simple and log returns	§1.3
W_t	Wealth at time t	§1.1
A	Periods per year: 252 for daily bars, 12 for monthly	§2.3
h	A horizon, in periods	§2.1
N; i; w_i	Number of assets; asset index; portfolio weights, with $\sum_i w_i = 1$	§2.1
R_p; r_p	Portfolio simple and log return	§2.1
$\mathrm{Var}_w(R)$	Weighted cross-sectional variance of returns across holdings	§2.3
Σ; $\mathbf{w}$	$N \times N$ covariance matrix of simple returns; vector of weights	§2.3
L	Leverage multiple	§3.2
SR	Sharpe ratio	§7.1, §8.3
R_f; r_f	Simple and log risk-free rate	§1.3, §6.3
$\mathcal{F}_t$	Information set at time t	§6.2
$\widehat{\sigma}_t$	Estimate of per-period volatility formed from returns up to and including r_t; a contemporaneous return is normalised as $r_t/\widehat{\sigma}_{t-1}$	§5.4
Φ	Standard normal CDF	§4.2
ϕ	A generic smooth, strictly increasing transform; never the normal density	§2.5
$\mathbb{1}\{\cdot\}$	Indicator of the condition in braces	§11.2

A hat denotes an estimate and a bar a sample average. Bold denotes a cross-sectional vector over the N assets. The chapter uses μ, σ for the moments of simple returns and m, s for the moments of log returns. This keeps the moments of the two conventions distinct, because the gap between them is the subject of the chapter.

Four symbols carry a second meaning. Each is standard in its own literature, and the text flags it where it occurs. σ is a return volatility everywhere except two places. In §6.3 it is the diffusion coefficient of a stochastic differential equation. In §4.6, $\sigma(\cdot)$ denotes a generated σ-algebra. The diffusion-coefficient use is harmless, because the two are the same object in the continuous-time limit. W_t is wealth everywhere except §6.3 and §A.14, where it is a standard Brownian motion. §6.3 writes the price as S_t. g is the geometric mean *simple* return. g^* in §7.1 is the maximised expected *log* growth rate, so the two are different objects. λ is the Box–Cox parameter in §1.4 and the GARCH persistence in §6.2.

Two departures from the collection's reserved letters are deliberate. Here r is the log return, as in the econometrics of returns, and the risk-free rate is r_f. The price is P_t, because the lowercase s denotes the log-return volatility.

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Part I — Foundations

1. What a return is, and why there are two of them

1.1 The primitive is wealth, not return

Every return describes a change in a quantity of money. An investor holds W_{t-1} dollars at the end of period $t-1$ and W_t dollars at the end of period t. Each convention in this chapter describes the map $W_{t-1} \mapsto W_t$.

That map is a **multiplication**. Wealth does not gain a fixed number of dollars per period. A factor scales it:

$$W_t = W_{t-1} \cdot G_t, \qquad G_t = \frac{W_t}{W_{t-1}} > 0$$

G_t is the **gross return**, and it is the primitive object. The two conventions in this chapter are two coordinate systems for the same G_t:

$$\underbrace{R_t = G_t - 1}_{\text{simple return}} \qquad \underbrace{r_t = \ln G_t}_{\text{log return}}$$

Neither coordinate is more fundamental than the other. Each is an invertible function of G_t, so both carry exactly the same information. Common belief about log returns assumes otherwise, so this point comes first. The conventions differ in which *algebraic operations* on the coordinate correspond to meaningful operations on wealth. §2 makes this precise.

The map between them. The two coordinates are related by a bijection from $(-1, \infty)$ onto $\mathbb{R}$:

$$r = \ln(1 + R) \qquad \Longleftrightarrow \qquad R = e^r - 1$$

The map is smooth, strictly increasing, and strictly concave, and each property has consequences. Because the map is strictly increasing, it preserves order, and §11.2 relies on this. Because it is strictly concave, Jensen's inequality applies in one direction everywhere, and §3 catalogues the results. Because it is smooth, it has a Taylor expansion around zero:

$$r = R - \frac{R^2}{2} + \frac{R^3}{3} - \cdots$$

The series converges for $|R| < 1$. At the return magnitudes seen in practice, the quadratic term accounts for almost all of the gap between r and R.

1.2 The wrong intuition, and why it costs money

Most readers start with this mental model:

> *Log returns are an approximation to simple returns that is convenient because it makes them add up. The approximation is good for small returns and bad for large ones.*

Each clause of this model is wrong, and each error has consequences.

Log returns are not an approximation. The definition $r = \ln(1 + R)$ is exact. The approximation is a separate claim, $r \approx R$, about a different pair of objects. Much of the confusion in this area comes from conflating the log return with the approximation $r \approx R$. The conflation suggests that the choice between conventions is a trade-off in precision. The choice is about which arithmetic operation should be exact. It involves no trade-off in precision.

Log returns do not "make returns add up" as a convenience. They make returns add up *across time*, and in exchange they stop adding up *across assets*. A theorem forces this exchange (§2). Simple returns have the complementary property. No third convention has both, and neither convention is the "right" one that the other approximates.

Whether the difference matters depends on the next operation, not on the size of the return. The choice does not affect a decision tree that takes a single 40% move as a raw feature (§11.2). The choice does affect a sequence of 0.5% moves that a pipeline compounds over ten years and then annualises. Here the per-period gap $R_t - r_t \approx R_t^2/2$ is 1.25×10^{-5}, too small to notice on its own. The gap accumulates linearly in the number of periods, however, and over 2,520 trading days it reaches about 3 percentage points.

A better mental model:

> **Log returns are the natural coordinate for a process that evolves by multiplication. Simple returns are the natural coordinate for a quantity that adds across positions. A financial data pipeline does both, at different stages. A practitioner therefore needs both conventions, and needs to know which one each number is in.**

1.3 Vocabulary and the things that share a name

Several distinct objects share overlapping names, and the ambiguity commonly causes reconciliation errors between systems. The table gives each object one definition and lists its other names.

Name	Definition	Also called
Gross return	$G_t = (P_t + D_t)/P_{t-1}$	Wealth relative, price relative, growth factor
Simple return	$R_t = G_t - 1$	Arithmetic return, linear return, discrete return, percentage return, holding-period return
Log return	$r_t = \ln G_t$	Continuously compounded return, compounded return, geometric return†, logarithmic return
Cumulative simple return	$\prod_{i=1}^{h}(1 + R_i) - 1$	Total return over h, buy-and-hold return
Cumulative log return	$\sum_{i=1}^{h} r_i$	—

Name	Definition	Also called
Arithmetic mean return	$\bar{R} = \frac{1}{h}\sum_i R_i$	Average return
Geometric mean return	$g = \left(\prod_i (1+R_i)\right)^{1/h} - 1$	CAGR (annualised), time-weighted return
Excess return	$R_t - R_{f,t}$	Risk premium (in expectation)

† "Geometric return" is ambiguous in practice. Some authors mean the log return r_t, and others mean the geometric mean g. The two are related: $1 + g = \exp(\bar{r})$ exactly, where $\bar{r}$ is the arithmetic mean of the log returns. They are still different objects, because r_t is a per-period quantity and g summarises a sample. This chapter does not use the term.

Two further distinctions cause more real-world damage than the choice between log and simple returns:

- **Price return versus total return.** The price return $P_t/P_{t-1}-1$ ignores dividends, and the total return $(P_t+D_t)/P_{t-1}-1$ includes them. For US equities at the index level, dividends are worth roughly 1.5–2% per year. That gap is larger than every second-order effect in this chapter, so settle it first, in either convention.
- **Adjusted versus unadjusted prices.** Data vendors apply split and dividend adjustments retroactively, so an adjusted "close" series depends on the date of download. A missed two-for-one split adjustment appears as a -50% simple return, or a -0.693 log return. The error is visible in either convention and equally damaging in both (§9.6).

1.4 The one-parameter family that contains both

Simple and log returns are the two endpoints of a continuum. The continuum explains why no convention in between is worth using. The Box–Cox family (Box & Cox, 1964) applied to the gross return $G = 1 + R$ is

$$f_\lambda(R) \;=\; \begin{cases} \dfrac{(1+R)^\lambda - 1}{\lambda}, & \lambda \neq 0 \\ \ln(1+R), & \lambda = 0 \end{cases}$$

The $\lambda = 0$ case is the limit of the others: $\lim_{\lambda\to 0} \frac{(1+R)^\lambda - 1}{\lambda} = \ln(1+R)$. L'Hôpital's rule gives the limit, and so does the expansion $(1+R)^\lambda = e^{\lambda \ln(1+R)} = 1 + \lambda \ln(1+R) + O(\lambda^2)$. At the other end, $\lambda = 1$ gives $f_1(R) = R$, the simple return exactly.

The two conventions are therefore the members $\lambda \in \{0, 1\}$ of a smooth family. Intermediate values such as $\lambda = 0.4$ go unused, because the useful properties of the endpoints do not vary continuously with λ:

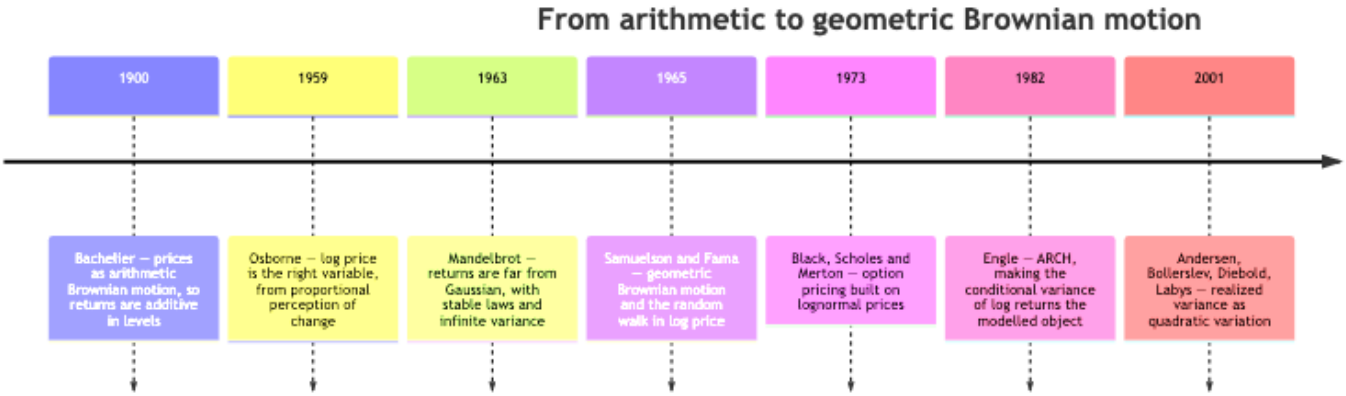
- Time additivity holds **only** at $\lambda = 0$.
- Portfolio linearity holds **only** at $\lambda = 1$.

An interior λ has neither property. It gives only a better fit to some marginal distribution, and a standardisation or a rank transform (§11.3) gives that fit more cheaply and more interpretably. [Fact] The trade-off in §2 therefore has a corner solution, not an interior one.

The Box–Cox framing is useful in one respect. It names what taking logs does. Taking logs applies a **variance-stabilising transform**. The name connects the log return to a large statistical literature on when such transforms help. §2.5 develops the argument, and §6.1 applies it.

1.5 Where the two conventions come from historically

The choice is old, and the field contested it for decades. The timeline marks the moments when the field's view changed.



Four lessons follow from this history.

Bachelier's choice was reasonable. [Fact] A model of the price itself as a Brownian motion gives a tractable theory, and it fits well at short horizons on liquid instruments. The model assigns positive probability to negative prices, and Bachelier himself noted this fatal defect. The same defect reappears in §4.2 as the "-100% floor," the reason simple returns cannot be Gaussian.

Osborne's argument was psychophysical and not mathematical. [Hypothesis] Writing in *Operations Research* in 1959, Osborne argued that investors perceive equal *ratios* of price as equally significant. This is the Weber–Fechner law from perception research, and it makes $\ln P$ the natural variable. The argument is not a proof, and the modern justification is different (variance stabilisation, §6.1). The field nonetheless moved because of it.

Mandelbrot's challenge was never fully resolved, and it still shapes practice. [Contested] His 1963 finding is that cotton price changes have far fatter tails than the Gaussian, possibly with infinite variance. The finding is empirically robust in the tails. The open question is whether the variance is infinite, as in a stable-Paretian law, or merely large and time-varying, as ARCH models would later describe it. Researchers still debate the question for high-frequency data. The answer matters, because infinite variance would break the $\sigma^2/2$ corrections that this chapter relies on. The modern consensus leans toward finite variance with a time-varying scale. It also leans toward a tail index of about 3–5 for daily equity returns. With that tail index, enough moments exist for the corrections to be meaningful. Too few exist for trustworthy inference based on fourth moments.

The log convention won in continuous-time finance, but not in the cross-section. Derivatives pricing, volatility modelling, and time-series econometrics work almost entirely in logs. Asset-pricing tests, portfolio construction, and performance reporting work almost entirely in simple returns. §12 argues that this split is correct.

§1 Key takeaways

1. The primitive is the gross return $G_t = W_t/W_{t-1}$. Simple and log returns are two invertible coordinates for it, and they carry identical information.
2. The log return is an exact definition. The approximation is the separate claim $r \approx R$.
3. The next operation (averaging, compounding, aggregating, or fitting) decides whether the choice matters. The size of the returns does not.
4. Simple and log returns are the $\lambda = 1$ and $\lambda = 0$ members of the Box–Cox family. No intermediate member is useful, because each desirable property holds only at one endpoint.
5. Total-versus-price returns and correct split adjustment are worth more basis points than the entire log-versus-simple question. Settle them first.
6. The negative-price defect of additive price models drove the historical shift from Bachelier to Osborne to Samuelson. A desire for additivity did not.

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2. The central trade-off: additivity or linearity, never both

This section states the idea from which the rest of the chapter follows. Sections 3 through 14 are consequences of it, instances of it, or practical accommodations to it.

2.1 The two properties you want

Analysts aggregate financial data along two axes. A return measure should behave well along both.

Axis 1: time. An investor holds one asset for h periods. The wealth relative over the whole span is the product of the per-period wealth relatives:

$$\frac{W_h}{W_0} \;=\; \prod_{i=1}^{h}(1+R_i)$$

The ideal return measure would make its h-period value the **sum** of its per-period values. Statistics relies on sums: means, variances, central limit theorems, linear regressions, and every horizon-scaling rule.

Axis 2: assets. An investor holds N assets with weights w_i at the start of the period. The portfolio's wealth relative is the weighted average of the constituents' wealth relatives:

$$1+R_p \;=\; \sum_{i=1}^{N} w_i(1+R_i) \qquad\Longrightarrow\qquad R_p \;=\; \sum_{i=1}^{N} w_i R_i$$

The implication uses $\sum_i w_i = 1$. Along this axis, the ideal return measure is **linear in the weights**. Linearity makes portfolio optimisation a quadratic program. It makes performance attribution add up. It also makes the left-hand side of a factor model well defined.

Call the first property **time additivity** and the second **portfolio linearity**. Each convention satisfies exactly one of them:

Log returns:	$r_{1:h} = \sum_{i=1}^{h} r_i$ (exact),	$r_p \neq \sum_i w_i r_i$
Simple returns:	$R_{1:h} \neq \sum_{i=1}^{h} R_i$,	$R_p = \sum_i w_i R_i$ (exact)

2.2 The impossibility theorem

A natural question is whether some other transform could have both properties. None can, and the proof takes three lines.

Proposition. Let $f : (-1,\infty) \to \mathbb{R}$ be continuous, and suppose both

(T) $f\big((1+R_1)(1+R_2)-1\big) = f(R_1)+f(R_2)$ for all $R_1, R_2 > -1$, and

(P) $f\big(wR_1+(1-w)R_2\big) = w\,f(R_1)+(1-w)\,f(R_2)$ for all $w \in [0,1]$, $R_1, R_2 > -1$.

Then $f \equiv 0$.

Proof. Property (P) says that f is affine on the interval. To see this, put $b = f(0)$ and take $R_2 = 0$ in (P). Then $f(wR) - b = w(f(R)-b)$ for every $w \in [0,1]$, so $(f(R)-b)/R$ is one constant on $R>0$ and one constant on $R \in (-1,0)$. Choosing $R_1 > 0 > R_2$, and the w that makes $wR_1+(1-w)R_2 = 0$, forces the two constants to agree. Hence $f(R) = aR+b$. Setting $R_1 = R_2 = 0$ in (T) gives $f(0) = 2f(0)$, so $f(0) = 0$. Hence $b = 0$ and $f(R) = aR$. Substituting into (T) gives

$$a\big[(1+R_1)(1+R_2)-1\big] = a(R_1+R_2+R_1R_2) \;\overset{!}{=}\; aR_1+aR_2$$

which forces $aR_1R_2 = 0$ for all R_1, R_2, hence $a = 0$. ■

The proof also works from the other end. Write property (T) in terms of the gross return $x = 1+R$, with $\tilde{f}(x) \equiv f(x-1)$. It becomes the multiplicative Cauchy equation $\tilde{f}(x_1x_2) = \tilde{f}(x_1)+\tilde{f}(x_2)$ for all $x_1, x_2 > 0$, whose only continuous solutions are $\tilde{f}(x) = c\ln x$ (Aczél, 1966). So (T) alone forces f to be a scalar multiple of the log return. Property (P) then requires

$$c\,\ln\big(wx_1+(1-w)x_2\big) \;=\; c\,\big(w\ln x_1+(1-w)\ln x_2\big)$$

The **strict** concavity of ln makes $\ln(wx_1 + (1-w)x_2) > w \ln x_1 + (1-w)\ln x_2$ whenever $x_1 \neq x_2$ and $w \in (0,1)$. Multiplying a strict inequality by a nonzero c keeps it strict, with the direction set by the sign of c. The equality can therefore hold only if $c = 0$.

The obstruction is the cross-term. Both proofs reach the contradiction at the same object, the product $R_1 R_2$. Compounding is multiplication, and multiplication produces a cross-term that no linear function can represent. Simple returns are linear, so they cannot represent the cross-term, and that is why they fail to compound. Log returns turn multiplication into addition, so the transform already includes the cross-term, and that is why they fail to average across a portfolio.

The result fits in one sentence: **a transform cannot linearise multiplication and preserve addition at the same time.**

2.3 The size of the error from ignoring the theorem

The theorem says that one property must be given up. The cost of giving it up is the practical question, and the theorem does not state it. Both gaps have simple second-order expressions. The two gaps are the same quantity.

Cost of using log returns across a portfolio. By the concavity of ln (Jensen's inequality),

$$r_p \;=\; \ln\Big(1 + \sum_i w_i R_i\Big) \;\geq\; \sum_i w_i \ln(1+R_i) \;=\; \sum_i w_i r_i$$

So a naive average of log returns across a portfolio always *understates* the portfolio's log return. Jensen's inequality needs the weights to form a probability distribution, so this subsection assumes a long-only book, $w_i \geq 0$. With short positions, the gap can have either sign. Expanding both sides to second order, with $\bar{R}_w \equiv \sum_i w_i R_i$, gives

$$r_p - \sum_i w_i r_i \;\approx\; \tfrac{1}{2}\Big(\underbrace{\textstyle\sum_i w_i R_i^2 - \bar{R}_w^2}_{\text{weighted cross-sectional variance}} \Big) \;=\; \tfrac{1}{2}\,\mathrm{Var}_w(R)$$

The error is half the *cross-sectional variance of returns across the holdings*. Take a diversified equity book whose daily cross-sectional return dispersion is 2%. The error is $\frac{1}{2}(0.02)^2 = 2 \times 10^{-4}$, or 2 basis points per day, which is about 5% per year.

The portfolio literature calls this quantity the **diversification return** or **rebalancing return** (Booth & Fama, 1992; Willenbrock, 2011). In continuous time, a portfolio rebalanced back to fixed weights w satisfies

$$g_p - \sum_i w_i g_i \;=\; \tfrac{1}{2}\Big(\sum_i w_i \sigma_i^2 - \sigma_p^2 \Big) \;\geq\; 0$$

Here g_p is the portfolio's geometric mean return, g_i are the constituents' geometric mean returns, σ_i^2 are their variances, and $\sigma_p^2 = \mathbf{w}^\top \Sigma \mathbf{w}$ is the portfolio variance. [Fact] The "error" from mis-aggregating log returns is therefore exactly the diversification return. Two equally weighted, uncorrelated assets with 30% volatility each generate $\frac{1}{2}(0.09 - 0.045) = 2.25\%$ per year of it.

Cost of using simple returns across time. Compare the compounded product of h simple returns with their sum:

$$\prod_{i=1}^{h}(1+R_i) - 1 \;=\; \sum_i R_i \;+\; \underbrace{\sum_{i<j} R_i R_j}_{\text{first cross-term}} \;+\; \cdots$$

For h uncorrelated returns with mean μ, the leading discrepancy has expectation $\binom{h}{2}\mu^2 \approx \frac{1}{2}h^2\mu^2$. It grows **quadratically** in the horizon. This mechanism underlies the most common reporting error in the industry. The error is quoting $A \cdot \bar{R}$ as an "annualised return." With a daily $\bar{R} = 0.0004$ and $A = 252$, $A\bar{R} = 10.08\%$, while $(1+\bar{R})^{252} - 1 = 10.60\%$. The cross-terms add 52 bp, so the linear figure is *below* the compounded one by that amount. Ignoring compounding therefore understates the return. $A\bar{R}$ is nonetheless the flattering number. It also ignores a second and larger effect, volatility drag. §3.5 combines the two effects.

2.4 Two exceptions

The theorem concerns exact properties for arbitrary inputs. Two special structures escape it, and both arise in practice.

Continuous rebalancing makes log returns portfolio-linear in the limit. Suppose continuous rebalancing holds the weights fixed. Over an infinitesimal interval, the portfolio's *instantaneous* log return then satisfies $dr_p = \sum_i w_i\, dr_i + \frac{1}{2}(\sum_i w_i \sigma_i^2 - \sigma_p^2)\, dt$. The relation is linear up to a deterministic Itô correction, and that correction is the diversification return again. This is why continuous-time portfolio theory can work in logs throughout, while discrete-time portfolio construction cannot.

Single-asset problems have no cross-section. If $N = 1$, portfolio linearity is vacuous. No argument then opposes log returns until the conversion to dollars, which still uses simple-return arithmetic (§7.2). A large share of practical work falls in this regime: volatility estimation, single-name signal research, and univariate time-series modelling. For that work, the debate does not arise.

2.5 The second job logs do, which has nothing to do with additivity

Additivity is the property usually cited for log returns. Another property makes the log transform indispensable in econometrics: **variance stabilisation**. The argument is separate, though it points the same way, so it gets its own subsection.

Suppose the price evolves multiplicatively, so that the conditional variance of the price *change* scales with the square of the price level:

$$\mathsf{Var}(P_t \mid P_{t-1}) \;=\; \sigma^2 P_{t-1}^2$$

The assumption follows from requiring that "a 1% move" mean the same thing at $P = 10$ and at $P = 1000$, so it is more than a convenience. The delta method identifies the transform $\phi(\cdot)$ that makes the transformed series homoskedastic. Expand about the conditional mean $\mathbb{E}[P_t \mid P_{t-1}]$, which equals P_{t-1} up to one period of drift. Then $\mathsf{Var}\left(\phi(P_t) \mid P_{t-1}\right) \approx \phi'(P_{t-1})^2\, \mathsf{Var}(P_t \mid P_{t-1}) = \phi'(P_{t-1})^2 \sigma^2 P_{t-1}^2$. For this variance to be free of P_{t-1}, the transform must satisfy

$$\phi'(P) \;\propto\; \frac{1}{P} \qquad \Longrightarrow \qquad \phi(P) = a \ln P + b$$

[Fact] **The logarithm is the unique variance-stabilising transform for a process whose volatility is proportional to its level.** It is unique up to the affine rescaling by a and b, which changes nothing. This is why analysts run unit-root tests, cointegration, ARIMA, and GARCH on log prices rather than on prices. The reason is independent of the additivity argument. Differencing raw prices instead would give a series whose variance grows with the price level. That unmodelled trend in the second moment would corrupt every standard error computed from the series.

This is also the accurate answer to the question of why econometricians prefer logs. They prefer logs because the residuals become homoskedastic, so the asymptotic theory applies. Returns adding up is not the reason.

2.6 The third generative idea: one correction term in many forms

To leading order, the gap between the two conventions is always the same quantity. For any smooth ϕ, the second-order delta method gives

$$\mathbb{E}[\phi(X)] \;\approx\; \phi(\mathbb{E}X) \;+\; \tfrac{1}{2}\,\phi''(\mathbb{E}X)\,\mathsf{Var}(X)$$

Apply it with $\phi = \ln(1 + \cdot)$, for which $\phi''(x) = -(1+x)^{-2}$:

$$\boxed{m \;=\; \mathbb{E}[\ln(1+R)] \;\approx\; \ln(1+\mu) - \frac{\sigma^2}{2(1+\mu)^2} \;\approx\; \mu - \frac{\sigma^2}{2}}$$

The term $\sigma^2/2$ recurs more often than any other quantity in the analysis of returns. It appears in at least ten places that look unrelated but are all the same Taylor term. The table lists them. The sections listed derive each entry and define the symbols local to each row.

Appearance	Statement	Where
Jensen gap	$m \approx \mu - \sigma^2/2$	§3.3
Volatility drag	$g \approx \mu - \sigma^2/2$	§3.4
Lognormal mean	$\mathbb{E}[e^r] = e^{m+s^2/2}$	§4.1
Itô drift	$d \ln S = (\mu - \frac{1}{2}\sigma^2)\,dt + \sigma\,dW$	§6.3
Black–Scholes d_2	$d_2 = \dfrac{\ln(S/K) + (r_f - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$	§7.1
Kelly growth rate	$g^* = \mu^2/(2\sigma^2) = \mathrm{SR}^2/2$	§7.1
Diversification return	$g_p - \sum_i w_i g_i = \frac{1}{2}(\sum_i w_i \sigma_i^2 - \sigma_p^2)$	§2.3
Sharpe-ratio gap	$\mathrm{SR}_{\text{simple}} - \mathrm{SR}_{\log} \approx \sigma_{\text{ann}}/2$	§8.3
CAPM alpha bias	$\alpha^{\log} \approx \alpha - \frac{1}{2}(\sigma_i^2 - \beta\sigma_M^2)$	§9.3
Retransformation bias	$\mathbb{E}[R \mid x] = e^{\hat{m}(x) + \hat{s}^2(x)/2} - 1$	§11.5

Each appearance of $\sigma^2/2$ has the same cause. An expectation passes through a concave function, and the curvature subtracts half the variance. Recognising the term on sight is the main skill this chapter teaches.

§2 Key takeaways

1. Log returns are additive across time but not across assets. Simple returns are linear across assets but not across time. This is a theorem. It is not a convention.
2. No transform achieves both properties. The obstruction is the cross-term $R_1 R_2$, which multiplication produces and linearity cannot represent.
3. Averaging log returns across a portfolio understates the portfolio's log return by about half the cross-sectional variance of returns. That gap is exactly the diversification return, a real economic quantity.
4. Adding simple returns across time omits cross-terms that grow quadratically in the horizon. Reporting $A\bar{R}$ as an annualised return is the standard instance.
5. Logs do a second, independent job. They are the unique variance-stabilising transform for a process whose volatility scales with its level. This property, not additivity, is why time-series econometrics uses log prices.
6. One Taylor term, $\sigma^2/2$, generates volatility drag, the Itô correction, the Kelly growth rate, the Black–Scholes d_2, the diversification return, the Sharpe gap, and the retransformation bias. They are one fact in ten forms.

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3. The identity catalogue

This section is reference material. It collects the relationships a practitioner needs, in one place and in consistent notation. Each relationship carries one of four labels: **exact** (holds for any returns), **exact under iid**, **exact under lognormality**, or **approximate**. Most errors in this area come from quoting an approximate result as if it were an identity, so the labels matter.

3.1 Conversion

The table lists the conversions between the two conventions, with the status of each.

Direction	Formula	Status
Simple $\to$ log	$r = \ln(1 + R)$	exact
Log $\to$ simple	$R = e^r - 1$	exact
Small-return expansion	$r = R - \frac{R^2}{2} + \frac{R^3}{3} - \cdots$	exact for $\lvert R\rvert < 1$
First-order	$r \approx R$	approximate
Second-order	$r \approx R - \frac{1}{2}R^2$	approximate

The domains are fixed: $R \in (-1, \infty)$ and $r \in (-\infty, \infty)$. A simple return of exactly -1, a total loss, maps to $r = -\infty$. A price series that touches zero or goes negative has no log return at all (§9.1).

3.2 Aggregation

Across time, over h consecutive periods:

$$r_{1:h} = \sum_{i=1}^{h} r_i \qquad \textbf{exact, no assumptions}$$

$$1 + R_{1:h} = \prod_{i=1}^{h} (1 + R_i) \qquad \textbf{exact, no assumptions}$$

$$R_{1:h} = \sum_{i=1}^{h} R_i + \sum_{i<j} R_i R_j + \cdots \qquad \textbf{exact expansion}$$

The first line is the whole practical case for log returns, and it holds unconditionally. It needs no independence, no stationarity, and no distributional assumption, because it restates $\ln(ab) = \ln a + \ln b$.

Across assets, for a portfolio with start-of-period weights w_i that sum to one:

$$R_p = \sum_{i=1}^{N} w_i R_i \qquad \textbf{exact, no assumptions}$$

$$r_p = \ln\Big(1 + \sum_i w_i e^{r_i} - \sum_i w_i\Big) \;=\; \ln\Big(\sum_i w_i e^{r_i}\Big) \qquad \textbf{exact, and not a sum}$$

$$r_p \geq \sum_i w_i r_i, \qquad r_p - \sum_i w_i r_i \approx \tfrac{1}{2}\,\mathrm{Var}_w(R) \qquad \textbf{Jensen (long-only); approximate gap}$$

The portfolio log return is the log of a weighted average of exponentials, a *log-sum-exp*. This object, a close relative of the softmax, appears whenever an analyst averages in log space. It is smooth and convex in $\mathbf{r}$, and it is not a weighted average of the r_i.

Multiplicative chains. Any decomposition of the form $1 + R = \prod_k (1 + R^{(k)})$ becomes exactly additive in logs. The table lists the common decompositions, together with two that do not chain.

Decomposition	Simple form	Log form
Currency translation	$R^{\mathrm{USD}} = R^{\mathrm{loc}} + R^{\mathrm{fx}} + R^{\mathrm{loc}} R^{\mathrm{fx}}$	$r^{\mathrm{USD}} = r^{\mathrm{loc}} + r^{\mathrm{fx}}$
Real from nominal (π the inflation rate)	$1 + R^{\mathrm{real}} = \frac{1+R^{\mathrm{nom}}}{1+\pi}$	$r^{\mathrm{real}} = r^{\mathrm{nom}} - \ln(1+\pi)$
Proportional fee c	$1 + R^{\mathrm{net}} = (1 + R^{\mathrm{gr}})(1 - c)$	$r^{\mathrm{net}} = r^{\mathrm{gr}} + \ln(1-c)$
Total from price return	$R^{\mathrm{TR}} = R^{\mathrm{PR}} + D_t / P_{t-1}$	not additive
Constant leverage L	$R^{L} = L\,R$	not additive

The last two rows are the ones practitioners forget. **Dividends and leverage are additive in simple returns, not in logs.** A 3× daily-rebalanced ETF has a simple return of exactly $3R_t$ before costs, and a log return of $\ln(1 + 3R_t) \neq 3r_t$. For leveraged products, and for splitting a total return into price and income components, simple returns are exact and logs are the approximation. The leverage case is the clearest counterexample to the claim that logs are always better.

3.3 Moments

Distribution-free (delta method, second order). For any return distribution with mean μ and variance σ^2:

$$m \;\approx\; \ln(1+\mu) - \frac{\sigma^2}{2(1+\mu)^2} \;\approx\; \mu - \frac{\sigma^2}{2}, \qquad s^2 \;\approx\; \frac{\sigma^2}{(1+\mu)^2} \;\approx\; \sigma^2$$

The two conventions disagree about the mean at first order in the variance, and they agree about the variance to the same order. This asymmetry explains why nearly every discrepancy in this chapter is a *level* discrepancy in a mean and not a scale discrepancy in a volatility. Volatility estimates barely depend on the convention. Return estimates depend on it strongly.

Exact, under lognormality ($r \sim \mathcal{N}(m, s^2)$, so $1 + R$ is lognormal):

$$\mu = e^{m+s^2/2} - 1 \qquad s^2 = \ln\left(1 + \frac{\sigma^2}{(1+\mu)^2}\right)$$
$$\sigma^2 = (1+\mu)^2(e^{s^2} - 1) \qquad m = \ln(1+\mu) - \tfrac{1}{2}s^2$$
$$\text{skew}(R) = (e^{s^2} + 2)\sqrt{e^{s^2} - 1} \quad \text{exkurt}(R) = e^{4s^2} + 2e^{3s^2} + 3e^{2s^2} - 6$$

The left column converts from log to simple, and the right column inverts it. Both directions are in constant use. The left column turns simulated log returns into economic quantities. The right column turns a historical mean and volatility in percentage terms into the parameters of a simulation.

A perfectly symmetric log-return distribution produces a right-skewed simple-return distribution, and the skew grows with the horizon. The table shows the size of the effect for equity-like volatility.

Horizon	s	skew(R)	exkurt(R)
Daily	0.0126	0.038	0.002
Monthly	0.058	0.174	0.054
Annual	0.20	0.614	0.678
Annual, high-vol asset	0.60	2.26	10.3

At daily frequency the induced skew is 0.04. It is invisible next to the *actual* skew of daily equity returns, which is 0.1–0.5 in size. At annual frequency the induced skew is 0.61, a substantial part of the observed right skew in long-horizon equity returns. [Fact] The Jacobian of the log transform therefore accounts for some of the well-known positive skew in long-horizon returns. That part is not a property of markets.

3.4 Geometric versus arithmetic mean, and volatility drag

This identity has the largest practical consequences in the chapter, because it quantifies the compound growth that investors lose to volatility.

Distribution-free, in-sample. For any sample of h returns:

$$1 + g \;=\; \left(\prod_{i=1}^{h}(1 + R_i)\right)^{1/h} \;=\; \exp\left(\frac{1}{h}\sum_{i=1}^{h} r_i\right) \;=\; \exp(\bar{r})$$

By the arithmetic–geometric mean inequality, $g \leq \bar{R}$ always, with equality if and only if every R_i is identical. **The arithmetic mean of a return series is an upper bound on what the investor actually earned.** The bound is strict whenever returns vary at all.

Second-order. Expanding, with $\hat{\sigma}^2 = \frac{1}{h}\sum_i (R_i - \bar{R})^2$ the in-sample variance:

$$g \;\approx\; \bar{R} - \tfrac{1}{2}\hat{\sigma}^2$$

Exact, under lognormality. The relationship becomes an identity in log space:

$$\ln(1+\mu) - \ln(1+g) \;=\; \frac{s^2}{2}$$

Here g is the **median** of the simple-return distribution, and μ is its mean. The gap between them comes entirely from the right skew of the lognormal.

The quantity $\mu - g \approx \sigma^2/2$ is called **volatility drag**, or *variance drain*. Its practical consequences are large and easy to miss, as the table shows.

Asset	μ	σ	Drag $\sigma^2/2$	$g \approx$
US large-cap equity	10%	20%	2.0%	8.0%
2× levered equity	20%	40%	8.0%	12.0%
3× levered equity	30%	60%	18.0%	12.0%
4× levered equity	40%	80%	32.0%	8.0%
High-vol single name	20%	60%	18.0%	2.0%

The leverage rows show the mechanism. Doubling leverage doubles the arithmetic mean and *quadruples* the drag. Compound growth is therefore a concave function of leverage: it rises, peaks, and then falls. The peak is at $L^* = \mu/\sigma^2$, the Kelly fraction, so the growth-optimal leverage follows from the same $\sigma^2/2$ term (§7.1). [Fact] For the 10%/20% asset, $L^* = 0.10/0.04 = 2.5$, and the table shows g peaking between $2\times$ and $3\times$, as predicted.

The table is also the clearest argument that **volatility is a direct subtraction from compound return, not merely a measure of risk.** An investor who maximises the long-run growth rate of wealth must account for volatility even with no risk aversion at all, because volatility reduces growth. Log-return arithmetic shows this fact immediately. Simple-return arithmetic does not.

3.5 Annualisation and horizon scaling

The table gives the horizon-scaling and annualisation rules in each convention.

Quantity	Log returns	Simple returns
Mean, h periods	$h\,m$ (exact under iid)	$(1+\mu)^h - 1$ (exact under iid)
Variance, h periods	$h\,s^2$ (exact under iid)	$(\sigma^2 + (1+\mu)^2)^h - (1+\mu)^{2h}$ (exact under iid)
Volatility scaling	$s\sqrt{h}$ (exact under iid)	$\sigma\sqrt{h}$ (approximate)
Annualised mean	$A\,m$	$(1+\mu)^A - 1$
Annualised volatility	$s\sqrt{A}$	$\approx \sigma\sqrt{A}$

Three points follow from the table. First, the square-root-of-time rule is an *exact* consequence of iid additivity for log returns and only an approximation for simple returns. Practitioners state the rule confidently because of that exactness. Second, every rule in the table fails under autocorrelation, in either convention. The variance ratio $\mathrm{Var}(r_{1:h})/(h\,s^2)$ departs from 1 exactly when returns are serially correlated. Momentum and mean-reversion testing study that departure. Third, the common shortcut of annualising a mean simple return as $A\bar{R}$ errs in a flattering direction. For a daily $\bar{R}$ of 4 bp, $A\bar{R} = 10.1\%$, below the compounded 10.6%. The compounded figure, however, does not subtract volatility drag. The accurate annualised *growth* figure is $\exp(A\bar{r}) - 1$, which is lower than both, so $A\bar{R}$ overstates growth.

3.6 The summary table

The table below is the most useful single reference in the chapter. It lists common operations and marks the convention in which each one is exact.

Operation	Exact in simple returns	Exact in log returns
Compound over time	✗ (product)	✓ (sum)
Aggregate across assets in a portfolio	✓ (weighted sum)	✗ (log-sum-exp)
Scale variance by $\sqrt{h}$ (iid)	✗	✓
Apply a proportional fee	✗	✓
Convert currency	✗	✓
Deflate by inflation	✗	✓
Add dividend income to a price return	✓	✗
Apply constant per-period leverage L	✓	✗
Compute dollar P&L on a position	✓	✗
Bounded below at -100%	✓	✗ (unbounded below)
Can be exactly Gaussian	✗	✓
Intraday values sum to the period return	✗	✓
Order-preserving relative to the other	✓	✓

The last row records that the two conventions are monotonically related, so any statement about *rankings* holds in both. §11.2 turns this fact into the central practical result for tree-based models.

§3 Key takeaways

1. Time additivity of log returns and portfolio linearity of simple returns are unconditional identities. Every other entry in the catalogue carries a distributional assumption. Check that assumption.
2. The two conventions disagree about means at order σ^2 and agree about variances to that order. Volatility estimates barely depend on the convention, but return estimates do.
3. Under lognormality, $\mu = e^{m+s^2/2} - 1$ and $g = e^m - 1$. The arithmetic mean is the mean of the simple-return distribution, and the geometric mean is its median.
4. Volatility drag $\mu - g \approx \sigma^2/2$ makes compound growth concave in leverage, with a peak at the Kelly fraction μ/σ^2. A 4× levered position on a 10%/20% asset compounds no faster than the unlevered one.
5. The log transform itself produces some of the observed right skew in long-horizon simple returns. Symmetric annual log returns with $s = 0.20$ imply a skew of 0.61 in simple returns.
6. Dividends and leverage are additive in simple returns, not in logs. Leveraged products are the standard counterexample to the claim that logs are unconditionally better.

Part II — Statistical theory

4. Distributional properties: what actually differs

A common claim is that “log returns are normally distributed.” As an empirical statement about daily data, the claim is false. As a statement about a *model*, it is true. This section explains the difference between those two readings.

4.1 The lognormal model and what it provides

Assume that log returns are iid Gaussian, $r_t \sim \mathcal{N}(m, s^2)$. Then

$$P_T \;=\; P_0 \exp\Big(\sum_{t=1}^{T} r_t\Big) \quad\Longrightarrow\quad \ln P_T \sim \mathcal{N}\big(\ln P_0 + Tm,\; Ts^2\big)$$

so the price is lognormal at every horizon. Three consequences follow, and together they explain why this model dominated finance for forty years:

1. **Prices stay positive.** $P_T = P_0 e^X$ with X real is strictly positive by construction, which removes Bachelier's defect.
2. **The model is closed under time aggregation.** A sum of independent Gaussians is Gaussian. A model calibrated on daily data therefore makes an internally consistent statement about monthly data, with parameters (Tm, Ts^2), and needs no refitting.
3. **Expectations of payoffs have closed forms.** $\mathbb{E}[e^r] = e^{m+s^2/2}$ is the moment generating function of the Gaussian evaluated at 1. Every Black–Scholes-type formula integrates a payoff against a lognormal density in the same way.

Point 2 is the deepest of the three, though analysts usually take it for granted. §4.2 develops it.

4.2 Why simple returns cannot be Gaussian, and why it is not a technicality

Two arguments show this. The first is weak, and the second is strong.

The weak argument: support. A limited-liability asset cannot lose more than everything, so $R \geq -1$ always. A Gaussian has support on all of $\mathbb{R}$, so it assigns positive probability to $R < -1$, that is, to negative prices. That probability depends only on how many standard deviations the -1 floor sits below the mean. Take a daily equity volatility of $\sigma \approx 1.27\%$ and a mean of zero. The floor is $1/0.0127 \approx 79$ standard deviations away, and the probability is $\Phi(-79) \approx 3 \times 10^{-1358}$. That is negligible, so this argument alone is unconvincing. The probability becomes material at long horizons and high volatilities. For annual returns with $\sigma = 60\%$ and a zero mean, the floor is only $1/0.6 = 1.67$ standard deviations away, and $\Phi(-1.67) \approx 4.8\%$. A Gaussian model would then assign a probability of nearly 1 in 20 to losing everything or more, a result of arithmetic rather than economics.

The strong argument: closure under aggregation. This argument is decisive. It is also the most precise answer to the question of which statistical relationship holds for log returns but not for simple returns.

A distributional model is *consistent across horizons* when its statements about daily and monthly data agree. Consistency constrains the law of the quantity that aggregates additively. Moving from a daily law *up* to a monthly law requires that law's family to be closed under convolution. Moving from a monthly law back *down* to a daily law requires the converse: the law must be expressible as a sum of n iid pieces. Requiring both at every horizon is exactly **infinite divisibility**. For every n, some distribution must exist whose n-fold convolution is the horizon-1 law. The infinitely divisible laws are precisely the time-1 marginals of Lévy processes. They make up the entire set of time-consistent models. Examples are Gaussian, Poisson, variance-gamma, normal-inverse-Gaussian, α-stable, and their mixtures.

Log returns are the additive quantity. An analyst can impose any of these laws on them, and the model then extends coherently to every horizon. Simple returns are *not* additive. Even if R_1 and R_2 were exactly Gaussian and independent, $R_{1:2} = R_1 + R_2 + R_1 R_2$ would not be Gaussian, because the cross-term from §2.2 moves it outside the Gaussian family.

The loose version of this claim is false, so the precise version matters. A modeller *can* write every time-consistent model in simple-return coordinates. Pushing an infinitely divisible law for r through $R = e^r - 1$ gives a family of simple-return laws that aggregates correctly to any horizon. Shifted-lognormal simple returns are the obvious example. A modeller cannot **choose the family in simple-return coordinates**. [Fact] The simple-return families that aggregate consistently are exactly the images under $R = e^r - 1$ of the infinitely divisible ones. Only logs can *state* the closure condition. None of the distributions a modeller would actually fit to a simple-return series (Gaussian, Student-t, skew-t, a mixture of normals) is among those images.

This theorem underlies the practitioner rule of thumb. **Log returns are the only coordinate in which "the distribution at horizon h" is determined by "the distribution at horizon 1."** The rule does not rest on log returns being more normal.

4.3 What the data actually looks like

[Fact] The table lists the "stylized facts" of Cont (2001), the standard reference list. Researchers have replicated each across markets, asset classes, and decades. Any candidate model has to reproduce them. The right-hand column records whether the convention affects each one.

Stylized fact	Statement	Convention-sensitive?
Heavy tails	Tail index roughly 3–5; far more mass beyond ± 4 standard deviations than Gaussian	No, at daily frequency
No linear autocorrelation	$\rho_k(r)$ indistinguishable from zero beyond a few minutes for liquid assets	No

Stylized fact	Statement	Convention-sensitive?
Volatility clustering	$\rho_k(\lvert r\rvert)$ positive and decaying slowly (hyperbolically) out to months	No
Aggregational Gaussianity	Distribution approaches Gaussian as the horizon grows	**Yes** — see below
Leverage effect	Returns negatively correlated with future volatility, asymmetrically	Slightly
Gain/loss asymmetry	Equity indices show negative skew; large drawdowns exceed large run-ups	**Yes**
Volume–volatility correlation	Trading volume correlates with all measures of volatility	No

Quotations of this list often omit the right-hand column. **At daily frequency, on liquid equities, switching conventions changes none of the first three facts qualitatively.** The sample excess kurtosis of daily S&P 500 returns typically lies between 5 and 25, depending on the sample window and on whether the window includes October 1987. On a window without a crash, the simple-return and log-return figures agree to within about 1%. On a window that contains October 1987, the log figure is visibly larger. §4.5 gives the reason: the transform stretches the single observation that dominates the fourth moment. Either way, the heavy tails are a property of markets. No transform removes them. A claim that log returns "fix" non-normality confuses the daily case with the annual case.

The convention does matter for the fourth row and for the two asymmetry rows below it.

4.4 Aggregational Gaussianity runs in opposite directions

This result most clearly demonstrates the longitudinal advantage of log returns.

Suppose per-period log returns are iid with finite variance. The assumption is a real restriction (see the Mandelbrot caveat in §1.5), but the data support it at daily frequency for liquid assets. Then

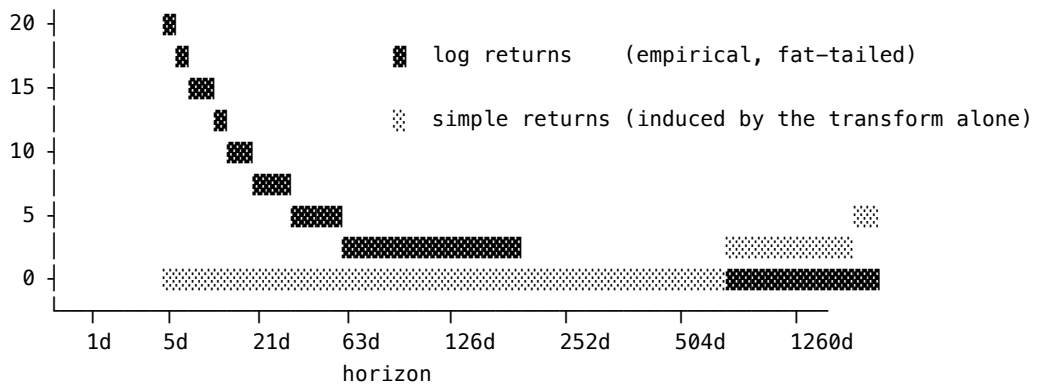
$$\frac{r_{1:h} - hm}{s\sqrt{h}} \;=\; \frac{1}{s\sqrt{h}} \sum_{i=1}^{h} (r_i - m) \quad \xrightarrow[h\to\infty]{d} \quad \mathcal{N}(0,1) \qquad \text{by the central limit theorem}$$

The standardisation isolates the *shape* of $r_{1:h}$. The shape becomes Gaussian, while location and scale both grow with h. For iid summands, the skewness of $r_{1:h}$ decays like $h^{-1/2}$, and its excess kurtosis decays like h^{-1}. Log returns become *more* Gaussian under aggregation.

The corresponding simple return is $R_{1:h} = e^{r_{1:h}} - 1$, the exponential of an increasingly Gaussian variable. So $1 + R_{1:h}$ approaches a lognormal, and the shape of a lognormal changes in the opposite direction. If log returns are exactly iid Gaussian, the horizon-h log variance is hs^2, and the formulae of §3.3 apply with s^2 replaced by hs^2:

$$\text{skew}\,(R_{1:h}) = (e^{hs^2} + 2)\sqrt{e^{hs^2} - 1}, \qquad \text{exkurt}\,(R_{1:h}) = e^{4hs^2} + 2e^{3hs^2} + 3e^{2hs^2} - 6$$

Both quantities grow without bound in h. As the horizon grows, the two conventions therefore diverge in opposite directions, as the chart shows.




```
Log returns:    empirical excess kurtosis falls toward 0 as the CLT bites.
Simple returns: only the excess kurtosis the transform itself manufactures,
                assuming Gaussian log returns; it rises from ~0 to about 0.7
                at one year and about 4.3 at five, and keeps climbing.

The two curves measure different things and their levels are not comparable:
real simple returns at 1 day carry the same fat tails as the log curve plus
the (negligible) induced amount. What the picture shows is which way each
effect runs. Past about six months the log curve sits on the floor at this
resolution and is hidden under the induced curve until that one lifts off.
```

The chart carries two lessons. **Longitudinally, log returns are decisively more comparable**, because a distributional statement made at one horizon transfers to another. **At short horizons, the choice does not matter here**, because both series are non-Gaussian for reasons unrelated to the transform.

4.5 The tail asymmetry the transform introduces

Because $\ln(1+R)$ is concave, it compresses the right tail and stretches the left tail. The table shows that the magnitudes are larger than most readers expect.

Event	R	r
Position doubles	$+100\%$	$+0.693$
Position up tenfold	$+900\%$	$+2.303$
Position halves	-50%	-0.693
Position down 90%	-90%	-2.303
Position down 99%	-99%	-4.605
Position wiped out	-100%	$-\infty$

In simple-return coordinates the largest possible loss is 1 and the largest possible gain is unbounded. The *simple* return therefore has a bounded left tail and an unbounded right tail. In log coordinates both tails are unbounded, and a halving is the exact negative of a doubling.

This asymmetry has a practical consequence that counts against log returns. §11.3 returns to it. [Practice] **For any method that squares its errors or assumes light tails, log returns can be the worse choice on crash-heavy data.** A -90% day is a -2.3 observation in log space and only a -0.9 observation in simple space. A Gaussian likelihood fitted in logs to a series that contains delistings, halts, or bankruptcies amplifies exactly the observations that will dominate the loss.

4.6 What does not differ: the information

How much the choice can matter is limited, and the limit constrains every claim that follows.

The map $R \mapsto \ln(1+R)$ is a smooth bijection from $(-1,\infty)$ onto $\mathbb{R}$ with a smooth inverse, so it is a measurable isomorphism. The σ-algebras generated by the two series are therefore identical. Here $\sigma(\cdot)$ denotes the generated σ-algebra, not a volatility:

$$\sigma(R_1, \dots, R_n) \;=\; \sigma(r_1, \dots, r_n)$$

Anything measurable with respect to one series is measurable with respect to the other. Likelihood ratios agree as well. Take any two hypotheses $\mathcal{H}_0, \mathcal{H}_1$ about the data-generating process. Use the *same* two laws, each written in whichever coordinate is in use. By the change of variables, the density in R-coordinates equals the density in r-coordinates times the Jacobian $\prod_{i=1}^{n}(1+R_i)^{-1}$. That factor does not depend on the hypothesis, so it appears in the numerator and the denominator alike and cancels:

$$\frac{L_{\mathcal{H}_1}(r_1, \dots, r_n)}{L_{\mathcal{H}_0}(r_1, \dots, r_n)} \;=\; \frac{L_{\mathcal{H}_1}(R_1, \dots, R_n)}{L_{\mathcal{H}_0}(R_1, \dots, R_n)}$$

[Fact] **A likelihood ratio test has identical power in either coordinate system.** This settles whether log returns can support more powerful conclusions. The transform itself cannot make them more powerful. Neither series contains information absent from the other, and no test is intrinsically more powerful in one coordinate.

Differences in power do occur. They enter through three channels, and the transform is not one of them:

1. **Model specification.** "Gaussian" is a different hypothesis in each coordinate, and the log-coordinate version fits financial data better. The gain in power comes from the better hypothesis, not from the logarithm.
2. **Estimator bias.** A sample mean is not equivariant under a non-linear transform: $\overline{\ln(1+R)} \neq \ln(1+\bar{R})$. If the estimand is a compound growth rate, computing it in the wrong coordinate biases it by $\sigma^2/2$ at any sample size. The error is an *inconsistency*, not noise.
3. **Aggregation exactness.** If a pipeline sums returns anywhere, the sum is exact in one coordinate and approximate in the other.

This filter applies to the rest of the chapter. Every claimed advantage of one convention should reduce to one of these three channels. A claimed advantage that does not reduce to a channel is probably unsupported.

§4 Key takeaways

1. Log returns can be exactly Gaussian, and simple returns cannot. Simple returns are bounded below at -100%, and the Gaussian family is not closed under their aggregation rule.
2. The decisive theorem is closure. A distributional model that is consistent across horizons requires infinite divisibility of the *additive* quantity, and log returns are the only additive coordinate. The simple-return families that do aggregate are exactly the images of the infinitely divisible ones under $R = e^r - 1$. Only logs can state the condition, and no family a modeller would naturally write down in simple-return coordinates satisfies it.
3. At daily frequency neither convention is close to Gaussian, and both show heavy tails and volatility clustering. The measured kurtosis differs only on samples that contain a crash, and then the log figure is the larger. Claims that logs "fix normality" confuse the daily case with the annual case.
4. As the horizon grows, log returns become more Gaussian, while simple returns become more lognormal and more skewed. This is the real longitudinal advantage of logs.
5. The log transform stretches the left tail and compresses the right tail: a -90% move is a -2.3 log observation. On crash-heavy data, this can make squared-error objectives *worse* behaved in logs.
6. The transform is a bijection, so it adds no information and no test power. Every real advantage reduces to better model specification, unbiased estimation of a compound quantity, or exact aggregation.

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5. Comparability, laterally and longitudinally

The question "are log returns more comparable?" contains three separate questions, and they have three different answers. Separating them resolves most of the problem. The table states each question and its answer.

Sense of "comparable"	Question	Winner
Scale invariance	Does the number depend on the price level or position size?	Tie — both are scale-free

Sense of "comparable"	Question	Winner
Horizon invariance	Does the number mean the same thing at 1 day and 1 year, and can it be converted exactly?	**Log**, decisively
Distributional comparability	Do the same numeric values in two series carry the same statistical meaning?	Neither — see §5.4

5.1 Scale invariance: a tie, and it is worth saying why

Both conventions are invariant to the price level and to position size. A move from \$3.00 to \$3.06 and a move from \$300 to \$306 are both $R = 2\%$ and both $r = 0.0198$. Scale invariance is sometimes offered as an argument for log returns. Simple returns have it too, so it is not an argument for log returns. The property comes from dividing by P_{t-1}, which both conventions do.

One real complication remains. The two conventions are scale-free in the price. They are **not** equally well behaved under *sign* changes, because the log is defined only for positive prices. Some quantities can cross zero: a calendar spread, a P&L series, a basis, or an interest rate in a negative-rate regime. For these, simple "returns" are meaningless, because the denominator can be near zero. Log returns are undefined. Neither convention applies. Use differences of levels instead of returns (§9.1).

5.2 Longitudinal comparability: log returns win, on four separate counts

Count 1: exact horizon conversion. The identity $r_{1:h} = \sum_i r_i$ makes the h-period return the same kind of object as the one-period return, with h times the mean and, under iid, h times the variance. To put a one-day move and a 20-day move on the same axis, divide the 20-day log return by $\sqrt{20}$. The corresponding simple-return operation has no exact form.

Count 2: distributional consistency. By §4.2, a distributional family imposed on log returns extends coherently to every horizon, and one imposed on simple returns does not. A statement such as "this is a three-sigma move" therefore transfers between horizons in log coordinates but not in simple coordinates.

Count 3: symmetry, which is a metric property. In log coordinates, a move up and the move that exactly undoes it are negatives of each other. In simple coordinates they are not, as the table shows.

Move	Up leg	Down leg that undoes it	Symmetric?
Simple	$100 \to 150$ is $+50\%$	$150 \to 100$ is -33.3%	No
Log	$100 \to 150$ is $+0.4055$	$150 \to 100$ is -0.4055	Yes

Stated precisely, $d(P_a, P_b) = |\ln P_a - \ln P_b|$ is a true metric on prices, and the log return is a signed distance in that metric. The distance is non-negative, zero only when $P_a = P_b$, and symmetric. It also adds up along a path that moves in one direction. Each property follows from $|\cdot|$ on $\mathbb{R}$, because ln is injective. The magnitude of the simple return has neither of the last two properties. First, it fails symmetry. A $+x$ move followed by a $-x$ move ends at $(1+x)(1-x) = 1-x^2$, not at 1. So $+10\%$ then -10% gives -1%, and $+50\%$ then -50% gives -25%. The $-x^2$ is the same cross-term as before. Second, it fails to add along a path. The path $100 \to 200 \to 400$ is $+100\%$ and then $+100\%$, but $+300\%$ end to end. In logs it is $0.693 + 0.693 = 1.386$ exactly.

This is why oscillator-style features built from simple returns carry a small negative bias, and their log-return counterparts do not. It also explains a common error in retail arithmetic: the belief that a stock down 50% needs to rise 50% to recover. It needs $+100\%$, which is $+0.693$ in logs, the exact negative of the -0.693 it lost.

Count 4: regime comparability is unaffected. Neither convention makes a 2020 return comparable to a 1995 return. The volatility regime breaks that comparison. The transform does not. Daily S&P 500 volatility ranged from roughly 0.35% in mid-2017 to roughly 5% in March 2020, a factor of 14. No choice of return convention changes that. §5.4 describes a method that does.

5.3 Lateral comparability: log returns do not win, with two exceptions

Log returns are not more comparable across assets at short horizons, and the usual argument for them is wrong. Both conventions are already scale-free. The naive argument that "logs put everything on the same scale" therefore fails, because simple returns already do that.

The claim holds in two places and fails in a third.

Where it is true: long-horizon cross-sections. The skewness table in §3.3 shows that the log transform removes a shape distortion, and the size of the distortion depends on the horizon log volatility $s\sqrt{h}$. The distortion therefore grows with both the asset's volatility and the horizon. Consider a cross-section that contains a utility with 15% volatility and a crypto asset with 120% volatility. These are annualised log volatilities, so at h = one year the two assets have $s\sqrt{h} = 0.15$ and 1.20. The §3.3 formulae then give simple-return distributions with induced skewness of 0.46 and 11.2, before any real economic skew. The induced excess kurtosis is 0.37 and 515. A regression of annual simple returns on characteristics across this cross-section has residuals whose shape varies across observations by a factor of 24 in skewness and 1,400 in kurtosis. Least squares behaves badly in exactly that setting. In log coordinates the shapes are far closer. [Practice] **Recommendation: use log returns for long-horizon cross-sectional regressions.** The practice rests on the arithmetic above.

Where it is true: exact multiplicative decompositions across a cross-section. For a holder of foreign assets, the USD return of asset i decomposes as

$$r_i^{\text{USD}} \;=\; r_i^{\text{loc}} \;+\; r^{\text{fx}} \qquad \text{exactly, for every } i$$

so the currency effect is a *common additive constant* across the whole cross-section. In simple returns the decomposition is $R_i^{\text{loc}} + R^{\text{fx}} + R_i^{\text{loc}} R^{\text{fx}}$, with an interaction term that differs by asset. [Fact] For decomposing a global book into local and currency contributions, or for hedging a currency overlay, log coordinates make the decomposition exact and simple coordinates do not. The same structure applies to inflation, fees, and any chain of gross returns.

Where it is false: leverage, income, and portfolio aggregation. This repeats §3.2, because readers commonly over-correct here. A leveraged product's return is exactly L times the underlying's *simple* return. Dividends add to the *simple* price return. The portfolio's *simple* return is the weighted average of its constituents' simple returns. Moving any of these into logs replaces an identity with an approximation.

5.4 What actually makes returns comparable, and the numbers that settle it

The effects above are second-order. The first-order effect is volatility, and neither convention addresses it.

The table shows what a $+3\%$ daily move means across a realistic cross-section.

Asset type	Typical daily $\hat{\sigma}$	A $+3\%$ day, in $\hat{\sigma}$ units
Short-duration Treasury ETF	0.10%	30.0
Utility sector ETF	1.0%	3.0
S&P 500 index	1.1%	2.7
Mega-cap technology single name	2.0%	1.5
Bitcoin	3.5%	0.86
Small-cap biotech	5.0%	0.60

The same number ranges in meaning from "unremarkable" to "off the historical record," a spread of a factor of **50** in $\hat{\sigma}$ units. Compare this with the effect of the convention on the same $+3\%$. Here $\ln(1.03) = 0.029559$, a difference of 4.4 basis points, or **1.5% of the value**.

> Volatility normalisation changes the meaning of a return by up to a factor of 50. The choice between log and simple returns changes its value by 1.5%. The former is roughly three orders of magnitude more important, and it is available in either convention.

This calibration is the most useful one in the chapter. It should set how much weight a reader gives the rest of the chapter. Consider a choice between two options:

1. simple returns, volatility-normalised, or
2. log returns, raw.

Option 1 wins by a wide margin in almost every cross-sectional application. The best choice is log returns *and* volatility normalisation, but the normalisation provides nearly all the benefit.

The standard forms are:

$$z_{i,t} \;=\; \frac{r_{i,t}}{\widehat{\sigma}_{i,t-1}}, \qquad u_{i,t} \;=\; \frac{\text{rank}_i\left(r_{i,t}\right) - \frac{1}{2}}{N_t}, \qquad \tilde{z}_{i,t} \;=\; \Phi^{-1}(u_{i,t})$$

The first is a time-series normalisation. It scales each asset by its own recent volatility, where $\hat{\sigma}$ is typically an EWMA or a rolling standard deviation over a 20–60 day window. The second is a cross-sectional rank transform. $\text{rank}_i(\cdot)$ is asset i's rank, from 1 to N_t, among the N_t assets in the cross-section at time t. Subtracting $\frac{1}{2}$ centres the resulting $u_{i,t}$ inside $(0, 1)$, so the third expression, the Gaussianised rank, never reaches $\pm\infty$. The rank transform is scale-free and immune to outliers. It is also **identical in both conventions**, because ranks are invariant under any strictly increasing map. §11 relies on this property.

If a pipeline ranks returns cross-sectionally at any point, everything upstream of the rank is convention-free. Much quantitative equity research has this structure. This is one reason cross-sectional equity practitioners treat the log-versus-simple debate as less urgent than derivatives or risk practitioners do.

§5 Key takeaways

1. "Comparable" splits into three senses: scale invariance (a tie), horizon invariance (log returns win decisively), and distributional comparability (neither convention wins, but volatility normalisation does).
2. Longitudinally, log returns win on exact horizon conversion, distributional consistency across horizons, and symmetry. The log-price distance is a metric on prices. The magnitude of the simple return is not.
3. A $+x$ move followed by a $-x$ move loses x^2. This is the cross-term from §2.2, and it is the arithmetic behind "down 50% needs up 100%."
4. Laterally, log returns do *not* win at short horizons, because both conventions are already scale-free. They win at long horizons, where the induced skew of simple returns varies enormously across a heterogeneous cross-section.
5. The exact additive decomposition of currency, inflation, and fee effects across a whole cross-section is a real lateral advantage of logs.
6. The same $+3\%$ move ranges from 0.6 to 30 standard deviations across a realistic cross-section, while the convention moves the number by 1.5%. Normalise by volatility before arguing about the transform.
7. Any pipeline that ranks returns cross-sectionally is convention-free upstream of the rank.

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6. The time-series econometrics of returns

In time-series econometrics the log transform is a requirement. Each subsection names a standard method and states exactly what breaks when the method receives simple returns.

6.1 Stationarity, unit roots, and the thing you should difference

Time-series econometrics usually assumes that the modelled series is covariance-stationary. Its mean and variance are constant, and its autocovariances depend only on the lag. Prices are not stationary, so analysts must transform them.

Differencing the price does not work. If the price evolves multiplicatively, $P_t = P_{t-1}e^{r_t}$, then

$$\Delta P_t \;=\; P_{t-1}(e^{r_t} - 1) \qquad \Longrightarrow \qquad \mathsf{Var}(\Delta P_t \mid P_{t-1}) \;=\; P_{t-1}^2\,\mathsf{Var}(e^{r_t})$$

The differenced price has a variance proportional to the square of the price level. In the 1980s the S&P 500 traded near 150, and today it trades near 6,000. A series of dollar changes over that period has a conditional variance 1,600 times larger at the end than at the start. That series is not stationary. Further differencing cannot fix it, because the problem is in the second moment, not the first.

Differencing the log price does work, for the reason developed in §2.5. The function ln is the variance-stabilising transform for a process whose volatility is proportional to its level. The canonical decomposition is therefore

$$p_t = \ln P_t \;\sim\; I(1), \qquad\qquad r_t = \Delta p_t \;\sim\; I(0)$$

Analysts run every unit-root and cointegration procedure (ADF, Phillips–Perron, KPSS, Johansen) on p_t, never on P_t. [Fact] An ADF test on raw prices is invalid. The derivation of the asymptotic distribution of its test statistic assumes homoskedastic innovations. Heteroskedasticity proportional to the level invalidates the critical values.

The cost of differencing, and fractional differencing as the compromise. A full first difference achieves stationarity by discarding the series' information about its own level. The return r_t carries no information about the level of p_t. Recovering the level would take an unbounded window of past returns. In forecasting, this loss is costly. Level-dependent quantities carry signal that the differenced series no longer holds. Examples are the log price relative to a moving average, the distance from a 52-week high, and the level of a valuation ratio. Fractional differencing is the principled compromise (Granger & Joyeux, 1980; Hosking, 1981). López de Prado (2018) applied it to finance:

$$\Delta^d p_t \;=\; \sum_{k=0}^{\infty} \binom{d}{k} (-1)^k \, p_{t-k}, \qquad 0 < d < 1$$

Here $\binom{d}{k} = \frac{d(d-1)\cdots(d-k+1)}{k!}$ is the generalised binomial coefficient. The expansion is therefore the formal series for $(1-L)^d$ in the lag operator. Written out, $\Delta^d p_t = p_t - d\,p_{t-1} - \frac{d(1-d)}{2} p_{t-2} - \cdots$. Because d is not an integer, the series never terminates. Its weights decay hyperbolically, like $k^{-(1+d)}$. At $d = 1$ the weights instead stop after one term. The slow decay is the retained memory. In practice, analysts truncate the sum once the weights fall below a tolerance. They set d to the smallest value that passes an ADF test. The result is stationary but keeps long memory. Feature engineering must often choose whether a feature is the price or the return. [Practice] The best choice is some fractional difference of the log price, and $d = 1$ is rarely optimal. This practice has a solid theoretical basis and modest published evidence.

6.2 Volatility models

GARCH. The standard specification (Engle, 1982; Bollerslev, 1986) uses log returns:

$$r_t = m + \varepsilon_t, \qquad \varepsilon_t = \sigma_t z_t, \quad z_t \sim \text{iid}(0,1), \qquad \sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

At daily frequency, estimating the model on simple returns gives parameter estimates that agree to two or three decimal places. Logs matter for **aggregating the forecast**, not for estimation. The multi-period variance forecast is

$$\text{Var}_t\left(r_{t+1:t+h}\right) \;=\; \sum_{j=1}^{h} \mathbb{E}_t[\sigma_{t+j}^2]$$

The sum is exact because log returns are additive and ε_t is a martingale difference. The cross-covariances therefore vanish. The corresponding sum in simple returns is off by the compounding cross-terms. In production, a GARCH model usually produces a horizon-h risk number, so the difference matters.

The sum has a useful closed form. Let $\lambda = \alpha + \beta$ be the persistence and $\bar{\sigma}^2 = \omega/(1-\lambda)$ the long-run variance. Take $\mathbb{E}_t$ of the recursion and use $\mathbb{E}_t[\varepsilon_{t+j-1}^2] = \mathbb{E}_t[\sigma_{t+j-1}^2]$. This gives $\mathbb{E}_t[\sigma_{t+j}^2] = \omega + \lambda\,\mathbb{E}_t[\sigma_{t+j-1}^2]$. Iterating from the $\mathcal{F}_t$-measurable σ_{t+1}^2 gives

$$\mathbb{E}_t[\sigma_{t+j}^2] \;=\; \bar{\sigma}^2 + \lambda^{j-1}(\sigma_{t+1}^2 - \bar{\sigma}^2) \qquad \Longrightarrow \qquad \sum_{j=1}^{h} \mathbb{E}_t[\sigma_{t+j}^2] \;=\; h\bar{\sigma}^2 \;+\; \frac{1-\lambda^h}{1-\lambda}(\sigma_{t+1}^2 - \bar{\sigma}^2)$$

The per-period forecast decays geometrically to $\bar{\sigma}^2$. The h-period forecast is h periods of long-run variance plus a shock term that *saturates* at $(\sigma_{t+1}^2 - \bar{\sigma}^2)/(1-\lambda)$. Current conditions therefore shift the long-horizon variance forecast by a bounded amount. They do not change its slope in h. For daily equity data, λ is typically 0.94–0.99. The shock half-life $\ln 2/\ln(1/\lambda)$ is then about 11 trading days at the low end and 69 at the high end, or two weeks to three months.

EGARCH and the second logarithm. Nelson (1991) models $\ln \sigma_t^2$ instead of σ_t^2. The motivation is the same as for prices. The variance is positive and evolves multiplicatively. The log therefore makes the parameter space unconstrained and the innovations closer to symmetric. Most logs in financial models have this explanation. To interpret a log in a model, identify which positive, multiplicatively evolving quantity it transforms.

Realized variance, where log returns are not optional. Sum the squared intraday log returns over a day with M intervals:

$$\mathrm{RV}_t \;=\; \sum_{j=1}^{M} r_{t,j}^2 \quad \xrightarrow[M\to\infty]{} \quad [\,p\,]_t \;=\; \int_{t-1}^{t} \sigma_u^2 \, du$$

[Fact] This limit is the **quadratic variation of the log-price semimartingale**. The convergence result (Andersen, Bollerslev, Diebold & Labys, 2001, 2003; Barndorff-Nielsen & Shephard, 2002) applies to the log price. The quadratic variation of the *price*, from summed squared dollar changes, has a different limit, $[\,P\,]_t = \int_{t-1}^{t} \sigma_u^2 P_u^2 \, du$. That quantity is in squared dollars and scales with the price level. It is not a volatility. Analysts cannot compare it across assets or across time.

Squared *simple* returns are an intermediate case. They need an exact treatment, not grouping with dollar changes. Since $R_{t,j} = r_{t,j} + O(r_{t,j}^2)$, the discrepancy $\sum_j (R_{t,j}^2 - r_{t,j}^2) = \sum_j r_{t,j}^3 + \cdots$ is third order. It vanishes as $M \to \infty$. Summed squared simple returns therefore converge to the same integrated variance. At five-minute sampling on liquid equities, the two estimators agree to about five decimal places. The case for logs here is definitional, not numerical.

The definitional case is the first non-negotiable one in this chapter. **Integrated variance is, by definition, the quadratic variation of the log price.** No simple-return object has it as its quadratic variation. The nearest candidate, $[\,P\,]_t$, is a different quantity. The whole asymptotic theory applies to p. It includes the consistency result, the mixed-normal limit law, and the jump-robust and noise-robust refinements. Working in logs keeps an analysis inside that theory.

A third logarithm appears here. Andersen, Bollerslev, Diebold, and Labys (ABDL) found that $\ln \mathrm{RV}_t$ is approximately Gaussian. It is much better behaved for modelling than RV_t, which is heavily right-skewed. Analysts therefore usually fit volatility forecasting models, HAR and its descendants, to log realized variance.

6.3 Continuous time: Itô, and where the $\sigma^2/2$ comes from

Geometric Brownian motion states that the *instantaneous simple return* is a diffusion with constant coefficients:

$$\frac{dS_t}{S_t} \;=\; \mu \, dt \;+\; \sigma \, dW_t$$

Here W_t is a standard Brownian motion. This is the one place in the chapter where W is not wealth. σ is the diffusion coefficient, not a per-period return volatility. The notation block flags this deliberate overload. Applying Itô's lemma to $f(S) = \ln S$, with $f' = 1/S$ and $f'' = -1/S^2$, gives

$$d \ln S_t \;=\; \frac{1}{S_t} dS_t \;-\; \frac{1}{2}\frac{1}{S_t^2} \, d\langle S \rangle_t \;=\; \Big(\mu - \frac{\sigma^2}{2}\Big) dt \;+\; \sigma \, dW_t$$

using $d\langle S \rangle_t = \sigma^2 S_t^2 \, dt$. Over $[0, T]$, therefore,

$$\ln \frac{S_T}{S_0} \;\sim\; \mathcal{N}\Big((\mu - \tfrac{\sigma^2}{2})T, \;\; \sigma^2 T \Big)$$

This result is the cleanest statement of the chapter's central theme. **In continuous time, the expected simple return per unit time is μ, and the expected log return per unit time is $\mu - \sigma^2/2$. The relation is exact.** The discrete-time $\sigma^2/2$ corrections in §3 are approximate versions of this identity. The Itô correction term *is* volatility drag, written in different notation.

Two further consequences follow:

- **Quadratic variation is level-free in logs.** $d\langle \ln S \rangle_t = \sigma^2 \, dt$ is a constant, while $d\langle S \rangle_t = \sigma^2 S_t^2 \, dt$ grows with the price. This restates the variance stabilisation of §2.5 in continuous time.
- **The risk-neutral drift of the log price is $r_f - \sigma^2/2$, not r_f.** Under the pricing measure $\mathbb{Q}$ the discounted price is a martingale, so $\mathbb{E}^{\mathbb{Q}}[S_T] = S_0 e^{r_f T}$. This is a statement about the *simple* return. The identity $\mathbb{E}[e^X] = e^{\mathbb{E}X + \mathrm{Var}(X)/2}$, applied to $X = \ln S_T$ with $\mathrm{Var}^{\mathbb{Q}}(X) = \sigma^2 T$, translates this statement to the log price: $\mathbb{E}^{\mathbb{Q}}[\ln S_T] = \ln S_0 + (r_f - \frac{1}{2}\sigma^2)T$. Setting the log drift to r_f in a Monte Carlo simulation is the most common derivatives-pricing bug. [Fact] The error has an exact size. If $\ln S_T \sim \mathcal{N}(\ln S_0 + r_f T, \, \sigma^2 T)$, the simulation gives $\mathbb{E}^{\mathbb{Q}}[S_T] = S_0 e^{r_f T} \cdot e^{\sigma^2 T/2}$. The simulated forward is too high by a factor of exactly $e^{\sigma^2 T/2}$. That is 2% for a one-year contract at 20% volatility. Every call price computed from it inherits an upward bias.

6.4 Cointegration: a ratio hypothesis versus a spread hypothesis

Two integrated series are **cointegrated** if some linear combination of them is stationary (Engle & Granger, 1987). For a pair of assets, the choice of coordinate changes the economic hypothesis under test. Practitioners do not widely appreciate this point. The table sets out the two hypotheses.

Coordinate	Stationary object	Economic claim	Corresponding position
Log prices	$\ln P_A - \beta \ln P_B$	The *ratio* P_A/P_B^{β} mean-reverts	Constant dollar ratio β, continuously rebalanced
Levels	$P_A - \beta P_B$	The dollar *spread* mean-reverts	Fixed share ratio, never rebalanced

Neither hypothesis implies the other. The right one depends on how the trader will trade the pair. If the hedge is "hold β shares of B for each share of A and never adjust," the level formulation matches it. If the hedge is "keep the dollar exposure ratio at β," the log formulation matches it. Most statistical-arbitrage implementations rebalance, so they should test the log relationship. [Contested] Much published pairs-trading work tests the level relationship and then trades the rebalanced version, which is a specification error, not a stylistic one. Practitioners disagree about how much the error matters empirically.

The log formulation has two further advantages, and both are decisive in practice. First, it survives a change of currency, and the level formulation does not. Quoting both legs in a new currency multiplies both prices by the same *time-varying* rate X_t. In logs at $\beta = 1$, the two $\ln X_t$ terms cancel. Then $\ln P_A - \ln P_B$ is the same series in any currency. Away from $\beta = 1$ a term $(1-\beta)\ln X_t$ survives, so the analyst must re-specify the relation. In levels the spread becomes $X_t(P_A - \beta P_B)$, a stationary spread multiplied by a unit-root process. That product is not stationary for any choice of β, so re-estimation cannot fix it. A constant change of *units*, such as pence to pounds or a share split, is harmless in both coordinates. The stochastic rate causes the damage. Second, the cointegrating relationships that hold up in economics are proportional relationships. Examples are purchasing power parity, money demand, the consumption–wealth ratio, and the term structure. Proportional relationships are linear in logs and non-linear in levels. Applied macroeconometrics therefore works almost entirely in logs.

6.5 The log-linear present-value identity

The return identity

$$1 + R_{t+1} \;=\; \frac{P_{t+1} + D_{t+1}}{P_t}$$

is exact but of little use for analysis. It is non-linear, so it does not iterate forward into a statement about long-run expectations. Campbell & Shiller (1988) solved this by log-linearising around the mean log dividend–price ratio. With lowercase letters for logs, a first-order expansion gives

$$r_{t+1} \;\approx\; \kappa + \rho\, p_{t+1} + (1-\rho)\, d_{t+1} - p_t, \qquad \rho \;=\; \frac{1}{1 + \exp\left(\overline{d-p}\right)}$$

Here $\overline{d-p}$ is the sample mean of the log dividend–price ratio. So $\exp(\overline{d-p})$ is the average dividend yield, and $\rho = 1/(1+\text{yield})$ is a discount factor slightly below one. κ is the constant that the expansion leaves. For US equities with an average dividend yield near 4%, $\rho = 1/1.04 \approx 0.96$ at annual frequency. Because the approximation is *linear*, iterating it forward solves it. The result is the identity that organises the return-predictability literature:

$$d_t - p_t \;\approx\; \text{const} \;+\; \mathbb{E}_t \sum_{j=0}^{\infty} \rho^j \Big(r_{t+1+j} \;-\; \Delta d_{t+1+j} \Big)$$

A high dividend yield must forecast either high future returns or low future dividend growth. No third possibility exists. The statement is an accounting identity, log-linearised, plus a transversality condition that rules out a bubble term $\lim_{j\to\infty} \rho^j (d_{t+j} - p_{t+j})$. It is not a model. Nothing in it is estimated. The question of whether the dividend yield is a valid predictor therefore differs in kind from other predictor questions. Campbell (1991) extends the same method to decompose realised return variance into cash-flow news and discount-rate news.

None of this exists in simple returns. The log-linearisation makes the present-value relation iterable, and it has no simple-return counterpart. This is the second non-negotiable case.

6.6 Long-horizon predictive regressions

The standard predictability regression is

$$r_{t+1:t+h} \;=\; a \;+\; b\,x_t \;+\; \varepsilon_{t+1:t+h}$$

with $r_{t+1:t+h} = \sum_{j=1}^{h} r_{t+j}$, and x_t a predictor such as the dividend yield or a valuation ratio. The left-hand side is a *sum*. That is the only reason the standard inference methods apply. These methods are the Hansen–Hodrick and Newey–West corrections for the MA($h-1$) error structure that overlapping observations create. If a compounded simple return replaces the sum, the autocovariance structure of the error is no longer the tractable moving average that the corrections assume.

The variance ratio (Lo & MacKinlay, 1988) rests on the same additivity:

$$\mathrm{VR}(q) \;=\; \frac{\mathrm{Var}\left(r_{t+1:t+q}\right)}{q\;\mathrm{Var}(r_{t+1})} \;=\; 1 + 2\sum_{k=1}^{q-1}\Big(1-\frac{k}{q}\Big)\rho_k$$

Here ρ_k is the lag-k autocorrelation of the one-period log return, and γ_k below is the lag-k autocovariance. The second equality follows from the additivity of log returns and the definition of autocovariance. The variance of a sum of q terms expands to $q\gamma_0 + 2\sum_{k=1}^{q-1}(q-k)\gamma_k$. Dividing by $q\gamma_0$ produces the triangular Bartlett weights $1-k/q$. No simple-return variance ratio exists. $\mathrm{Var}(R_{t+1:t+q})$ is the variance of a product, and it has no expansion in the ρ_k alone.

Two cautions apply to anyone running these regressions. Neither concerns the convention. Overlapping observations inflate apparent t-statistics severely. A persistent predictor whose innovations correlate with the return innovation produces the Stambaugh bias in $\widehat{b}$. Long-horizon predictability results have a poor replication record, and the log transform does not improve it.

§6 Key takeaways

1. Difference the log price, never the price. The differenced price has a variance that scales with the level. This invalidates every unit-root critical value.
2. Full differencing discards the series' information about its own level. Fractional differencing with the smallest d that achieves stationarity is the principled compromise, and $d = 1$ is rarely optimal for forecasting.
3. GARCH parameter estimates barely depend on the convention. *Aggregating* a GARCH forecast to horizon h requires additivity, and therefore log returns.
4. Realized variance is, by definition, the quadratic variation of the log price, and the asymptotic theory applies to p. Summed squared simple returns also converge to the same limit. Summed squared dollar changes converge to a level-dependent quantity that is not a volatility. The case for logs here is definitional and not negotiable.
5. The Itô correction $-\sigma^2/2$ in $d\ln S$ *is* volatility drag. In continuous time, the drifts of the two conventions differ by exactly $\sigma^2/2$.
6. Cointegration in logs tests a ratio hypothesis and corresponds to a rebalanced position. Cointegration in levels tests a spread hypothesis and corresponds to a fixed-share position. Test the hypothesis that matches the intended trade.
7. The Campbell–Shiller decomposition, and with it the present-value approach to return predictability, exists only in logs. So does the variance ratio.

Part III — Practice

7. Where the choice is forced

The two tables in this section put the results of Parts I and II to practical use. Each row names an application, states the convention it requires, and says what breaks under the other convention.

7.1 Applications that require log returns

The first table lists the applications that require log returns.

Application	Why	Breaks how
Continuous-time models (SDEs, diffusions)	Itô calculus applies to ln S, and the drift is $\mu - \sigma^2/2$	Nothing in simple-return space has both constant SDE coefficients and positive prices. Additive innovations in P go negative, and $dS = \mu S\,dt + \sigma S\,dW$ has state-dependent coefficients
Option pricing	Every formula is a function of $\ln(S/K)$ and $\sigma\sqrt{T}$	Setting the log drift to r_f mis-prices by a factor $e^{\sigma^2 T/2}$ in the forward
Realized variance	$\sum_j r_j^2$ estimates the quadratic variation of ln P	Squared *price changes* converge to $\int \sigma_u^2 P_u^2\,du$, a level-dependent quantity. Intraday simple returns do not sum to the day's return, so $\sum_j R_j^2$ estimates the variability of no quantity that aggregates
Cointegration, VECM	Economic equilibria are proportional relationships	A level cointegration test asks a different question and is not currency-invariant
Campbell–Shiller decomposition	Log-linearisation is what makes the present-value relation iterable	No simple-return counterpart exists
Long-horizon predictive regressions	The LHS must be an additive sum for the standard error corrections to apply	Overlapping-return error structure is no longer MA($h-1$)
Variance ratios	VR(q) expands in the autocorrelations only because returns add	No expansion exists in simple returns
Multi-horizon price simulation	Additive Gaussian innovations in ln P keep prices positive	Additive innovations in P produce negative prices
Growth-optimal (Kelly) sizing	The objective is $\mathbb{E}[\ln W_T]$, which is additive over periods	The multi-period problem stops being separable
Long-horizon VaR / stress	Losses map back through $e^r - 1 > -1$	A Gaussian simple-return model assigns mass to losses beyond -100%
Currency, inflation and fee decomposition	Chains of gross returns are additive in logs	Interaction terms appear per asset and do not aggregate
Term-structure work	Continuously compounded yields make forward rates additive. Here $y_{0,T}T = \sum_k f_k \Delta_k$, for forward rates f_k over sub-intervals of length Δ_k	Discretely compounded forwards chain multiplicatively. The yield is built from a product of $(1+f_k)^{\Delta_k}$, not a sum

The Kelly case needs more detail. Here the log appears for a reason other than the convenience of additivity or variance stabilisation.

An investor who maximises expected terminal log wealth over T periods faces the problem

$$\max\ \mathbb{E}\big[\ln W_T\big] \;=\; \ln W_0 \;+\; \sum_{t=1}^{T} \mathbb{E}\big[\ln(1 + f_t R_t)\big]$$

where f_t is the fraction of wealth at risk. Because the log turns the product into a sum, the multi-period problem **decomposes into T independent one-period problems**. The optimal f_t depends only on the current period's return distribution. The investor is *myopic*, so no dynamic programming is needed. This separability is specific to log utility. It explains structurally why the growth-optimal literature uses logs. The reason is not a stylistic preference.

For a small edge, the solution is

$$f^* \approx \frac{\mu}{\sigma^2}, \qquad g^* = \frac{\mu^2}{2\sigma^2} = \frac{\mathrm{SR}^2}{2}$$

where μ and σ are the moments of the *excess* return, so that $\mathrm{SR} = \mu/\sigma$. Both results come from the expansion $\mathbb{E}[\ln(1 + fR)] \approx f\mu - \frac{1}{2}f^2\sigma^2$. This is a downward parabola in f, with its peak at $f^* = \mu/\sigma^2$ and a height of $g^* = \mu^2/(2\sigma^2)$.

The second formula deserves more attention. **A strategy with an annual Sharpe ratio of 1.0, run at full Kelly, compounds at 50% per year in excess of the risk-free rate.** The 50% is a log growth rate. It corresponds to a wealth multiple of $e^{0.5} = 1.65\times$ a year. This number surprises most readers, so state it together with the leverage it requires. For $\mu = 10\%$ and $\sigma = 10\%$, the same Sharpe-1.0 strategy requires $f^* = 0.10/0.01 = 10\times$. Full Kelly is not a practical prescription. It is the *upper* end of a trade-off between growth and drawdown. Its drawdowns are severe. Under full Kelly, the probability of halving wealth at some point is 50%. [Practice] Practitioners run fractional Kelly, typically a quarter to a half. Growth is quadratic near the peak, so half-Kelly keeps 75% of g^* while cutting the variance substantially.

The contested part. [Contested] Kelly (1956) and Latané (1959) proposed maximising expected log wealth as a general criterion. They argued that a log-optimal strategy almost surely outgrows any other strategy in the long run. Samuelson attacked the proposal repeatedly. The most memorable attack was a 1979 note written in words of one syllable, except the last. He argued that "almost surely ends up richer" does not imply "preferred," because the limiting probability of dominance does not determine expected utility. Merton & Samuelson (1974) formalised the objection. Markowitz (1976) defended the criterion's practical relevance, and Thorp has made the applied case for decades.

This chapter takes the view that Samuelson is right on the mathematics and Kelly's proponents are right about practice. Maximising $\mathbb{E}[\ln W]$ is optimal if and only if utility is logarithmic. No theorem makes log utility mandatory. As an *engineering* heuristic, however, fractional Kelly is hard to beat for sizing a repeated bet over a long horizon when ruin is unacceptable. Its failure modes are well understood. **Recommendation: use Kelly as a ceiling on leverage, not as a target.**

7.2 Applications that require simple returns

The second table lists the applications that require simple returns.

Application	Why	Breaks how
Portfolio construction / mean-variance	The objective $\mathbf{w}^\top\mu - \frac{\gamma}{2}\mathbf{w}^\top\Sigma\mathbf{w}$, with γ the risk-aversion coefficient, is linear and quadratic in simple returns	The log-return objective is $\ln\big(\sum_i w_i e^{r_i}\big)$, a log-sum-exp. The problem stops being a QP
Factor models, CAPM, Fama–MacBeth	The dependent variable is a portfolio return, which must aggregate linearly	Intercepts absorb a $-\frac{1}{2}(\sigma_i^2 - \beta_i\sigma_M^2)$ bias that grows with idiosyncratic volatility (§9.3)
Performance attribution	Contributions must sum to the total	Log contributions do not sum to the log total. The residual is the diversification return
Index construction	An index is a portfolio	Same as portfolio construction
Dollar P&L, position sizing, margin	P&L is notional $\times\, R$	P&L is not notional $\times\, r$
Transaction costs	Costs are proportional to notional traded, so they are linear in prices	A log-space cost is not a dollar cost
Leveraged / inverse products	The product's return is exactly $L\,R$ per period	$\ln(1 + LR) \neq L\ln(1 + R)$
Total-return construction	$R^{\mathrm{TR}} = R^{\mathrm{PR}} + D_t/P_{t-1}$ is additive in simple returns	Income and price components do not add in logs
Client and regulatory reporting	Convention and, in many jurisdictions, requirement	Reported figures will not reconcile to account statements

The unifying principle is short. **Anything that adds up across positions, and anything denominated in dollars, uses simple returns.** Anything that compounds through time uses log returns. So does anything that a model treats as a stochastic process.

7.3 The two-space discipline

Well-built quantitative systems do not pick one convention. They state explicitly which space each stage works in. Meucci (2010) frames this as a three-step pipeline. First, find the invariants. Second, project them to the horizon. Third, map back to simple returns for pricing and aggregation. The pipeline is the most useful organising idea for production code. The diagram shows its stages.



The **invariants** stage matters most. Model the transformation of the raw data that is closest to independent and identically distributed across time. Meucci calls it the **invariant**. For equity prices the invariant is the log return. For interest rates it is the *yield change*, not the log return of the yield, because rates mean-revert and can be negative. For options it is the change in implied volatility, or in log implied volatility. For credit it is the change in log spread.

[Practice] The general rule is **"find the invariant for the instrument, model it, and map back to simple returns before aggregating or reporting."** It is not "use log returns." The log return is the invariant for equity-like instruments, so it dominates the discussion. The principle generalises to other instruments.

Bugs occur mostly at the two boundary crossings. On the way into invariant space, check the domain, which excludes zero and negative prices. On the way out, remember that $\mathbb{E}[e^r] \neq e^{\mathbb{E}[r]}$. This is the retransformation problem. §11.5 treats it in detail, because it is the most damaging error in machine-learning pipelines.

§7 Key takeaways

1. Log returns are required wherever a continuous-time model, a quadratic variation, an iterated present-value identity, or an additive multi-period sum appears. For the present-value identity and the additive sum, the alternative has no formulation at all. For the others, it exists but sits outside the theory that supplies the estimators, standard errors, and bias corrections.
2. Simple returns are required wherever a calculation aggregates positions or counts dollars. Portfolio optimisation, attribution, P&L, costs, leverage, and reporting are all simple-return domains.
3. Log utility makes the multi-period portfolio problem separable into independent one-period problems. That structural fact, not convenience, is why growth-optimal theory works in logs.
4. $g^* = \mathrm{SR}^2/2$: a Sharpe-1.0 strategy at full Kelly compounds at 50% a year, and needs roughly 10× leverage to do it. Run a fraction of Kelly. Half-Kelly keeps 75% of the growth at a quarter of the variance.
5. Economists contest whether "maximise expected log wealth" is a normative criterion. Samuelson's mathematical objection is correct, and fractional Kelly remains an excellent engineering heuristic. The two claims are distinct.
6. Build systems around three explicit stages (invariants, projection, pricing), and convert deliberately at the boundaries instead of picking one convention globally.
7. The general rule is "model the invariant," not "use logs." The invariant is the log return for equities, the yield change for rates, and the change in implied volatility for options.

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8. Where it does not matter, quantified

Most of the time, on most data, in most pipelines, the choice has no visible effect. A catalogue of differences alone would make a reader over-cautious. This section states precisely when the choice has no effect, so that the reader can focus on the questions that matter.

8.1 The divergence, tabulated

The table compares each simple return with its log return and gives the error of the approximation $r \approx R$.

R	$r = \ln(1+R)$	$r - R$	Relative error $\lvert r-R\rvert/\lvert R\rvert$
$+0.1\%$	$+0.0009995$	-0.005 bp	0.05%
$+1\%$	$+0.009950$	-0.50 bp	0.50%
$+3\%$	$+0.029559$	-4.4 bp	1.5%
$+5\%$	$+0.048790$	-12 bp	2.4%
$+10\%$	$+0.095310$	-47 bp	4.7%
$+20\%$	$+0.182322$	-1.77 pp	8.8%
$+50\%$	$+0.405465$	-9.45 pp	18.9%
$+100\%$	$+0.693147$	-30.7 pp	30.7%
-1%	-0.010050	-0.50 bp	0.50%
-10%	-0.105361	-54 bp	5.4%
-50%	-0.693147	-19.3 pp	38.6%
-90%	-2.302585	-140 pp	156%

Memorise this rule: **the relative error of the approximation $r \approx R$ is about $|R|/2$.** The approximation is accurate to within half a percent of itself for a 1% return, and to within 5% for a 10% return. For a 50% return the rule predicts 25%, but the true figure, 19%, is smaller, because the cubic term reduces the error.

8.2 The correlation between the two series

For a return series with volatility σ and an approximately symmetric distribution, the correlation between the simple and log versions is

$$\mathrm{Corr}(R, r) \;\approx\; \frac{1}{\sqrt{1+\sigma^2/2}} \;\approx\; 1 - \frac{\sigma^2}{4}$$

Derivation. Write $r \approx R - \tfrac{1}{2}R^2$. Take R symmetric about zero with variance σ^2, and Gaussian where the fourth moment is needed. Then $\mathrm{Cov}(R, r) = \sigma^2 - \tfrac{1}{2}\mathbb{E}[R^3] = \sigma^2$ by symmetry. Also $\mathrm{Var}(r) = \sigma^2 + \tfrac{1}{4}\,\mathrm{Var}(R^2) - \mathrm{Cov}(R, R^2) = \sigma^2(1 + \sigma^2/2)$, since the cross-term vanishes by the same symmetry and $\mathrm{Var}(R^2) = 2\sigma^4$ for a Gaussian. Dividing gives $\sigma^2 \big/ \sqrt{\sigma^2 \cdot \sigma^2(1+\sigma^2/2)}$.

The table evaluates the correlation at typical volatilities.

Frequency	Typical σ	$\mathrm{Corr}(R, r)$
Daily equity	1.3%	0.999958
Monthly equity	5.8%	0.99916
Annual equity	20%	0.990
Daily crypto	3.5%	0.99969
Annual, 60%-vol asset	60%	0.921

At daily frequency the correlation between the two series is 1 to five significant figures. **Any statistic that is a smooth function of the series agrees across conventions to a precision far below the sampling error of any achievable sample.** This covers an autocorrelation, a beta, an information coefficient, and a signal's rank correlation with forward returns. For scale, an information coefficient estimated on a decade of daily cross-sections has a standard error in the *second* decimal place. The convention moves it in the fifth decimal place, three orders of magnitude below the noise in the estimate. A rank-based statistic moves by exactly zero (§11.6).

8.3 The one daily-frequency statistic that does differ: the Sharpe ratio

The Sharpe ratio is an important exception. It differs between conventions even at daily frequency, where every other statistic agrees. Practitioners overlook this.

The annualised Sharpe ratio computed on simple returns exceeds the one computed on log returns by

$$\mathrm{SR}_{\text{simple}} - \mathrm{SR}_{\log} \approx \frac{(\mu - m)\sqrt{A}}{\sigma} = \frac{(\sigma^2/2)\sqrt{A}}{\sigma} = \frac{\sigma\sqrt{A}}{2} = \frac{\sigma_{\text{ann}}}{2}$$

[Fact] **Switching from simple to log returns lowers the reported Sharpe ratio by half the annualised volatility.** The numerator causes the whole difference. The denominators agree to order σ^2 (§3.3). The means differ by $\sigma^2/2$, and annualisation multiplies that difference by $\sqrt{A}/\sigma$. The table gives the size of the gap.

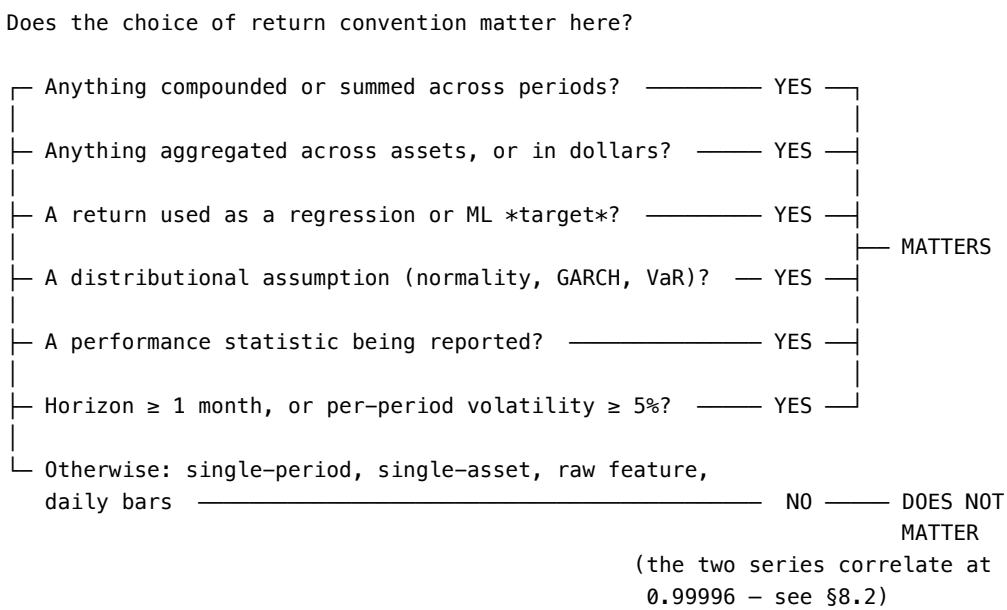
Strategy annualised volatility	Sharpe difference
5% (market-neutral, low gross)	0.025
10% (typical hedge-fund target)	0.05
20% (long-only equity)	0.10
60% (levered or crypto)	0.30
100%	0.50

For a 60%-volatility strategy, the gap is a third of a Sharpe unit, enough to separate a fundable track record from an unfundable one. The Sharpe ratio is the chapter's clearest case of a *reporting* statistic that needs a stated convention. The two numbers differ materially, and both computations are defensible.

Of the two, the log-return Sharpe is the more conservative and the more meaningful, because its numerator is the compound growth rate an investor actually experiences. The industry convention, however, is simple returns. An unlabelled log Sharpe makes a manager look worse than peers for no economic reason. [Practice] **Recommendation: report the simple-return Sharpe, state the convention, and remember that compound growth is $\sigma^2/2$ lower than the arithmetic mean suggests.**

8.4 The decision rule

The results of this section combine into a mechanical rule:



The last line covers a large share of day-to-day signal research on daily equity bars. Experienced practitioners therefore often dismiss this question. The dismissal is wrong as soon as the work moves into any of the six branches above.

§8 Key takeaways

1. The relative error of $r \approx R$ is about $|R|/2$. At 1% it is half a percent. At 50% it is a fifth.
2. $\mathrm{Corr}(R, r) \approx 1 - \sigma^2/4$. For daily equity data the correlation is 0.99996, so the two series are indistinguishable for any smooth statistic at any achievable sample size.
3. The Sharpe ratio is the exception at daily frequency. Its difference between conventions, $\mathrm{SR}_{\text{simple}} - \mathrm{SR}_{\log} \approx \sigma_{\text{ann}}/2$, is 0.10 for a long-only equity book and 0.30 for a 60%-volatility one.

4. Report the simple-return Sharpe because it is the convention, state the convention, and remember that the compound figure is lower.
5. Six triggers make the choice matter: compounding, cross-sectional aggregation, use as a target, distributional assumptions, performance reporting, and horizons past a month or volatilities past 5% per period. Outside them, the choice does not matter.

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9. Failure modes

Each entry below is a failure that occurs in production systems. Each gives the mechanism, the symptom a practitioner would observe, and the fix.

9.1 The domain error: zero, negative, and sign-crossing quantities

Mechanism. $\ln(x)$ requires $x > 0$. The log return $\ln(1 + R)$ therefore requires $R > -1$, that is, a strictly positive price. Several financial quantities routinely violate this requirement, as the table shows.

Quantity	Why it breaks
Futures on a physically-settled commodity	WTI settled at $-\$37.63$ on 20 April 2020
Calendar and inter-commodity spreads	Cross zero by construction
Interest rates in a negative-rate regime	EUR, JPY, CHF policy rates were negative for years
P&L series and account equity for a levered book	Can go to zero or below
Basis, carry, and any difference of two prices	Sign-crossing is the normal state
Delisted equity, bankruptcy	$R = -1$ exactly, so $r = -\infty$

Symptom. `RuntimeWarning: invalid value encountered in log`, followed by `NaN` values that propagate silently through every downstream rolling window. A worse case gives no warning at all. The library returns `-inf`, which corrupts a rolling mean for the length of the window and produces a plausible-looking but wrong feature.

Fix. For sign-crossing quantities, do not use returns at all. Model the *difference* of the level. Normalise it by a volatility estimate of that difference, not by the level. For delistings, use the exchange's delisting return where available (CRSP provides one). Otherwise cap the log return at a finite floor. A -100% simple return should become something like $\ln(0.01) = -4.6$, not $-\infty$. Record the chosen floor, because it affects the tail statistics.

9.2 `log` where you meant `log1p`

Mechanism. The code applies `np.log(R)` to a *return* series instead of a gross-return series. Roughly half of all returns are negative, so roughly half the output is `NaN`.

Symptom. This is the most common code-level bug in the area. It is hard to spot because gradient-boosting libraries accept `NaN` natively and treat it as a learnable branch. The model trains and produces reasonable-looking metrics. The bug silently turns the return feature into a *sign indicator*, because every negative return becomes the same value. Even the feature importance looks plausible.

Fix. Use `np.log1p(R)` for returns and `np.log(P).diff()` for prices. Also assert on feature matrices. A return-derived column whose fraction of `NaN` matches the fraction of negative returns indicates this bug.

The inverse error is applying `log1p` to a quantity already in log space. This error is less visible and worse, because it produces no `NaN` at all. It applies $\ln(1 + r) \approx r - r^2/2$ to an already-logged quantity. The resulting distortion is small and gives no warning. Column names should make this error impossible. If the repository convention is `ret(1,close)` for simple returns and `log1p(ret(1,close))` for log returns, the name records the convention, and a double application is visible in the column name.

9.3 Regressions run in the wrong space

Mechanism. The CAPM and every factor model use simple excess returns, because the left-hand side must be a portfolio return that aggregates linearly. If the same regression uses log excess returns, the intercept absorbs a Jensen term:

$$\alpha^{\log} \;\approx\; \alpha \;-\; \tfrac{1}{2}(\sigma_i^2 - \beta_i \sigma_M^2) \;=\; \alpha \;-\; \tfrac{1}{2}\Big(\underbrace{\beta_i(\beta_i - 1)\sigma_M^2}_{\text{small unless } \beta \text{ extreme}} \;+\; \underbrace{\sigma_{\varepsilon,i}^2}_{\text{idiosyncratic variance}} \Big)$$

Derivation. Take expectations through $m \approx \mu - \sigma^2/2$ (§3.3) for both the asset and the market. Use $r_f \approx R_f$, because the risk-free rate has no variance, and collect terms. The second equality uses $\sigma_i^2 = \beta_i^2 \sigma_M^2 + \sigma_{\varepsilon,i}^2$, which turns $\sigma_i^2 - \beta_i \sigma_M^2$ into $\beta_i(\beta_i - 1)\sigma_M^2 + \sigma_{\varepsilon,i}^2$.

Symptom. For $\beta = 1$ the bias is $-\frac{1}{2}\sigma_\varepsilon^2$, a mechanical negative alpha proportional to idiosyncratic variance. A stock with 50% annualised idiosyncratic volatility gets -12.5% per year of spurious alpha, and one with 15% gets -1.1%. [Fact] **The arithmetic alone produces a low-volatility anomaly of roughly 11 percentage points a year.**

Fix. Run asset-pricing regressions on simple excess returns. This is *not* a claim about the published idiosyncratic-volatility literature. Ang, Hodrick, Xing & Zhang and their successors use simple returns. This bias does not affect them. The warning applies to pipelines that "log everything" on the theory that logs are always better.

The same caution applies to any regression whose coefficient the analyst interprets in economic units. A beta estimated on log returns is a beta of log returns. It is not the hedge ratio for a dollar-neutral position.

9.4 Averaging log returns and calling it a return

Mechanism. $\bar{r}$ is a sound estimate of the mean log return, and $e^{\bar{r}} - 1 = g$ is the geometric mean return. But $\bar{r}$ itself is not a return in any economic sense. Reported as the average return, it understates that average by $\sigma^2/2$.

Symptom. In a performance report, the arithmetic "average daily return" times 252 does not match the equity curve. The gap is almost exactly the annualised variance divided by two.

Fix. Decide which question the report answers. The average earned per period is $\bar{R}$. The rate at which wealth compounded is $e^{\bar{r}} - 1$. Report both if the audience might care. They are different, and the difference is the volatility drag.

9.5 Aggregation errors in backtests

Mechanism. Two related errors occur:

- The backtest computes the portfolio return as $\sum_i w_i r_i$, the weighted average of constituent *log* returns, and treats it as the portfolio log return. This **understates** the portfolio log return by $\frac{1}{2}\,\mathrm{Var}_w(R)$ per period, where $\mathrm{Var}_w(R) = \sum_i w_i R_i^2 - \big(\sum_i w_i R_i\big)^2$ is the weighted cross-sectional dispersion of the constituent returns.
- The backtest computes the strategy's multi-period return as $\sum_t R_t$, the sum of *simple* returns. This drops the compounding cross-terms $\sum_{i<j} R_i R_j$, and so it **understates** the buy-and-hold return $\prod_t (1 + R_t) - 1$ by roughly $\frac{1}{2} h^2 \mu^2$.

Symptom. On a 50-name book with 2% daily cross-sectional dispersion, the first error is about 2 bp per day, or roughly 5% per year of systematically missing return. That is larger than most alpha. The second error grows with the square of the horizon: a decade of 10% annual returns compounds to 159% but sums to 100%.

Fix. Always aggregate across assets in **simple** space first, and then convert to logs to compound. Per period, $R_{p,t} = \sum_i w_{i,t} R_{i,t}$, then $r_{p,t} = \ln(1 + R_{p,t})$, and finally cumulative $= \exp(\sum_t r_{p,t}) - 1$. Never use the reverse order.

9.6 Data hygiene, which the log transform amplifies

Mechanism. A missed split adjustment turns a two-for-one split into a -50% return. In simple space that return is -0.5, and in log space it is -0.693. Squared for a variance estimate, it contributes 0.25 in simple space and 0.48 in log space. The log transform nearly doubles the damage that a single bad print does to a volatility estimate.

More generally, the concavity of the log **amplifies bad data on the downside and compresses bad data on the upside**. A spurious $10\times$ price print contributes 9.0 to a simple-return series and 2.3 to a log series. A spurious 99% drop contributes -0.99 and -4.6 respectively.

Symptom. Volatility estimates spike on specific historical dates. Data from a different vendor does not reproduce the spikes.

Fix. Clean the data before transforming it. Screen the *simple* return for implausible magnitudes, since a $\pm 50\%$ daily move in a large-cap stock is nearly always a data error. Cross-check against a second source, and only then take logs. Winsorise

in the modelling space. A symmetric clip in one space is asymmetric in the other. Clipping r at ± 0.5 allows simple returns in $[-39\%, +65\%]$, while clipping R at $\pm 50\%$ allows log returns in $[-0.69, +0.41]$.

9.7 Exponentiating a forecast

Mechanism. The analyst fits a model to $\mathbb{E}[r \mid x]$ but wants $\mathbb{E}[R \mid x]$. Applying $e^{\hat{r}} - 1$ gives the conditional *median*, not the conditional *mean*. It is biased low by $\hat{s}^2(x)/2$.

Symptom. Forecast returns sit systematically below realised returns, and the gap is largest for the most volatile names. The *ranking* is therefore distorted, not just the level.

Fix. §11.5 covers this in full. In brief, three options exist. First, add $\hat{s}^2(x)/2$ from a companion volatility model. Second, use the smearing estimator of Duan (1983). Third, and usually best, stay in log space and evaluate the model on the quantity it estimates.

9.8 Mixing conventions across a system boundary

Mechanism. The research code computes log returns, the execution system computes simple returns, the risk system computes log returns, and the reporting layer computes simple returns. No component is wrong, yet the numbers do not reconcile.

Symptom. A persistent, small, unexplained difference appears between backtest and live P&L and scales with volatility. This pattern is the classic symptom. A discrepancy of roughly $\sigma^2/2$ per period indicates a convention mismatch, not a slippage problem.

Fix. Name columns for their convention (`ret(...)` versus `log1p(ret(...))`), and make the conversion an explicit, named transformation in the pipeline instead of an implicit assumption in each module. A one-line reconciliation test catches the mismatch permanently. Compute the same period's return both ways in both systems, and assert that they agree after conversion.

§9 Key takeaways

1. Log returns require strictly positive prices. Spreads, negative rates, negative futures prices, and P&L series have no log return. Model their level differences instead.
2. `np.log` in place of `np.log1p` silently converts a return feature into a sign indicator, because gradient-boosting libraries absorb the `NaN`s without raising an error.
3. Factor regressions run on log returns manufacture a negative alpha proportional to idiosyncratic variance. Between a stock with 50% idiosyncratic volatility and one with 15%, the gap is about 11 percentage points a year.
4. Always aggregate in this order: sum across assets in simple space, then take logs, then sum across time. Reversing the order costs about the diversification return.
5. Concavity amplifies bad downside data and compresses bad upside data. Clean and screen in simple space, then transform.
6. A backtest and a live P&L that differ by roughly $\sigma^2/2$ per period indicate a convention mismatch across a system boundary, not slippage.

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10. Building intuition

This section teaches the reader to read a log return at a glance, without conversion.

10.1 The three anchors

Memorise these three values and interpolate from them:

$$\ln(1.01) = 0.00995 \qquad \ln 2 = 0.693 \qquad \ln 10 = 2.303$$

Everything else follows by arithmetic. $\ln 4 = 2\ln 2 = 1.386$. $\ln 100 = 2\ln 10 = 4.605$. $\ln 5 = \ln 10 - \ln 2 = 1.609$. $\ln 1.5 = \ln 3 - \ln 2 = 1.099 - 0.693 = 0.405$. This case alone needs a fourth number, $\ln 3 = 1.099$.

The table lists the reference points that matter for daily work.

Log return	Simple return	Reads as
0.01	$+1.005\%$	“one percent”
0.05	$+5.13\%$	“five percent”
0.10	$+10.5\%$	“ten percent”
0.25	$+28.4\%$	“a quarter”
0.405	$+50\%$	“half up”
0.693	$+100\%$	“a double”
1.0	$+172\%$	“an e-fold”
2.303	$+900\%$	“a ten-bagger”
-0.693	-50%	“a halving”
-2.303	-90%	“down 90%”

10.2 The mental model: log return counts doublings

A log return counts multiplications, in units of e. This is the most useful reframing. Dividing by 0.693 gives a count of *doublings*, which most readers find more natural:

$$\text{doublings} \;=\; \frac{r}{\ln 2} \;=\; \frac{r}{0.693}$$

A cumulative log return of $+2.08$ is exactly three doublings, an $8\times$ gain. A cumulative -1.386 is two halvings, leaving a quarter of the starting value. Because log returns add, summing them and dividing by 0.693 gives the number of doublings a position went through. Compounding calculations are therefore trivial in log space and awkward in simple space.

The classic *rule of 70* follows from this identity. Money that grows at $x\%$ per period doubles in $\ln 2 / \ln(1+x/100) \approx 69.3/x$ periods.

10.3 The symmetry drill

Log space is symmetric, and simple space is not. Working through the pairs below builds the intuition.

price path	simple returns	log returns	net
100 → 110 → 100	+10.0%, −9.09%	+0.0953, −0.0953	exactly 0
100 → 110 → 99	+10.0%, −10.00%	+0.0953, −0.1054	−1.0%
100 → 150 → 100	+50.0%, −33.33%	+0.4055, −0.4055	exactly 0
100 → 150 → 75	+50.0%, −50.00%	+0.4055, −0.6931	−25.0%
100 → 50 → 100	−50.0%, +100.0%	−0.6931, +0.6931	exactly 0

The pattern is general. **Equal-and-opposite simple returns always lose x^2, and equal-and-opposite log returns always return exactly to the start.** A whipsawing market with $\pm 2\%$ daily moves and no trend loses $0.02^2 = 4$ bp per round trip, which annualises to roughly 5%. This is the volatility drag of §3.4, derived a second way.

10.4 The drag calculator

One formula handles most mental arithmetic. **Compound growth equals the arithmetic mean minus half the variance.** Variance is volatility squared, and practitioners think in volatility, so the practical version is:

$$\text{drag} \;=\; \frac{\sigma^2}{2} \qquad\Longrightarrow\qquad \sigma = 20\% \Rightarrow 2\%, \quad \sigma = 40\% \Rightarrow 8\%, \quad \sigma = 60\% \Rightarrow 18\%$$

The relation is quadratic. Doubling volatility quadruples the drag. This one number answers several questions that would otherwise need a simulation. A 3× levered ETF does not track three times the index over a year. It lags by $\frac{1}{2}((3\sigma)^2 - 3\sigma^2) = 3\sigma^2$, which for a 20%-volatility index is 12 percentage points a year. A strategy with a 25% return and 50% volatility still beats one with a 10% return and 15% volatility. Compound growth is $25 - 12.5 = 12.5\%$ for the first and $10 - 1.1 = 8.9\%$ for the second. The margin is about a quarter of what the headline numbers suggest.

The levered-ETF figure is the most useful instance, so this paragraph verifies it. A daily-rebalanced $L\times$ fund has a simple return of LR_t per day, so its log drift is $L\mu - \frac{1}{2}L^2\sigma^2$, against the index's $\mu - \frac{1}{2}\sigma^2$. Multiplying the index's *compound* growth by L gives $L\mu - \frac{L}{2}\sigma^2$. [Fact] The shortfall is therefore $\frac{1}{2}(L^2 - L)\sigma^2$, which for $L = 3$ and $\sigma = 20\%$ is $\frac{1}{2}(9-3)(0.04) = 12\%$ per year.

10.5 Reading a chart

Two habits improve chart reading.

Plot prices on a log axis. On a log axis, equal vertical distances are equal *returns*. A trend line then has a constant growth rate, and its visual slope shows that rate directly. On a linear axis, a stock that went from 10 to 20 looks like a smaller move than one that went from 100 to 120. This reverses the true order of the two returns. A linear axis visually understates the moves in the early part of any long price history.

Read vertical distance as a log return. On a log-price chart, the distance from a low to a high is the cumulative log return of that leg, with no arithmetic required. Two legs of equal visual height are the same multiplicative move.

10.6 Three drills

Each drill takes a few minutes. Together they build the intuition.

1. **The reconciliation.** Take one asset's daily bars for a year. Compute (a) $\prod(1 + R_t) - 1$, (b) $\exp(\sum r_t) - 1$, (c) $252 \times \bar{R}$, and (d) $\exp(252\bar{r}) - 1$. Confirm that (a), (b), and (d) agree to machine precision, since over a full year of bars they are three expressions for one number. Then confirm that (c), the figure most performance reports quote, exceeds all three by roughly the annualised variance divided by two. This exercise shows the identity holding exactly and the approximation failing by the predicted amount. It teaches more than reading about it.
2. **The drag demonstration.** Simulate a series with a zero mean log return and 2% daily volatility. Its expected simple return is $+2$ bp per day, and its expected terminal wealth is above the starting value. Yet its median terminal wealth equals the starting value, and the probability of ending below the start is one half at every horizon. A positive expected return and zero expected growth coexist.
3. **The leverage curve.** For $\mu = 10\%$ and $\sigma = 20\%$, plot $L\mu - \frac{1}{2}L^2\sigma^2$ against L from 0 to 6. The peak is at $L = 2.5$, and $L = 5$ gives exactly the same growth as $L = 0$. The plot is the Kelly criterion, found by drawing a parabola.

§10 Key takeaways

1. Three anchors, $\ln(1.01) = 0.00995$, $\ln 2 = 0.693$, and $\ln 10 = 2.303$, combine with additivity to give every conversion needed for mental arithmetic.
2. A log return counts e-foldings. Dividing by 0.693 counts doublings. A decade's cumulative log return divided by 0.693 is the number of times the money doubled.
3. Equal-and-opposite *simple* returns lose x^2, while equal-and-opposite *log* returns return exactly to par. The x^2 is volatility drag, derived a second way.
4. Use $\sigma^2/2$ as a mental subtraction. Drag is quadratic in volatility, so doubling volatility quadruples it.
5. A $3\times$ daily-rebalanced fund lags three times the index's compound growth by $\frac{1}{2}(L^2 - L)\sigma^2$, which is 12 percentage points a year on a 20%-volatility index. The lag is arithmetic, not fees.
6. Plot prices on a log axis. A linear axis understates the early moves in any long price history.

Part IV — Machine learning, convention, synthesis

11. Returns as features and targets for GBT and DNN

This section stands alone. Its answers differ most from what practitioners expect. For tree models, the dominant consideration is a theorem that makes the question moot. For every other model, the dominant consideration is a bias that has nothing to do with features.

11.1 Four decisions, not one

The question "should a model use log returns?" contains four independent questions with different answers. The table lists them.

Decision	The question
(a) Raw features	What convention for a single-period return used directly as an input?
(b) Derived features	What convention for moving averages, volatility estimates, z-scores, and spreads?
(c) Target	What convention for the quantity being predicted?
(d) Objective	What loss function, and does it align with the economic decision?

The answer to (a) is usually "it does not matter." The answer to (b) is usually "log." The answer to (c) is "it depends on the economic objective, and the wrong choice introduces a bias that changes rankings." Decision (d) has the largest financial consequences, because it is the only one of the four that can set the model's objective against the decision the model feeds. The subsections take the four decisions in order.

11.2 Gradient-boosted trees: the invariance result

This is the central practical result of the section. It is stronger than most practitioners assume.

> **Proposition (monotone invariance of axis-aligned trees).** Let ϕ be strictly increasing on the range of a feature x_j. Consider a decision-tree learner with three properties. Its splits have the form $\mathbb{1}\{x_j \le \theta\}$. Its candidate thresholds come from the order statistics of x_j. Its split criterion sees a candidate only through which training rows fall on each side of it. That learner induces exactly the same partition of the training rows on $\{x_j\}$ as on $\{\phi(x_j)\}$. The fitted trees are therefore identical, up to the relabelling $\theta \mapsto \phi(\theta)$ of the printed thresholds. The predictions and the gain-based feature importances are identical too.

Proof sketch. A threshold split on x_j can produce only those partitions of the training rows that are prefixes of the sort order of x_j. At most $n-1$ such partitions exist, one per boundary between adjacent *distinct* values, and every threshold falls into one. A strictly increasing ϕ preserves the sort order and preserves which values are tied. Exactly the same partitions are therefore achievable, in the same order, with the thresholds relabelled as $\phi(\theta)$. The split criterion depends on the targets on each side of the candidate, not on the feature values themselves. The criterion is squared-error reduction, Gini, entropy, or the gradient-and-Hessian gain of a boosted tree. Every candidate therefore has the same gain in both coordinates, and the greedy search makes identical choices at every node. Because the trees are identical, the residuals passed to the next boosting round are identical too. The argument then repeats over the whole ensemble. ■

Since $R \mapsto \ln(1+R)$ is strictly increasing on $(-1, \infty)$, **a gradient-boosted tree cannot distinguish a simple-return feature from its log-return counterpart.** [Fact] The difference is zero. The splits, predictions, and importances are the same. Only the printed threshold values change.

The result survives the histogram-based implementations used in practice. XGBoost's `hist` method and LightGBM both bin features before searching, and both derive bin boundaries from order statistics. XGBoost uses a weighted quantile sketch, and LightGBM uses the sorted distinct values and their counts. Quantiles and value counts are defined by rank, and ϕ preserves rank exactly. Every training row therefore lands in the same bin in both coordinates, and the argument above holds bin by bin. It still holds when `max_bin` is small enough to force distinct values to share a bin, because rank also decides which values merge. Equal-**width** binning *would* break invariance. It cuts the observed range into intervals of equal size, so it depends on the values, not on their order. Neither library uses it by default. Monotone constraints also hold, since they constrain the sign of a fitted relationship and ϕ changes no signs. XGBoost's learned default direction for missing values also stays the same. The set of missing rows is the same in both coordinates. The gain comparison that picks the direction is one of the gains the proposition has already fixed.

Three caveats, all small and all real. First, the invariance is exact in exact arithmetic. In floating point, two values that are distinct in one coordinate can round together in the other. That breaks a tie and can move a split. Second, bin edges lie *between* observed values (LightGBM uses the midpoint of an adjacent pair), and ϕ does not preserve midpoints. An unseen observation that falls inside that gap can land on different sides of the same split in the two coordinates. The training partition does not change. The effect reaches only points within a bin width of a threshold. Third, $\phi = \ln(1+\cdot)$ is strictly increasing only on $(-1, \infty)$. A delisting return of exactly -1 leaves the domain and becomes `-inf` or `NaN`. That changes *which rows*

are missing, and therefore the default direction the tree learns. It is not a monotone transform of anything. It is the one route by which the log step really alters a tree model (§9.1, §12.3).

Where the invariance stops. The proposition concerns a *single feature entering a single split*. It does not extend to features that combine observations. The table shows the boundary.

Feature	Convention-invariant for a GBT?	Why
R_t versus r_t	**Yes**	Strictly monotone
Cumulative h-period return: $\prod(1+R_i)-1$ versus $\sum r_i$	**Yes**	$\prod_i(1+R_i)-1 = e^{\sum_i r_i}-1$: each is a strictly increasing function of the other
Mean return over a window: $\frac{1}{h}\sum R_i$ versus $\frac{1}{h}\sum r_i$	**No**	They differ by about half the mean squared return in the window, which varies window to window
EWMA of returns	**No**	Same reason
Rolling standard deviation	**No**	Averaging of squares
Z-score $r/\hat{\sigma}$	**No**	A ratio of two non-invariant quantities
Difference of two returns, $r_t - r_{t-k}$	**No**	Differences do not commute with ϕ
Portfolio or basket return	**No**	Cross-sectional aggregation
Cross-sectional rank of returns	**Yes**	Ranks are invariant under any increasing map
Cross-sectional z-score	**No**	Mean and standard deviation are not
Sign or direction indicator	**Yes**	Thresholds map through

The second row is counterintuitive and useful. **A 20-day cumulative return is convention-free for a tree model**, because the log version is the log of the simple version. The 20-day *average* return is not convention-free, because averaging the logs and averaging the simple returns give different orderings across observations. Momentum measured as a cumulative return over the last quarter is therefore unaffected by the convention. The convention does affect the average daily return over the last quarter.

The practical implication. For a GBT pipeline whose features are raw and cumulative returns, ranks, and indicators, the question deserves no time. Use whatever the data layer produces, which is usually simple returns (§12.3). Skip the `log1p` step entirely. It is a source of bugs (§9.2) and has no offsetting benefit. Switch to logs as soon as a feature averages, differences, or aggregates.

11.3 Neural networks: the transform matters, but not for the reason usually given

The first layer of a neural network computes $Wx + b$, an affine map. Affine maps are not invariant to monotone transformations of their inputs. So a network, unlike a tree, sees a different problem under each convention. The usual explanation is that "log returns are more normal, and networks work best with normal inputs." At daily frequency that explanation is wrong (§4.3). Daily returns are heavy-tailed and volatility-clustered in *both* coordinates. The sample excess kurtosis of the two series differs only in the second decimal place. The real considerations point in a mixed direction.

What actually changes. After the near-universal standardisation step, a monotone transform changes only the *shape* of the input distribution, meaning its skewness and its tail weight. The log transform of returns has two effects:

- It compresses the right tail. A $+100\%$ move becomes $+0.69$, and a $+900\%$ move becomes $+2.30$. [Practice] This helps for assets with occasional very large gains. It is why log returns are the right default for crypto, small caps, biotech, and anything with binary event risk.
- It **stretches the left tail.** A -50% move becomes -0.69, and a -90% move becomes -2.30. Equity returns are already left-skewed, so the log transform makes them *more* left-skewed, not less.

Practitioners often miss the second point. On 19 October 1987 the S&P 500 fell 20.47%, a simple return of -0.2047 and a log return of -0.2290. [Fact] A pre-crash volatility of about 1% standardises that move to a 20-sigma input in simple coordinates and a 23-sigma input in log coordinates. [Practice] **When the concern is a handful of extreme observations dominating the gradient, log returns make the problem slightly worse for equity indices, not better.**

What to do instead. Fix the conditioning of network inputs with one of the following transformations. The logarithm does not do this:

$$\underbrace{\frac{r - \mathrm{med}(r)}{\mathrm{IQR}(r)}}_{\text{robust scaling}} \qquad \underbrace{\frac{r_t}{\hat{\sigma}_t}}_{\text{volatility normalisation}} \qquad \underbrace{\Phi^{-1}\left(\frac{\mathrm{rank}(r) - \frac{1}{2}}{n}\right)}_{\text{rank-Gauss}} \qquad \underbrace{\mathrm{clip}(z, -c, c)}_{\text{winsorisation}}$$

In these formulas, n is the number of observations in the ranking. Φ is the standard normal CDF. z is the standardised feature that the earlier steps produced. c is a clip level of three or four.

The rank-Gauss transform replaces a feature with the standard-normal quantiles of its own ranks. The marginal is therefore Gaussian by construction. The transform is completely insensitive to outliers. Because it depends only on ranks, it is also **identical for simple and log returns**. A network pipeline that ends in a rank-Gauss step is therefore indifferent to the convention chosen upstream, exactly as a tree is.

Volatility normalisation carries real information, beyond improving conditioning. The ratio $r_t/\hat{\sigma}_t$ is approximately stationary across regimes and comparable across assets. Raw returns have neither property (§5.4). Use it as the default for both model families.

11.4 Where the convention sits in the hierarchy of things that matter

Practitioners consistently misallocate feature-engineering effort. The table ranks the main decisions by the size of their effect on out-of-sample performance. [Practice] The ranking reflects practitioner experience and is consistent with published financial ML work. It is a judgement, not a measured result, so the tag covers the whole table.

Rank	Decision	Typical magnitude of effect
1	Point-in-time correctness; no lookahead in any feature or label	Decides whether results are real at all
2	Purged, embargoed cross-validation respecting label overlap	Can swing measured performance by more than 100%
3	Volatility normalisation of features and targets	Large; often the single biggest modelling gain
4	Feature stationarity (fractional differencing over raw levels)	Moderate to large
5	Outlier treatment (winsorisation, robust scaling)	Moderate
6	Cross-sectional versus time-series feature construction	Moderate
7	**Log versus simple returns**	Small for features, moderate for targets
8	Hyperparameter tuning beyond a sensible default	Small

Items 1 and 2 separate a real result from an artefact. Item 3 is roughly the size of everything below it combined. Deliberating over item 7 while leaving item 3 unaddressed optimises the wrong thing. The one exception is the *target*, where the log-versus-simple choice has real consequences. That half of item 7 is the subject of the next two subsections.

11.5 The target, and the retransformation bias

In the machine-learning question, a wrong choice here costs real money. The finance ML literature discusses the problem far too little. Health econometrics solved it forty years ago.

The setup. A model estimates the conditional mean log return, $\hat{m}(x) = \hat{\mathbb{E}}[r \mid x]$. Downstream, a portfolio optimiser or a P&L calculation needs an expected simple return. The naive conversion is $e^{\hat{m}(x)} - 1$, and it is wrong:

$$\mathbb{E}[R \mid x] \;=\; \mathbb{E}\big[e^{r} \mid x\big] - 1 \;\neq\; e^{\mathbb{E}[r|x]} - 1$$

Under conditional lognormality the correct expression is

$$\mathbb{E}[R \mid x] \;=\; \exp\Big(\hat{m}(x) + \tfrac{1}{2}\hat{s}^2(x)\Big) - 1$$

In this expression, $\hat{s}^2(x)$ is the *conditional* variance of the log return at x. It equals the model's residual variance at that point, not the unconditional variance of the series. The naive $e^{\hat{m}(x)} - 1$ therefore gives the conditional **median**, not the mean. This is the retransformation problem of Goldberger (1968) and Duan (1983). Manning (1998) gives the standard treatment of the case where the residual variance is not constant.

Why it is worse in finance than elsewhere. In most applications the correction $\frac{1}{2}s^2$ is roughly constant, so the bias is a level shift that changes no decision. In finance, $s^2(x)$ varies by an order of magnitude across the cross-section. **The bias therefore varies across observations, and it changes the ranking.**

Consider two stocks, both with a predicted log return of exactly zero. Stock A has 1% daily volatility, and stock B has 5%. The table shows the two conversions.

	$\hat{m}(x)$	$\hat{s}(x)$	$e^{\hat{m}} - 1$	$e^{\hat{m}+\hat{s}^2/2} - 1$
Stock A (low vol)	0	1%	0.0 bp	$+0.5$ bp
Stock B (high vol)	0	5%	0.0 bp	$+12.5$ bp

Ranked by predicted log return, the two stocks are tied. Ranked by expected *simple* return, B exceeds A by 12 bp per day. That is roughly 30% per year if summed rather than compounded and if it were tradeable. A long-short book built on predicted log returns and evaluated on realised simple returns will appear to have a systematic short bias in high-volatility names. The apparent bias can lead researchers to a long search for a signal that does not exist.

What to do. The options, in descending order of preference:

1. **Stay in log space.** Fit the model on log returns and evaluate it on log returns. Let the position-sizing layer make the conversion once, explicitly, with a volatility model in hand. This is the cleanest option. The three-stage discipline of §7.3 prescribes it.
2. **Predict the volatility-normalised return** $r_t/\hat{\sigma}_t$ instead. This fix dominates the others. It addresses the retransformation bias and the heteroskedasticity that causes it at the same time. §5.4 already recommends it. When $\hat{\sigma}$ scales the target, the residual variance becomes roughly constant across the cross-section. The correction then reduces to one constant times $\hat{\sigma}^2$ and needs no second model.
3. **Apply the correction explicitly** with a companion conditional-variance model: $\hat{\mathbb{E}}[R \mid x] = \exp(\hat{m}(x) + \frac{1}{2}\hat{s}^2(x)) - 1$. The correction is right if the conditional distribution is close to lognormal. For daily returns it is not close to lognormal in the tails.
4. **Use the smearing estimator of Duan (1983).** It replaces the parametric correction with the empirical residual distribution: $\hat{\mathbb{E}}[R \mid x] = \frac{1}{n}\sum_j \exp\left(\hat{m}(x) + \hat{\varepsilon}_j\right) - 1$. It makes no assumption about the residual shape. It does assume homoskedastic residuals, and finance violates that assumption. Manning (1998) shows that heteroskedasticity breaks smearing too and recommends modelling the variance directly.
5. **Fit the simple return directly** if the economic objective is a one-period dollar return. The targets are worse conditioned, but the estimate of the quantity actually wanted is unbiased.

The deeper point. Choosing the "right" target is a question about the objective, not a statistical question:

- If the decision is *"how much dollar P&L will this position make over the next period,"* the estimand is $\mathbb{E}[R \mid x]$, and simple returns are the aligned target.
- If the decision is *"what allocation maximises long-run compound growth,"* the estimand is $\mathbb{E}[r \mid x]$, and log returns are the aligned target. This is the Kelly objective of §7.1. It needs no correction, because the log return *is* the objective.

The two targets rank assets differently, by exactly $\frac{1}{2}s^2(x)$. The difference reflects the risk aversion that growth optimality implies, and it acts as a volatility penalty. It is not a bug. The choice depends on which quantity the downstream system consumes.

11.6 Loss functions and economic alignment

A squared-error loss weights each observation by the square of its residual. A return model explains almost none of the variance, so the residual is essentially the target itself. The table shows that the two conventions give different squares in the tails.

Event	R^2	r^2	Ratio
$+100\%$	1.00	0.48	log weights it $0.48\times$
-50%	0.25	0.48	log weights it $1.92\times$

Event	R^2	r^2	Ratio
-90%	0.81	5.30	log weights it $6.5\times$

MSE on log returns is therefore a crash-sensitive loss, and MSE on simple returns is a rally-sensitive one. If the economic objective is dollar P&L, the simple-return loss is aligned, since a $+100\%$ winner really is worth twice a -50% loser in dollars. If the objective is survival and compounding, the log loss is aligned. In growth terms a -90% move has $3.3\times$ the magnitude of a $+100\%$ move, because undoing it takes a tenfold recovery. A halving undoes a doubling. Squaring turns that $3.3\times$ into an $11\times$ weight, which is 5.30 against 0.48 in the table.

Huber and quantile losses shrink the difference substantially, because they down-weight the tails in both conventions. That is one reason they are a good default in this domain. Evaluation should mostly use the **rank information coefficient**, $\text{Corr}_{\text{Spearman}}(\widehat{y}_{i,t}, y_{i,t+1})$. It is invariant to the convention on both sides, so it avoids the question. That is a valuable property in a metric compared across many experiments.

For calibration, a rank IC of 0.03 on daily cross-sectional equity returns is a real, tradeable signal. A sustained 0.10 is high enough that leakage (lookahead, survivorship, or a label that overlaps its own features) should be the first hypothesis, not the last. A rank IC does not change at all when the convention changes. Even a *Pearson* IC, which does change, moves by less than 10^{-4} at daily frequency (§8.2). That is more than two orders of magnitude below the effect a researcher wants to detect, and far below its sampling error.

11.7 Recommendations

The table collects the recommendations of this section by role in the pipeline.

Role in the pipeline	Recommendation	Rationale
Raw 1-period return, GBT feature	Either; use whatever the data layer gives	Monotone invariance; avoid the `log1p` bug surface
Raw 1-period return, DNN feature	Log, then robust-scale or rank-Gauss	Affine layers respond to shape. If the pipeline ends in rank-Gauss, the log step changes nothing
Cumulative h-period return, either model	Either	Trees: exactly, since $\sum r_i$ is a monotone map of $\prod(1+R_i)-1$. Networks: the scaling dominates, as above
Mean or EWMA of returns	**Log**	The mean of logs is the compound growth rate; the mean of simple returns is not a growth rate at all
Volatility / dispersion estimates	**Log**	Matches the quadratic-variation target; horizon-consistent
Z-scored return $r_t/\hat{\sigma}_{t-1}$	**Log**, numerator and denominator consistently	Highest-value feature transform available; do this first
Cross-sectional rank or quantile	Either	Rank-invariant
Basket, portfolio, or spread return	**Simple** to aggregate, then log	Aggregation order from §9.5
Regression target, horizon $\geq$ 1 month	**Log**, or volatility-normalised log	Additivity and closure across horizons
Regression target, horizon 1 day	Volatility-normalised (either convention)	The normalisation dominates the transform
Classification target (direction, quintile)	Either	Thresholds map through
Target consumed by a P&L or optimiser	**Simple**, or log with an explicit variance correction	Retransformation bias changes rankings
Target for a growth or Kelly objective	**Log**	The log return is the objective
Reported metric	Rank IC (convention-free); state the convention on any Sharpe	§8.3

The one-paragraph version. For tree models, the log-versus-simple question does not arise for raw and cumulative return features. It becomes a real question as soon as a feature averages, differences, or aggregates. There, use logs. The average of log returns is a compound growth rate, and the average of simple returns is not a growth rate of anything. For the target, the choice depends on the objective: log for growth, simple for dollar P&L. A conversion between them needs the $\frac{1}{2}s^2(x)$ correction. Without it, the model misranks volatile assets systematically. Before any of this, normalise by volatility. Volatility normalisation is worth more than every other decision in this section combined.

§11 Key takeaways

1. Axis-aligned trees are exactly invariant to strictly monotone feature transforms. A GBT cannot distinguish a simple-return feature from its log counterpart. The splits, predictions, and importances are the same.
2. The invariance extends to cumulative h-period returns, because $\sum r_i$ is the log of $\prod(1+R_i)$. It does *not* extend to averages, EWMAs, volatilities, z-scores, differences, or cross-sectional aggregates.
3. For neural networks the transform matters, but for a reason other than the advertised one. Logs compress the right tail and *stretch the left tail*, so already left-skewed equity returns become more extreme. The 1987 crash is a 23-sigma input in logs and a 20-sigma input in simple returns.
4. Rank-Gauss and volatility normalisation are the transforms that fix conditioning, and both are convention-invariant. A pipeline that ends in a rank transform makes the question moot.
5. Among feature-engineering decisions, log versus simple ranks seventh. Point-in-time correctness, purged cross-validation, and volatility normalisation each matter more, and volatility normalisation matters more than everything below it combined.
6. The retransformation bias is the real ML hazard. $e^{\hat{m}(x)}-1$ is a conditional median, not a mean. The missing $\frac{1}{2}s^2(x)$ varies across the cross-section, so it changes rankings. Two stocks with identical predicted log returns and volatilities of 1% and 5% differ by 12 bp per day in expected simple return.
7. The target convention depends on the objective: the log return for growth maximisation, the simple return for one-period dollar P&L. Choose deliberately, and convert once, explicitly.

12. Academic and practitioner conventions

The academic and practitioner communities use different conventions, and the split is not sloppiness. It matches the theorem in §2 almost exactly. Fields that aggregate across positions use simple returns. Fields that model processes through time use logs.

12.1 Who uses what

The table lists the convention each field uses and the reason.

Field or function	Convention	Reason
Asset pricing (CAPM, Fama–French, Fama–MacBeth)	**Simple**	The dependent variable is a portfolio return that must aggregate linearly
Time-series econometrics (ARIMA, GARCH, VAR, VECM)	**Log**	Stationarity and variance stabilisation
Derivatives pricing and volatility modelling	**Log**	Itô calculus; everything is a function of $\ln(S/K)$
Macroeconometrics (PPP, money demand, growth)	**Log**	Economic relationships are proportional; elasticities are log derivatives
Market microstructure	**Log**	Price impact and spreads are modelled proportionally
Performance measurement (GIPS and equivalents)	**Simple**, geometrically linked	Reported figures must reconcile to account statements
Portfolio construction and optimisation	**Simple**	Mean-variance is a quadratic program in simple returns
Risk management, scenario generation	**Log** for price-like factors, **absolute changes** for rate-like ones	The invariant principle of §7.3
Risk reporting (VaR, ES in dollars)	**Simple**	The output is a currency amount
Execution and transaction-cost analysis	**Simple**, in basis points	Costs are proportional to notional
Financial machine learning	Mixed, often unstated	See §11

[Practice] The RiskMetrics Technical Document (1996) models log changes in risk factors and then maps them to dollar P&L for reporting. It is the founding document of modern market-risk practice. That is the three-stage discipline of §7.3, written down thirty years ago.

12.2 Where the two communities talk past each other

Three frictions recur.

"Returns are approximately normal." An econometrician who says this means monthly or lower-frequency log returns, where the claim is defensible. A risk manager who hears it may apply it to daily simple returns for a tail estimate. There it is badly wrong, for reasons unrelated to the transform. The claim is meaningful only with both a frequency and a convention attached.

Arithmetic versus geometric expected returns in portfolio optimisation. Mean-variance optimisation takes arithmetic expected simple returns as inputs. Suppose a practitioner estimates long-run expected returns from historical *compound* growth and feeds them to an optimiser. That practitioner has silently subtracted $\sigma^2/2$ from each asset. The subtraction is asset-specific, so it tilts the optimiser toward low-volatility assets by an amount unrelated to the investor's risk aversion. The error is real and common. It is hard to see because both numbers are called "the expected return." [Practice] The correct input is μ. Given an estimate of g, add back $\hat{\sigma}^2/2$.

What "the market returned 10%" means. The figure could be the arithmetic mean of annual simple returns, the compound annual growth rate, or something in between. For US equities over the long run, the first two differ by roughly two percentage points, a large fraction of the equity risk premium. A statement about long-run returns that does not say which figure it quotes is uncertain by about the size of the effect under discussion.

12.3 A note on data vendors

Most vendor return data is **simple** by default. CRSP holding-period returns, Bloomberg's `CHG_PCT`, and standard total-return series are all simple returns. They include distributions, and delistings follow a convention. The log transform is therefore a step the user applies, at a point the user chooses. That choice is the subject of this chapter. The domain errors of §9.1 enter there.

The delisting convention needs specific attention. CRSP provides delisting returns, which can be exactly -1. Their logs are $-\infty$. Any pipeline that transforms returns to logs needs an explicit policy for this case. Dropping the row introduces survivorship bias and is not a policy.

§12 Key takeaways

1. The academic split maps onto the theorem. Cross-sectional asset pricing uses simple returns because portfolios must aggregate. Time-series work and derivatives use logs because they model processes.
2. Risk management already follows the invariant, projection, and pricing discipline. It models log changes and reports dollars. It has done so since RiskMetrics in 1996.
3. Feeding compound growth rates to a mean-variance optimiser silently subtracts an asset-specific $\sigma^2/2$ and tilts the solution toward low-volatility assets. The optimiser needs arithmetic expected simple returns.
4. "The market returned 10%" is uncertain by about two percentage points until it says arithmetic or geometric.
5. Vendor data is simple by default, so the log transform is always a deliberate step. Have an explicit policy for $R = -1$.

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13. Foundational references

This section groups the references by kind and annotates them. Every entry has a link where one exists.

13.1 Books

Modern, and the ones worth buying.

- **Campbell, J. Y., Lo, A. W. & MacKinlay, A. C. (1997).** *The Econometrics of Financial Markets.* Princeton University Press. — Chapter 1 is still the best single treatment of the definitions in §1 and §3. The book is the standard reference for the empirical methodology in §6.
- **Tsay, R. S. (2010).** *Analysis of Financial Time Series*, 3rd ed. Wiley. — Chapter 1 covers the return definitions carefully. Chapter 3 covers the volatility models of §6.2. This is the most directly usable book on this list for someone implementing.
- **Meucci, A. (2005).** *Risk and Asset Allocation.* Springer. — Develops the invariant/projection/pricing pipeline of §7.3 systematically. The book is dense and worth the effort.
- **Ruppert, D. & Matteson, D. S. (2015).** *Statistics and Data Analysis for Financial Engineering*, 2nd ed. Springer. — Covers the transformation and distributional issues of §4 best, with R code.
- **McNeil, A. J., Frey, R. & Embrechts, P. (2015).** *Quantitative Risk Management*, revised ed. Princeton University Press. — The authority on tail modelling. It states the convention question in risk factor mapping explicitly.
- **Shreve, S. E. (2004).** *Stochastic Calculus for Finance II: Continuous-Time Models.* Springer. — Covers §6.3 in full. Chapter 4 covers Itô calculus.
- **López de Prado, M. (2018).** *Advances in Financial Machine Learning.* Wiley. — Chapter 5 is the fractional-differencing treatment cited in §6.1. Chapter 7 covers the purged cross-validation of §11.4. The book is idiosyncratic and occasionally over-claims, but the two chapters named are genuinely useful.
- **Bouchaud, J.-P. & Potters, M. (2003).** *Theory of Financial Risk and Derivative Pricing*, 2nd ed. Cambridge University Press. — Gives the physicist's view of §4.3. It is the most direct book on the list about how badly the Gaussian assumption fails.

Classic, for historical background rather than current fact.

- **Cootner, P. H., ed. (1964).** *The Random Character of Stock Market Prices.* MIT Press. — The collection brought the random-walk literature together. It includes the first English translation of Bachelier.

13.2 Papers specifically on the return-notion question

This literature is small and directly relevant. The authors wrote these four papers to answer the question of which convention to use.

- **Meucci, A. (2010).** "Quant Nugget 2: Linear vs. Compounded Returns — Common Pitfalls in Portfolio Management." *GARP Risk Professional*, April 2010, 49–51. — The paper has three pages and gives the clearest statement anywhere of the "linear aggregates across securities, compounded aggregates across time" dichotomy. Start here.
- **Hudson, R. S. & Gregoriou, A. (2015).** "Calculating and Comparing Security Returns Is Harder Than You Think: A Comparison Between Logarithmic and Simple Returns." *International Review of Financial Analysis* 38, 151–162. [DOI] — Works through the empirical consequences of the choice on real data. These include the non-monotone relationship between mean log and mean simple returns across a cross-section. This is the most thorough single treatment.
- **Dorfleitner, G. (2003).** "Why the Return Notion Matters." *International Journal of Theoretical and Applied Finance* 6(1), 73–88. — Shows that empirical conclusions about return distributions can reverse depending on the notion used. It is underread.
- **Blume, M. E. & Stambaugh, R. F. (1983).** "Biases in Computed Returns: An Application to the Size Effect." *Journal of Financial Economics* 12(3), 387–404. [[paywalled]] — The canonical demonstration that the method of computing and compounding returns can create an anomaly. Roll (1983), in the same issue, is its companion.

13.3 Foundational, in reading order

- **Bachelier, L. (1900).** "Théorie de la spéculation." *Annales scientifiques de l'École Normale Supérieure* (3) 17, 21–86. — Arithmetic Brownian motion on prices. Its negative-price defect motivated every model after it.
- **Osborne, M. F. M. (1959).** "Brownian Motion in the Stock Market." *Operations Research* 7(2), 145–173. [[paywalled]] — The paper argued for $\ln P$ as the state variable, on psychophysical grounds.
- **Mandelbrot, B. (1963).** "The Variation of Certain Speculative Prices." *Journal of Business* 36(4), 394–419. [[paywalled]] — Fat tails, and the challenge to finite variance that §1.5 flags as still unresolved.
- **Samuelson, P. A. (1965).** "Proof That Properly Anticipated Prices Fluctuate Randomly." *Industrial Management Review* 6(2), 41–49. [[paywalled]] The link is to the World Scientific reprint. The original journal is not online. — The martingale formalisation of efficiency, and the geometric-Brownian-motion framing.
- **Fama, E. F. (1965).** "The Behavior of Stock-Market Prices." *Journal of Business* 38(1), 34–105. [[paywalled]] — The empirical companion, and where the log-return convention became standard in empirical finance.
- **Black, F. & Scholes, M. (1973).** "The Pricing of Options and Corporate Liabilities." *Journal of Political Economy* 81(3), 637–654. [[paywalled]] — Lognormal prices, and the $\sigma^2/2$ in d_2.
- **Merton, R. C. (1973).** "Theory of Rational Option Pricing." *Bell Journal of Economics and Management Science* 4(1), 141–183.

13.4 Time-series econometrics

- **Engle, R. F. (1982).** "Autoregressive Conditional Heteroscedasticity with Estimates of the Variance of United Kingdom Inflation." *Econometrica* 50(4), 987–1007. [[paywalled]]
- **Bollerslev, T. (1986).** "Generalized Autoregressive Conditional Heteroskedasticity." *Journal of Econometrics* 31(3), 307–327. [[paywalled]]
- **Nelson, D. B. (1991).** "Conditional Heteroskedasticity in Asset Returns: A New Approach." *Econometrica* 59(2), 347–370. [[paywalled]] — EGARCH, and the second logarithm of §6.2.
- **Engle, R. F. & Granger, C. W. J. (1987).** "Co-integration and Error Correction: Representation, Estimation, and Testing." *Econometrica* 55(2), 251–276. [[paywalled]]
- **Lo, A. W. & MacKinlay, A. C. (1988).** "Stock Market Prices Do Not Follow Random Walks: Evidence from a Simple Specification Test." *Review of Financial Studies* 1(1), 41–66. — The variance ratio of §6.6.
- **Campbell, J. Y. & Shiller, R. J. (1988).** "The Dividend-Price Ratio and Expectations of Future Dividends and Discount Factors." *Review of Financial Studies* 1(3), 195–228. — The log-linearisation of §6.5. Also NBER w2100.
- **Granger, C. W. J. & Joyeux, R. (1980).** "An Introduction to Long-Memory Time Series Models and Fractional Differencing." *Journal of Time Series Analysis* 1(1), 15–29. [[paywalled]] — With Hosking (1981), the basis for §6.1's fractional differencing.

13.5 Volatility and realized variance

- **Andersen, T. G., Bollerslev, T., Diebold, F. X. & Labys, P. (2003).** "Modeling and Forecasting Realized Volatility." *Econometrica* 71(2), 579–625. — The realized-variance framework of §6.2, including the log-realized-variance finding.

- **Barndorff-Nielsen, O. E. & Shephard, N. (2002).** "Econometric Analysis of Realised Volatility and Its Use in Estimating Stochastic Volatility Models." *Journal of the Royal Statistical Society, Series B* 64(2), 253–280. — The asymptotic theory for realized variance as an estimator of quadratic variation.
- **Cont, R. (2001).** "Empirical Properties of Asset Returns: Stylized Facts and Statistical Issues." *Quantitative Finance* 1(2), 223–236. — The canonical stylized-facts list used in §4.3, and the best single paper on what return data actually looks like.

13.6 Growth optimality, and the debate about it

This subsection is worth reading on its own. It is the clearest instance in finance of an unresolved disagreement between first-rate researchers.

- **Kelly, J. L., Jr. (1956).** "A New Interpretation of Information Rate." *Bell System Technical Journal* 35(4), 917–926. — The origin. It is an information-theory paper, not a finance paper.
- **Latané, H. A. (1959).** "Criteria for Choice Among Risky Ventures." *Journal of Political Economy* 67(2), 144–155. [[paywalled]] — The independent finance-side derivation.
- **Samuelson, P. A. (1971).** "The 'Fallacy' of Maximizing the Geometric Mean in Long Sequences of Investing or Gambling." *Proceedings of the National Academy of Sciences* 68, 2493–2496. — The criticism of the criterion.
- **Merton, R. C. & Samuelson, P. A. (1974).** "Fallacy of the Log-Normal Approximation to Optimal Portfolio Decision-Making Over Many Periods." *Journal of Financial Economics* 1(1), 67–94. [[paywalled]] — The formal version of that criticism.
- **Markowitz, H. M. (1976).** "Investment for the Long Run: New Evidence for an Old Rule." *Journal of Finance* 31(5), 1273–1286. [[paywalled]] — The defence of the criterion.
- **Samuelson, P. A. (1979).** "Why We Should Not Make Mean Log of Wealth Big Though Years to Act Are Long." *Journal of Banking and Finance* 3(4), 305–307. — Written entirely in words of one syllable, except the last. Read it for the argument and for the performance.

13.7 Retransformation and the log-dependent-variable problem

This literature comes from health economics, not finance, and §11.5 draws on it. Finance has largely not imported it.

- **Duan, N. (1983).** "Smearing Estimate: A Nonparametric Retransformation Method." *Journal of the American Statistical Association* 78(383), 605–610. — The smearing estimator.
- **Manning, W. G. (1998).** "The Logged Dependent Variable, Heteroscedasticity, and the Retransformation Problem." *Journal of Health Economics* 17(3), 283–295. — Shows heteroskedasticity breaks smearing too. This paper maps most directly onto the finance problem, because finance is heteroskedastic by construction.

13.8 Compounding, diversification, and skewness

- **Booth, D. G. & Fama, E. F. (1992).** "Diversification Returns and Asset Contributions." *Financial Analysts Journal* 48(3), 26–32. [[paywalled]] — The diversification return of §2.3. The authors have an interest in the topic: Booth co-founded Dimensional Fund Advisors, which sells diversified portfolios. [Contested] The dispute concerns emphasis, not arithmetic. The identity is not in question. Its interpretation as a "return" is.
- **Willenbrock, S. (2011).** "Diversification Return, Portfolio Rebalancing, and the Commodity Return Puzzle." *Financial Analysts Journal* 67(4), 42–49. — The clearest exposition of the same identity. It states explicitly that the diversification return is a rebalancing artefact and not a gain obtained at no cost.
- **Bessembinder, H. (2018).** "Do Stocks Outperform Treasury Bills?" *Journal of Financial Economics* 129(3), 440–457. — The most vivid illustration of compound-return skewness. The majority of individual US stocks have lifetime buy-and-hold returns below Treasury bills. A small minority accounts for the entire net gain. This is §3.3's induced skewness at a 30-year horizon.

13.9 Machine learning

- **Gu, S., Kelly, B. & Xiu, D. (2020).** "Empirical Asset Pricing via Machine Learning." *Review of Financial Studies* 33(5), 2223–2273. — The benchmark study. Their target construction and preprocessing choices are more consequential than their model choices.
- **Grinsztajn, L., Oyallon, E. & Varoquaux, G. (2022).** "Why Do Tree-Based Models Still Outperform Deep Learning on Typical Tabular Data?" *NeurIPS Datasets and Benchmarks.* — The inductive-bias analysis behind §11.2 and §11.3. Their rotation-invariance argument is the converse of the one used here. An axis-aligned tree is invariant to per-feature monotone transforms and sensitive to rotations of the feature basis. An MLP is invariant to rotations and sensitive to

the transform. Tabular data has meaningful individual columns and no meaningful rotations. So the tree's bias is the better match.

- **Shwartz-Ziv, R. & Armon, A. (2022).** "Tabular Data: Deep Learning Is Not All You Need." *Information Fusion* 81, 84–90. — A companion benchmarking study.

13.10 If you only read five things

In order:

1. **Meucci (2010)**, three pages, for the dichotomy.
2. **Campbell, Lo & MacKinlay (1997), Chapter 1**, for the definitions done carefully.
3. **Cont (2001)**, for what the data actually looks like.
4. **Hudson & Gregoriou (2015)**, for the empirical consequences of the choice.
5. **Manning (1998)**, for the retransformation problem, which finance papers repeatedly re-derive.

For work on machine-learning features, substitute Grinsztajn et al. (2022) for item 3.

§13 Key takeaways

1. The literature written directly on this question is small: Meucci (2010), Hudson & Gregoriou (2015), Dorfleitner (2003), and Blume & Stambaugh (1983) are most of it.
2. Health economists solved the retransformation problem in the 1980s. Finance has largely not imported the solution. Duan (1983) and Manning (1998) are worth the hour.
3. The growth-optimality debate is an open disagreement. Kelly and Latané are on one side, Samuelson and Merton on the other, and Markowitz defends the criterion. Reading both sides is more useful than reading either.
4. Practitioner sources on the diversification return come from firms that sell diversification. The arithmetic is not in dispute. The framing as a "return" is.

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14. Synthesis

14.1 The framework, restated

The rest of this chapter follows from three ideas.

One. The gross return $G_t = W_t/W_{t-1}$ is the primitive. Simple and log returns are two coordinates for it. The strictly increasing, strictly concave bijection $r = \ln(1 + R)$ relates them. They carry identical information. The transform adds no power to any test (§4.6).

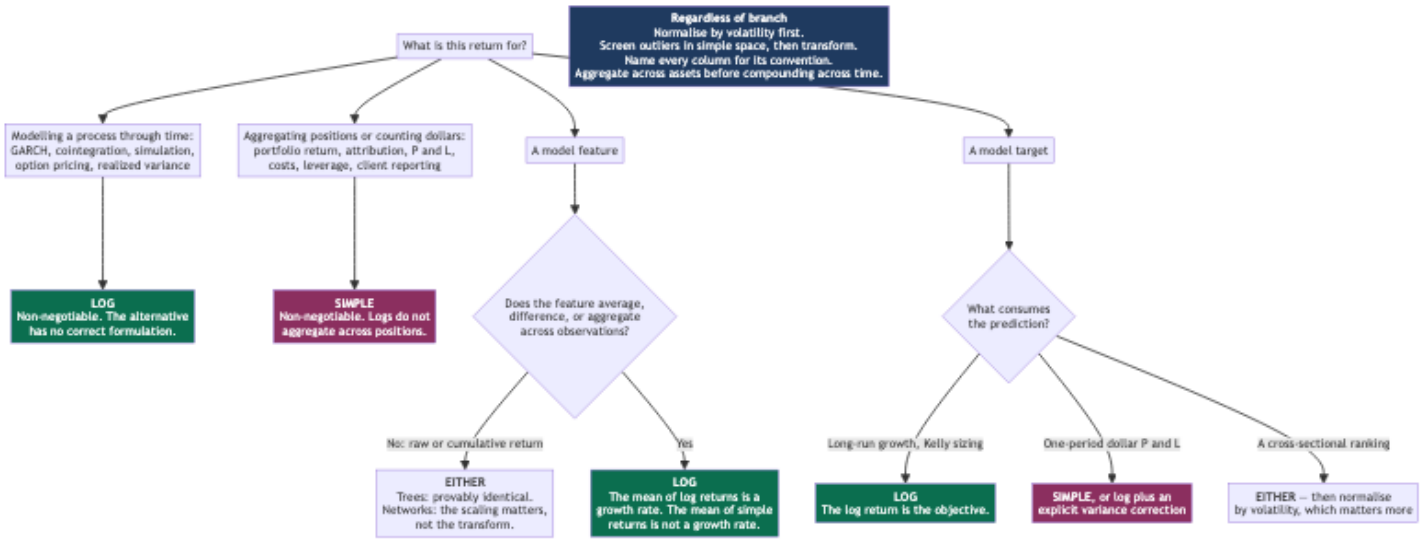
Two. Log returns add across time, and simple returns are linear across assets. **No transform does both**, because compounding generates a cross-term that linearity cannot represent (§2.2). The axis of aggregation therefore forces the choice. A well-built system converts deliberately at the boundary instead of picking one convention globally.

Three. To leading order, the gap between the two conventions is always $\sigma^2/2$. This is one Taylor term from passing an expectation through a concave function. The same term appears as volatility drag, the Itô correction, the Kelly growth rate, and the Black–Scholes d_2. It also appears as the diversification return, the Sharpe-ratio gap, the CAPM alpha bias, and the retransformation bias. Most of the skill in this area is recognising the term on sight (§2.6).

The other results follow from these three. Variance stabilisation (§2.5) is why econometrics uses logs, independently of additivity. Closure under aggregation (§4.2) is why only log returns admit a time-consistent distributional model. Monotone invariance (§11.2) is why tree models cannot distinguish the two at all.

14.2 The decision tree

The decision tree turns the framework into named choices, one per use of a return.



14.3 A roadmap for building this correctly

The roadmap has six stages, with a gate at the end of each. Stages 0 through 2 are infrastructure, not the interesting part. Skipping them is the usual cause of failure.

Stage 0: data correctness. Use total returns rather than price returns, and verify the split and dividend adjustments. Write an explicit policy for delistings and $R = -1$. Enforce point-in-time discipline, so that no feature can see a later revision. *Gate:* reconcile a full year of one instrument's returns against a second vendor to within a basis point. Confirm that the delisting policy is a real policy and not a dropped row.

Stage 1: choose the invariant for each instrument class. Use log returns for equities, ETFs, futures on positive-priced underlyings, and FX. Use absolute changes for interest rates, spreads, and anything that crosses zero. Use changes in log implied volatility for options. Record the choices in a table, not as an unwritten convention. *Gate:* one named transform function per instrument class, with a round-trip test asserting that inverse-of-forward is the identity to machine precision.

Stage 2: normalisation. Build a volatility estimator for each instrument (an EWMA or rolling standard deviation over 20–60 periods) and a normalised return $r_t / \hat{\sigma}_{t-1}$. Lag the estimator, because a $\hat{\sigma}_t$ that uses r_t is lookahead. Use cross-sectional ranks where the application is cross-sectional. *Gate:* the distribution of the normalised feature in a high-volatility year (2008, 2020) is close to its distribution in a calm one (2017). If it is not, the normalisation is not working, and nothing downstream will be stable.

Stage 3: aggregation discipline. Convert between conventions in exactly one place in the codebase, a named function. Aggregate across assets in simple space and across time in log space, never the reverse. *Gate:* compute a portfolio's cumulative return by both routes. The first route weights simple returns and then compounds. The second compounds the constituents and then weights. Confirm that the difference equals the predicted diversification return and is not an unexplained number.

Stage 4: targets tied to the objective. Decide what the prediction is for before choosing its convention. After any conversion, apply the $\frac{1}{2}\hat{s}^2(x)$ correction or document the reason for omitting it. *Gate:* a calibration check of predicted against realised returns, bucketed by volatility decile. A monotone tilt across the deciles is the retransformation bias, the most likely defect in a finance ML pipeline.

Stage 5: reporting. Every published statistic states its convention. *Gate:* the annualised arithmetic mean behind the Sharpe ratio in the tearsheet exceeds the compound growth of the equity curve by $\hat{\sigma}^2_{\text{ann}}/2$. Equivalently, recomputing that Sharpe on log returns lowers it by $\hat{\sigma}_{\text{ann}}/2$ (§8.3). The team can state this without checking.

14.4 Advice for someone starting today

These recommendations condense the chapter for a practitioner starting work in this area.

1. **Normalise by volatility before arguing about the transform.** Volatility normalisation matters roughly three orders of magnitude more than the transform (§5.4), and it is available in either convention.
2. **For tree models on raw or cumulative return features, the question does not exist.** The invariance is exact. Do not spend a week on it, and do not add a `log1p` step whose only effect is a new class of bug.

3. **As soon as a feature averages, the question exists again.** The mean of log returns is a compound growth rate. The mean of simple returns is not a growth rate of anything.
4. **Know which axis the aggregation runs along.** Across assets, use simple returns. Across time, use logs. Write the conversion as one named function and route everything through it.
5. **Subtract $\sigma^2/2$ when converting an arithmetic mean to compound growth.** Compound growth is the arithmetic mean minus half the variance. Drag is quadratic in volatility. A $3\times$ levered fund lags $3\times$ the index by $3\sigma^2$ a year.
6. **Never exponentiate a forecast of a log return and call it an expected return.** The result is a median, and the missing correction varies across the cross-section, so it distorts rankings as well as levels.
7. **State the convention on every reported Sharpe ratio.** The two versions differ by half the annualised volatility, which is a third of a unit for a levered book.
8. **Have an explicit policy for $R = -1$.** Dropping the row introduces survivorship bias, even when a `NaN` filter does it silently.
9. **Plot prices on a log axis.** A linear axis understates the early moves in any long price history.
10. **Fix total-return construction and split adjustment first.** They are worth more basis points than everything else in this chapter combined.

14.5 What is and is not known

This subsection states the epistemic status of the chapter as a whole.

Settled. All of Part I. The impossibility theorem, the identity catalogue, the moment relations, the closure argument, and the monotone-invariance result for trees are mathematics, and they will not change. The stylized facts of §4.3 are as well replicated as anything in empirical finance.

Contested. Three questions remain open. The first question is whether the return variance is finite. This matters because every $\sigma^2/2$ correction in this chapter assumes it is (§1.5). The tail-index estimates that support finiteness are themselves fragile. The defensible position is that finite variance is a working assumption that fits daily liquid-equity data adequately. No one should rely on it for tail risk. The second question is whether maximising expected log wealth is a normative criterion (§7.1). Samuelson's objection is mathematically correct. The practical case for fractional Kelly is nonetheless strong. The two positions do not contradict each other. The third question is whether the log-versus-level distinction in cointegration matters empirically for pairs trading (§6.4). Practitioners disagree, and the published evidence is thin.

Unknown, and the largest gap. No systematic empirical study of the log-versus-simple choice in machine-learning pipelines for finance appears in the published literature. Section 11 draws on three sources of support. The first is a theorem, monotone invariance, which is airtight. The second is an established econometric result imported from another field, the retransformation bias, which is also airtight. The third is practitioner reasoning about everything in between, and practitioner reasoning is not evidence. The experiment that would settle the question is easy to run. Fix a feature set, a model, and an evaluation protocol. Vary only the convention across raw features, derived features, and the target, on several asset classes and several horizons. [Hypothesis] The expected result is that raw features do not matter, derived features matter modestly, and the target matters a great deal. The retransformation term fully explains the effect on the target. This chapter reports no such experiment, and a search found no published version of it.

In summary, the mathematics is complete, and the econometric consequences are well understood. The machine-learning question, which is the one most practitioners are asking in 2026, rests more on inference than on evidence.

§14 Key takeaways

1. The subject rests on one bijection, one impossibility theorem, and one correction term.
2. Do not choose a convention globally. Choose one for each stage, and convert deliberately in exactly one named place.
3. Four steps apply on every branch of the decision tree. Normalise by volatility. Screen outliers in simple space. Name columns for their convention. Aggregate across assets before compounding across time.
4. For tree models the question is provably moot on raw and cumulative return features, and it returns as soon as a feature averages.
5. For targets, the convention depends on the objective: log for growth, simple for dollar P&L. Converting between them without the $\frac{1}{2}s^2(x)$ term distorts rankings, not just levels.
6. The mathematics is settled. The finite-variance assumption and the normative status of Kelly are not, and the machine-learning question has almost no direct published evidence behind it.

Appendix A: Concepts and prerequisites

The main text uses many concepts without defining them. This appendix defines each concept from first principles. It assumes a reader who is mathematically strong but not a specialist in quantitative finance. Each entry gives the idea in plain language first and the formal statement second. It then says where the chapter relies on the concept and where to read more.

The entries follow dependency order, not alphabetical order. Later entries use earlier ones, so the appendix builds from the first entry to the last. The index below gives direct links.

Index

Concept	Where the main text uses it
A.1 Bijections and information	§1.1, §4.6
A.2 Taylor expansion and the delta method	§1.1, §2.5, §2.6, §3.3
A.3 Concavity and Jensen's inequality	§1.1, §2.2, §2.3, §3.4
A.4 The Cauchy functional equations	§2.2
A.5 The AM–GM inequality	§3.4
A.6 Log-sum-exp	§3.2, §7.2
A.7 Order statistics, ranks, and monotone invariance	§5.4, §11.2, §11.3
A.8 Skewness and excess kurtosis	§3.3, §4.3, §4.5
A.9 The lognormal distribution	§3.3, §3.4, §4.1, §11.5
A.10 Heavy tails and the tail index	§1.5, §4.3, §14.5
A.11 Infinite divisibility and Lévy processes	§4.2, §14.1
A.12 Likelihood ratios and the power of a test	§4.6
A.13 Heteroskedasticity and variance-stabilising transforms	§2.5, §6.1, §11.5
A.14 Brownian motion, arithmetic and geometric	§1.5, §4.1, §6.3
A.15 Itô's lemma and the Itô correction	§2.4, §6.3, §7.1
A.16 Quadratic variation and realized variance	§6.2, §7.1
A.17 Risk-neutral pricing and the forward	§6.3, §7.1
A.18 Stationarity, unit roots, and I(0) versus I(1)	§6.1
A.19 Fractional differencing and long memory	§6.1, §11.4
A.20 ARCH and GARCH	§6.2, §8.4
A.21 Cointegration and error correction	§6.4, §7.1
A.22 Overlapping observations and long-horizon regressions	§6.6
A.23 The variance ratio	§3.5, §6.6
A.24 Price, total, and delisting returns	§1.3, §9.1, §12.3
A.25 Volatility drag	§3.4, §10.4
A.26 The diversification return	§2.3, §9.5
A.27 The Sharpe ratio and annualisation	§7.1, §8.3
A.28 CAPM, factor models, alpha, beta, and idiosyncratic volatility	§7.2, §9.3
A.29 Mean-variance optimisation	§7.2, §12.2
A.30 The Kelly criterion	§3.4, §7.1
A.31 Value at Risk and expected shortfall	§7.1, §12.1
A.32 Leveraged and inverse products	§3.2, §10.4
A.33 Market invariants: the three-stage pipeline	§7.3, §12.1
A.34 Conditional means, medians, and retransformation bias	§9.7, §11.5
A.35 Gradient-boosted trees	§11.2
A.36 Feature scaling: robust, rank-Gauss, and volatility normalisation	§5.4, §11.3
A.37 Point-in-time data and purged cross-validation	§11.4
A.38 The information coefficient	§8.2, §11.6
A.39 Robust loss functions	§11.6

A.1 Bijections and information

Conceptually. A bijection is a relabelling. It pairs each input with exactly one output and each output with exactly one input. So the pairing merges nothing and loses nothing. Each output maps back to exactly one input, so the two descriptions contain the same facts. Only the language changes. The content stays the same. So the chapter holds that neither return convention is "more informative" than the other.

Formally. $f : X \to Y$ is a bijection if it is injective ($f(a) = f(b) \Rightarrow a = b$) and surjective (every $y \in Y$ is some $f(x)$). Suppose X, Y carry σ-algebras, and both f and f^{-1} are measurable. Then f is a **measurable isomorphism**. For random variables, $\sigma(f(X)) = \sigma(X)$. The two generate the same collection of events.

Why it appears here. $R \mapsto \ln(1 + R)$ is a smooth bijection from $(-1, \infty)$ onto $\mathbb{R}$ with a smooth inverse (§1.1). §4.6 uses the equality of σ-algebras to argue that no test can be more powerful in one coordinate than in the other. That argument limits every claim the rest of the chapter makes.

Deeper. Any measure-theoretic probability text; Williams, *Probability with Martingales*, ch. 3.

A.2 Taylor expansion and the delta method

Conceptually. Near a point, a smooth function is close to a straight line, and closer still to a parabola. The delta method uses this fact in statistics. Given the mean and variance of X, it approximates the mean and variance of $g(X)$. It expands g around $\mathbb{E}X$ and keeps two terms. Every $\sigma^2/2$ in this chapter comes from the second term.

Formally. For g twice differentiable and X with mean μ_X and variance σ_X^2,

$$\mathbb{E}[g(X)] \approx g(\mu_X) + \tfrac{1}{2} g''(\mu_X)\, \sigma_X^2, \qquad \text{Var}\left(g(X)\right) \approx g'(\mu_X)^2\, \sigma_X^2$$

The first relation is the second-order expansion of the mean. The second is the first-order (classical) delta method. Both are asymptotic. They improve as $\sigma_X^2 \to 0$, and neither is exact for a fixed variance.

Why it appears here. Take $g = \ln(1 + \cdot)$ and $g''(x) = -(1 + x)^{-2}$. The mean relation then gives $m \approx \mu - \sigma^2/2$. This is the master correction of §2.6, which tabulates it in ten forms. The variance relation gives $s^2 \approx \sigma^2$, so volatility barely depends on the convention (§3.3). §2.5 uses the variance relation to derive the log as the variance-stabilising transform.

Deeper. Casella & Berger, *Statistical Inference*, 2nd ed., section 5.5.4; van der Vaart, *Asymptotic Statistics*, ch. 3.

A.3 Concavity and Jensen's inequality

Conceptually. A concave function bends downward. The chord between two points on its graph lies below the curve. So averaging the inputs and then applying the function gives a larger answer than applying the function and then averaging. The logarithm is concave, so the log of the average exceeds the average of the logs. This is also why an arithmetic mean return always exceeds the compound growth rate it delivers.

Formally. φ is concave on an interval if $\varphi(\lambda x + (1-\lambda)y) \geq \lambda\varphi(x) + (1-\lambda)\varphi(y)$ for all $\lambda \in [0, 1]$. It is **strictly** concave if the inequality is strict whenever $x \neq y$ and $\lambda \in (0, 1)$. Jensen's inequality extends this from two points to any probability distribution. For concave φ and integrable X:

$$\mathbb{E}\big[\varphi(X)\big] \;\leq\; \varphi\big(\mathbb{E}X\big)$$

Equality holds, in the strictly concave case, if and only if X is almost surely constant. For convex φ the inequality reverses.

Why it appears here. ln is strictly concave, and the chapter uses that fact four times. First, it rules out the second branch of the impossibility proof in §2.2. Second, it fixes the sign of the portfolio-aggregation gap in §2.3. That use needs the weights to form a probability distribution, which is the reason for the long-only caveat. Third, it gives $g \leq \bar{R}$ in §3.4. Fourth, it implies $\mathbb{E}[e^r] \neq e^{\mathbb{E}[r]}$ in §11.5.

Deeper. Hardy, Littlewood & Pólya, *Inequalities*, ch. 3; Boyd & Vandenberghe, *Convex Optimization*, section 3.1.

A.4 The Cauchy functional equations

Conceptually. Consider which functions turn one arithmetic operation into another: addition into addition, or multiplication into addition. Under a mild regularity condition, the answer is essentially unique. Linear functions turn addition into addition. Logarithms turn multiplication into addition. So the log return is *the* function that linearises compounding, not one convention among many.

Formally. The table lists the four classical equations, each with its continuous solutions on the relevant domain.

Equation	Continuous solutions
$f(x+y) = f(x) + f(y)$	$f(x) = cx$
$f(xy) = f(x) + f(y), x, y > 0$	$f(x) = c \ln x$
$f(x+y) = f(x)f(y)$	$f(x) = e^{cx}$
$f(xy) = f(x)f(y), x, y > 0$	$f(x) = x^c$

Weaker conditions can replace continuity. Measurability suffices, as does boundedness on a set of positive measure. Without any regularity condition, pathological (non-measurable) solutions exist by the axiom of choice.

Why it appears here. The impossibility proposition of §2.2 uses the second equation. Time additivity alone forces $f(R) = c \ln(1+R)$, and portfolio linearity then forces $c = 0$.

Deeper. Aczél, *Lectures on Functional Equations and Their Applications* (1966), ch. 2.

A.5 The AM–GM inequality

Conceptually. The ordinary average of a set of positive numbers is at least their geometric average. The two coincide only when all the numbers are equal. Applied to gross returns, the inequality says that compound growth never exceeds the arithmetic mean return. The shortfall grows with dispersion. The result is volatility drag with no distributional assumption.

Formally. For $x_1, \dots, x_n > 0$,

$$\frac{1}{n}\sum_{i=1}^{n} x_i \;\geq\; \left(\prod_{i=1}^{n} x_i\right)^{1/n}$$

Equality holds if and only if $x_1 = \cdots = x_n$. The inequality is Jensen's inequality for ln, applied to the uniform distribution on $\{x_i\}$.

Why it appears here. §3.4 applies it with $x_i = 1 + R_i$ to get $g \leq \bar{R}$ as an unconditional, in-sample fact. This differs from the lognormal identity $\ln(1+\mu) - \ln(1+g) = s^2/2$, which needs a distributional assumption.

Deeper. Steele, *The Cauchy–Schwarz Master Class*, ch. 2.

A.6 Log-sum-exp

Conceptually. Some quantities are logs of the *levels* that matter. Averaging the levels requires exponentiating, averaging, and taking the log again. The resulting function is smooth and convex. It behaves like a soft maximum and is dominated by the largest input. It is not the average of the inputs.

Formally. $\mathrm{LSE}(z_1, \dots, z_n) = \ln \sum_i e^{z_i}$, and the weighted version is $\ln \sum_i w_i e^{z_i}$. The function is convex. It satisfies $\max_i z_i \leq \mathrm{LSE}(z) \leq \max_i z_i + \ln n$, and its gradient is the softmax. Compute it numerically as $z^* + \ln \sum_i e^{z_i - z^*}$ with $z^* = \max_i z_i$. Otherwise it overflows.

Why it appears here. §3.2 shows that the portfolio log return is exactly $r_p = \ln \sum_i w_i e^{r_i}$, a weighted log-sum-exp. It therefore differs greatly from $\sum_i w_i r_i$. §7.2 notes that the same fact explains why a log-return objective removes the quadratic-program structure of mean-variance optimisation.

Deeper. Boyd & Vandenberghe, *Convex Optimization*, section 3.1.5.

A.7 Order statistics, ranks, and monotone invariance

Conceptually. Sort the observations. The sorted values are the order statistics, and each observation's position in the sort is its rank. A quantity computed only from ranks cannot detect a transformation that preserves the sorted order. A strictly increasing function preserves the sorted order. This one fact is why gradient-boosted trees cannot distinguish simple from log returns.

Formally. For a sample $x_1, \dots, x_n$, the order statistics $x_{(1)} \le \cdots \le x_{(n)}$ are the sorted values, and $\text{rank}(x_i) = \#\{j : x_j \le x_i\}$. If ϕ is strictly increasing, then $\phi(x_{(k)}) = \big(\phi(x)\big)_{(k)}$ and $\text{rank}(\phi(x_i)) = \text{rank}(x_i)$. Ranks and sort order are unchanged, including which values are tied. Ranks determine Spearman correlation, quantiles, medians, and the empirical CDF, so all of them are invariant.

Why it appears here. The monotone-invariance proposition of §11.2 depends entirely on this fact. §5.4 and §11.3 apply it to rank transforms. A cross-sectional rank transform is convention-free. So any choice made before the rank does not affect a pipeline that ranks.

Deeper. David & Nagaraja, *Order Statistics*, 3rd ed., ch. 1.

A.8 Skewness and excess kurtosis

Conceptually. Skewness and excess kurtosis summarise the shape of a distribution beyond its mean and variance. Skewness measures asymmetry, and a positive value means a long right tail. Kurtosis measures how much probability lies far from the centre, relative to a Gaussian. *Excess* kurtosis subtracts the Gaussian's value, so zero means Gaussian-like tails. Financial returns have large excess kurtosis and mildly negative skew.

Formally. With $\mu_X = \mathbb{E}X$ and $\sigma_X^2 = \text{Var}\, X$,

$$\text{skew}(X) = \mathbb{E}\left[\left(\frac{X-\mu_X}{\sigma_X}\right)^3\right], \qquad \text{exkurt}(X) = \mathbb{E}\left[\left(\frac{X-\mu_X}{\sigma_X}\right)^4\right] - 3$$

Both are dimensionless, and both require the corresponding moment to exist. That requirement is a real caveat for return data (§A.10). The sample kurtosis of a heavy-tailed series does not converge. It grows with the sample size.

Why it appears here. §3.3 shows that the log transform produces skewness in simple returns even when log returns are perfectly symmetric. It also quantifies the effect by horizon. §4.3 and §4.5 use both measures to describe the shape of return data and the effect of the transform on the tails.

Deeper. Cont (2001) for the empirical values; Kim & White, "On more robust estimation of skewness and kurtosis," *Finance Research Letters* 1(1), 2004, for why the classical estimators mislead here.

A.9 The lognormal distribution

Conceptually. A lognormal variable is a positive random variable whose logarithm is Gaussian. It arises when a quantity is the product of many small independent factors. This parallels the central limit theorem, which produces the Gaussian from additive accumulation. The mean of a lognormal variable is above its median, by a factor that grows with the variance. That gap is the whole content of volatility drag.

Formally. X is lognormal with parameters (m, s^2) if $\ln X \sim \mathcal{N}(m, s^2)$. Then

$$\mathbb{E}[X] = e^{m+s^2/2}, \quad \text{Med}[X] = e^m, \quad \text{Var}[X] = e^{2m+s^2}\big(e^{s^2}-1\big)$$

$$\text{skew}(X) = \big(e^{s^2}+2\big)\sqrt{e^{s^2}-1}, \qquad \text{exkurt}(X) = e^{4s^2} + 2e^{3s^2} + 3e^{2s^2} - 6$$

The mean formula is the Gaussian moment generating function $\mathbb{E}[e^{tZ}] = e^{mt+s^2t^2/2}$ evaluated at $t = 1$.

Why it appears here. The lognormal is the model of §4.1 and the source of every "exact under lognormality" row in §3.3. The gap between its mean and median is the gap between μ and g in §3.4. Its conditional version gives the retransformation correction of §11.5.

Deeper. Aitchison & Brown, *The Lognormal Distribution* (1957); Johnson, Kotz & Balakrishnan, *Continuous Univariate Distributions*, vol. 1, ch. 14.

A.10 Heavy tails and the tail index

Conceptually. A distribution has heavy tails if extreme values are far more likely than a Gaussian allows. The usual description is a power law. The probability of exceeding a large threshold falls as a power of the threshold. The exponent, called the tail index, determines how many moments exist. Every $\sigma^2/2$ correction in this chapter assumes that the variance is finite, so the tail index matters here.

Formally. X has a regularly varying right tail with tail index $\alpha > 0$ if

$$\mathbb{P}(X > x) = x^{-\alpha} L(x), \qquad L \text{ slowly varying } \big(L(cx)/L(x) \to 1\big)$$

Then $\mathbb{E}[|X|^p] < \infty$ exactly for $p < \alpha$. Finite variance therefore needs $\alpha > 2$, and finite kurtosis needs $\alpha > 4$. Estimates for daily equity returns are typically $\alpha \approx 3$–5. The variance is finite, and the fourth moment is marginal.

Why it appears here. §1.5 presents Mandelbrot's stable-Paretian challenge, under which $\alpha < 2$ and the variance is infinite. §4.3 gives the modern estimates, and §14.5 flags finite variance as the chapter's most consequential contested assumption.

Deeper. Embrechts, Klüppelberg & Mikosch, *Modelling Extremal Events*; Hill, "A simple general approach to inference about the tail of a distribution," *Annals of Statistics* 3(5), 1975, 1163–1174, for the standard estimator and its fragility.

A.11 Infinite divisibility and Lévy processes

Conceptually. Suppose a model gives the distribution of a monthly return and should also imply a coherent model for daily returns. This is possible only if the monthly law is the sum of 21 independent copies of *some* daily law. If such a decomposition works for every subdivision, the distribution is infinitely divisible. Infinitely divisible laws are exactly the laws that extend to a continuous-time process with independent, stationary increments.

Formally. A law ν is **infinitely divisible** if, for every $n \in \mathbb{N}$, ν is the n-fold convolution of some law ν_n. Equivalently, its characteristic function has the Lévy–Khintchine representation

$$\ln \mathbb{E}\big[e^{i\theta X}\big] = ib\theta - \tfrac{1}{2}c\theta^2 + \int_{\mathbb{R}} \Big(e^{i\theta x} - 1 - i\theta x\, \mathbb{1}\{|x| < 1\}\Big)\, \Pi(dx)$$

Here b is a drift, $c \geq 0$ is a diffusion coefficient, and Π is a Lévy measure. The Gaussian, Poisson, variance-gamma, normal-inverse-Gaussian and α-stable laws all qualify. Every infinitely divisible law is the time-1 marginal of a **Lévy process** (a process with stationary, independent increments).

Why it appears here. §4.2 uses it as the decisive argument. Consistency across horizons requires the *additive* quantity to be infinitely divisible, and only log returns are additive. §14.1 restates the closure result among the consequences of the chapter's three generative ideas.

Deeper. Sato, *Lévy Processes and Infinitely Divisible Distributions*; Cont & Tankov, *Financial Modelling with Jump Processes*, ch. 3.

A.12 Likelihood ratios and the power of a test

Conceptually. A test's power is its probability of detecting an effect that is really there. The Neyman–Pearson lemma says that the most powerful test between two simple hypotheses compares their likelihood ratio with a threshold. A change of variables that leaves the ratio unchanged also leaves the best achievable power unchanged. Relabelling the data cannot add statistical strength.

Formally. For simple hypotheses $\mathcal{H}_0, \mathcal{H}_1$ with densities p_0, p_1, the Neyman–Pearson test rejects when $p_1(x)/p_0(x) > k$. It is uniformly most powerful at its size. Under a smooth bijection $y = \phi(x)$, densities transform by the Jacobian: $\tilde{p}_j(y) = p_j(\phi^{-1}(y))\,\big|\frac{d}{dy}\phi^{-1}(y)\big|$. The Jacobian factor is common to numerator and denominator, so it cancels. The likelihood ratio, and hence the test, is unchanged.

Why it appears here. §4.6 uses it to show that log returns do not permit "more powerful conclusions." It also isolates the three channels through which real differences in power arise.

Deeper. Lehmann & Romano, *Testing Statistical Hypotheses*, 3rd ed., ch. 3.

A.13 Heteroskedasticity and variance-stabilising transforms

Conceptually. A homoskedastic series has constant variance. A heteroskedastic series has a variance that changes with the level of the series, with time, or with a covariate. Most classical inference assumes homoskedasticity. A variance-stabilising transform is a change of variable chosen to give the transformed quantity roughly constant variance. This restores the assumption.

Formally. Suppose $\text{Var}(X \mid \theta) = V(\theta)$ with $\mathbb{E}[X \mid \theta] = \theta$. The delta method then gives $\text{Var}\big(\phi(X) \mid \theta\big) \approx \phi'(\theta)^2 V(\theta)$. So the stabilising transform solves

$$\phi'(\theta) \;\propto\; \frac{1}{\sqrt{V(\theta)}} \qquad \Longrightarrow \qquad \phi(\theta) = \int^{\theta} \frac{du}{\sqrt{V(u)}}$$

Three familiar cases follow. $V \propto \theta$ (Poisson) gives $\sqrt{\cdot}$. $V \propto \theta^2$ (multiplicative) gives ln. $V \propto \theta^2(1-\theta)^2$ gives the logit. The Box–Cox family (§1.4) is the standard parametric search over such transforms.

Why it appears here. §2.5 derives the logarithm as the unique stabiliser for a process whose volatility is proportional to its level. This is the econometric case for logs, as distinct from the additivity case. It is also why §6.1 differences the log price instead of the price. The retransformation bias of §11.5 arises when heteroskedasticity remains after the transform.

Deeper. Bartlett, "The use of transformations," *Biometrics* 3(1), 1947, 39–52; Box & Cox (1964).

A.14 Brownian motion, arithmetic and geometric

Conceptually. Brownian motion is the continuous-time limit of a random walk. Its paths are continuous everywhere and differentiable nowhere. Its increments are independent and Gaussian, with a variance that grows linearly in time. *Arithmetic* Brownian motion adds these increments to the price, so the price can go negative. *Geometric* Brownian motion adds them to the log price, so the price is an exponential and stays positive. §1.5 traces the historical move of the field from the first model to the second.

Formally. A standard Brownian motion W_t has $W_0 = 0$, independent increments, $W_t - W_u \sim \mathcal{N}(0, t-u)$, and continuous paths. Arithmetic Brownian motion is $dP_t = \mu\, dt + \sigma\, dW_t$. Geometric Brownian motion is

$$dS_t = \mu S_t\, dt + \sigma S_t\, dW_t \qquad \Longleftrightarrow \qquad S_t = S_0 \exp\Big((\mu - \tfrac{1}{2}\sigma^2)t + \sigma W_t\Big)$$

Itô's lemma (§A.15) gives the equivalence. In this appendix and in §6.3, W_t is a Brownian motion, not wealth.

Why it appears here. The lognormal model of §4.1 is the discrete-time counterpart of geometric Brownian motion. §6.3 derives the $-\sigma^2/2$ from it, and §1.5 explains why it superseded Bachelier's arithmetic version.

Deeper. Shreve, *Stochastic Calculus for Finance II*, ch. 3.

A.15 Itô's lemma and the Itô correction

Conceptually. The ordinary chain rule fails for Brownian paths. A Brownian path is so irregular that its squared increments accumulate at a *finite* rate. $(dW)^2$ behaves like dt and is not negligible. The second-order term of a Taylor expansion therefore remains in the differential. Itô's lemma is the corrected chain rule. Its second-order term is the source of the $-\sigma^2/2$ that recurs throughout this chapter.

Formally. For X_t with $dX_t = a\, dt + b\, dW_t$ and f twice continuously differentiable,

$$df(X_t) \;=\; \underbrace{f'(X_t)\, dX_t}_{\text{ordinary chain rule}} \;+\; \underbrace{\tfrac{1}{2} f''(X_t)\, b^2\, dt}_{\text{Itô correction}}$$

For $f = \ln$ and geometric Brownian motion, $f' = 1/S$, $f'' = -1/S^2$, and $b = \sigma S$. This gives $d \ln S = (\mu - \tfrac{1}{2}\sigma^2)dt + \sigma\, dW$.

Why it appears here. §6.3 shows that the Itô correction *is* volatility drag. In continuous time, the expected simple and log returns differ by exactly $\sigma^2/2$. §2.4 uses the same correction to show that continuous rebalancing makes log returns portfolio-linear up to the diversification return.

Deeper. Shreve, *Stochastic Calculus for Finance II*, ch. 4; Øksendal, *Stochastic Differential Equations*, ch. 4.

A.16 Quadratic variation and realized variance

Conceptually. Divide a path into small intervals, square each increment, and add the squares. For a smooth function the sum tends to zero. For a Brownian-like path it tends to a finite, positive limit that measures how much the path moved. That limit is the quadratic variation. "Realized variance" is its estimate from high-frequency data. The result depends on which path is used. Applied to the log price, it gives a volatility. Applied to the price, it gives a volatility times a squared price level.

Formally. For a semimartingale X and partitions whose mesh tends to zero,

$$[X]_T \;=\; \lim \sum_j \left(X_{t_j} - X_{t_{j-1}}\right)^2$$

For $dX = a\,dt + b\,dW$, $[X]_T = \int_0^T b_u^2\,du$. So with $p = \ln P$ and $b = \sigma$, $[p]_T = \int_0^T \sigma_u^2\,du$. This is the **integrated variance**, the estimand. With P itself and $b = \sigma P$, $[P]_T = \int_0^T \sigma_u^2 P_u^2\,du$. This is in squared currency units.

Why it appears here. §6.2 and §7.1 rely on it. One point needs care. Summed squared *simple* returns converge to the same $\int \sigma_u^2\,du$ as log returns. This holds because $R_j = r_j + O(r_j^2)$ and the discrepancy is third order. Squared *price changes* give the level-dependent limit. The case for logs here is definitional and theoretical. The limits do not differ.

Deeper. Andersen, Bollerslev, Diebold & Labys (2003); Barndorff-Nielsen & Shephard (2002); Protter, *Stochastic Integration and Differential Equations*, ch. 2 for the general theory.

A.17 Risk-neutral pricing and the forward

Conceptually. In an arbitrage-free market, some change of probability measure makes every discounted traded price a martingale (a fair game). Derivative prices are then discounted expectations under that measure. The measure is not a forecast. It encodes the absence of arbitrage. The imposed drift is a drift on the *price*. Translating it into a drift on the *log* price brings in the usual $-\sigma^2/2$. Omitting this term is the practical error.

Formally. Under the risk-neutral measure $\mathbb{Q}$, with a constant risk-free rate r_f, $e^{-r_f t}S_t$ is a $\mathbb{Q}$-martingale, so $\mathbb{E}^{\mathbb{Q}}[S_T] = S_0 e^{r_f T}$. This is the **forward price**. Under geometric Brownian motion this forces $\mathbb{E}^{\mathbb{Q}}[\ln S_T] = \ln S_0 + (r_f - \frac{1}{2}\sigma^2)T$. Any derivative with payoff $H(S_T)$ prices at $e^{-r_f T}\mathbb{E}^{\mathbb{Q}}[H(S_T)]$.

Why it appears here. §6.3 and the option-pricing row of §7.1 rely on it. Setting the log drift to r_f instead of $r_f - \frac{1}{2}\sigma^2$ inflates the simulated forward by $e^{\sigma^2 T/2}$, about 2% at 20% volatility over a year.

Deeper. Shreve, *Stochastic Calculus for Finance II*, ch. 5; Björk, *Arbitrage Theory in Continuous Time*.

A.18 Stationarity, unit roots, and I(0) versus I(1)

Conceptually. Most time-series inference assumes that the series has the same mean, variance, and autocorrelations in every subsample. Prices do not behave this way, because their level does not settle around a fixed mean. A series that becomes stationary after one difference is called integrated of order one. Analysts attribute its behaviour to a "unit root" in its autoregressive representation. A unit-root test tells the analyst whether to difference.

Formally. X_t is **covariance stationary** if $\mathbb{E}X_t$, Var X_t, and Cov(X_t, X_{t-k}) are all finite and independent of t. $X_t \sim I(d)$ if $\Delta^d X_t$ is stationary and $\Delta^{d-1}X_t$ is not. An AR(1) process $X_t = \phi X_{t-1} + \varepsilon_t$ has a unit root when $\phi = 1$. It is then a random walk and $I(1)$. The **ADF** and **Phillips–Perron** tests take a unit root as the null hypothesis. **KPSS** takes stationarity as the null. The critical values of these tests are non-standard (Dickey–Fuller distributions). The theory derives them under homoskedastic innovations.

Why it appears here. In §6.1, $p_t = \ln P_t$ is $I(1)$ and $r_t = \Delta p_t$ is $I(0)$. Differencing the raw price instead leaves heteroskedasticity proportional to the level, which invalidates those critical values.

Deeper. Hamilton, *Time Series Analysis*, ch. 15–17.

A.19 Fractional differencing and long memory

Conceptually. One difference achieves stationarity, but it discards the information the series carried about its level. Differencing a *fractional* number of times is a compromise. It removes just enough of the trend to pass a stationarity test. It keeps

a slowly decaying dependence on the distant past. This helps forecasting, because the level information that full differencing discards is often predictive.

Formally. Using the lag operator L, define for real d

$$\Delta^d = (1-L)^d = \sum_{k=0}^{\infty} \binom{d}{k} (-L)^k, \qquad \binom{d}{k} = \frac{d(d-1)\cdots(d-k+1)}{k!}$$

The weights decay like $k^{-(1+d)}$, hyperbolically rather than geometrically. This slow decay is what "long memory" means. A process is stationary with long memory for $0 < d < \frac{1}{2}$. In practice, truncate the expansion once the weights fall below a tolerance, and set d to the smallest value that passes an ADF test.

Why it appears here. §6.1 offers it as the principled answer to whether a feature should be the price or the return. §11.4 ranks feature stationarity fourth in the hierarchy of things that matter.

Deeper. Granger & Joyeux (1980); Hosking (1981); López de Prado (2018), ch. 5.

A.20 ARCH and GARCH

Conceptually. Return volatility changes over time. Quiet periods cluster, and so do turbulent ones. ARCH models make today's variance a function of yesterday's squared surprise. GARCH adds a term for yesterday's variance. This gives a long memory of past shocks with very few parameters. Both models use the additive quantity, the log return, because they forecast the variance of a sum.

Formally. GARCH(1,1):

$$r_t = m + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad z_t \sim \text{iid}(0,1), \qquad \sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$$

The parameters satisfy $\omega > 0$ and $\alpha, \beta \geq 0$. The persistence is $\lambda = \alpha + \beta$, and stationarity needs $\lambda < 1$. The long-run variance is $\bar{\sigma}^2 = \omega/(1-\lambda)$, and shocks decay with half-life $\ln 2 / \ln(1/\lambda)$. **EGARCH** models $\ln \sigma_t^2$ instead. This removes the positivity constraints and allows an asymmetric response to negative returns.

Why it appears here. §6.2 relies on it, and §8.4 lists "a distributional assumption is being made" among the six triggers that make the convention matter.

Deeper. Engle (1982); Bollerslev (1986); Nelson (1991); Hansen & Lunde (2005) for how hard GARCH(1,1) is to beat.

A.21 Cointegration and error correction

Conceptually. Two series can each lack a fixed level and still be tied to one another, so that some combination of them is stable. This is cointegration. It is the formal content of "these two things move together in the long run." That statement is much stronger and more useful than correlation, which says nothing about levels. Cointegration is the statistical basis of pairs trading and of most long-run macroeconomic relationships.

Formally. $X_t, Y_t \sim I(1)$ are cointegrated if some $\beta \neq 0$ gives $Y_t - \beta X_t \sim I(0)$. The **Granger representation theorem** says such a system has an error-correction form

$$\Delta Y_t = \gamma(Y_{t-1} - \beta X_{t-1}) + (\text{lags}) + \varepsilon_t, \qquad \gamma < 0$$

In this form, deviations from the equilibrium relation reverse over time. The Engle–Granger procedure tests cointegration in two steps. Johansen's procedure handles several series and several cointegrating vectors at once.

Why it appears here. §6.4 shows that testing cointegration in logs and testing it in levels ask *different economic questions.* The log test is a ratio hypothesis and corresponds to a rebalanced position. The level test is a dollar-spread hypothesis and corresponds to a fixed-share position.

Deeper. Engle & Granger (1987); Johansen (1991); Hamilton, *Time Series Analysis*, ch. 19.

A.22 Overlapping observations and long-horizon regressions

Conceptually. Regressing the next twelve months' return on today's predictor, once every month, produces overlapping observations. Consecutive observations share eleven months of data. The residuals are mechanically correlated. Ordinary standard errors are far too small. The overlap inflates t-statistics, often by a factor of two or three. This is the most common reason long-horizon predictability results fail to replicate.

Formally. With $r_{t+1:t+h} = \sum_{j=1}^{h} r_{t+j}$ sampled at every t, the error term follows an MA($h-1$) process even under the null of no predictability. The Hansen–Hodrick and Newey–West estimators produce heteroskedasticity- and autocorrelation-consistent covariance estimates by weighting sample autocovariances out to a chosen lag. Newey–West uses Bartlett (triangular) weights $1 - k/(q+1)$, which guarantee positive semi-definiteness. Separately, the **Stambaugh bias** arises when the predictor is persistent and its innovation correlates with the return innovation. It biases the slope in small samples.

Why it appears here. §6.6 relies on it. This is also why the left-hand side must be an additive sum, which is a property of log returns.

Deeper. Hansen & Hodrick (1980); Newey & West (1987); Stambaugh (1999); Boudoukh, Richardson & Whitelaw, "The myth of long-horizon predictability," *Review of Financial Studies* 21(4), 2008.

A.23 The variance ratio

Conceptually. If returns were unpredictable, variance would grow linearly with the horizon, so two days of variance would be twice one day's. Measuring how variance actually grows shows directly whether the series trends or mean-reverts, and at which horizon. A ratio above one indicates trending, and a ratio below one indicates reversion.

Formally.

$$\mathrm{VR}(q) \;=\; \frac{\mathrm{Var}\left(r_{t+1:t+q}\right)}{q\ \mathrm{Var}(r_{t+1})} \;=\; 1 + 2\sum_{k=1}^{q-1}\Big(1 - \frac{k}{q}\Big)\rho_k$$

where ρ_k is the lag-k autocorrelation of r. The second equality follows from expanding the variance of the sum and collecting terms. The triangular weights are the Bartlett weights again. Under a random walk, $\mathrm{VR}(q) = 1$ for all q.

Why it appears here. §3.5 and §6.6 use it as the clearest example of a statistic that exists only because log returns add. $\mathrm{Var}(R_{t+1:t+q})$ has no expansion in the ρ_k alone.

Deeper. Lo & MacKinlay (1988).

A.24 Price, total, and delisting returns

Conceptually. A stock pays its holder in two ways. The price moves, and the company distributes cash. A price return counts only the first. A total return counts both. Getting this distinction wrong costs more than everything else in this chapter. Separately, companies disappear through bankruptcy, merger, or delisting. The return recorded in the final period is a convention, and it has real consequences for measured performance.

Formally. The price return is $R_t^{\mathrm{PR}} = P_t/P_{t-1} - 1$. The total return is $R_t^{\mathrm{TR}} = (P_t + D_t)/P_{t-1} - 1 = R_t^{\mathrm{PR}} + D_t/P_{t-1}$, where the second term is the dividend yield over the period. This decomposition is **additive in simple returns and not in logs**. A **delisting return** is the return recorded for the final partial period, based on the last traded or liquidation value. CRSP supplies one, and it can be exactly -1. The log return of that value is $-\infty$.

Why it appears here. §1.3 flags the distinction between total and price returns as more consequential than the whole log question. §9.1 and §12.3 treat the case $R = -1$. Dropping the row introduces survivorship bias, so it is not an acceptable policy.

Deeper. Shumway, "The delisting bias in CRSP data," *Journal of Finance* 52(1), 1997, 327–340; Bessembinder (2018) for what the full lifetime distribution looks like.

A.25 Volatility drag

Conceptually. Volatility measures risk, and it also directly reduces the rate at which money compounds. Two portfolios with the same average return but different volatilities end with different wealth. The more volatile one compounds more slowly, by about half the variance per period. So an investor who cares about long-run growth cares about volatility even without any risk aversion.

Formally. With μ the arithmetic mean simple return and g the geometric mean,

$$g \;\approx\; \mu - \frac{\sigma^2}{2}$$

The relation is exact under lognormality in the form $\ln(1+\mu) - \ln(1+g) = s^2/2$, and $g \le \bar{R}$ holds unconditionally by AM–GM (§A.5). The drag is quadratic in σ. So compound growth as a function of leverage L is the downward parabola $L\mu - \frac{1}{2}L^2\sigma^2$. It peaks at $L^* = \mu/\sigma^2$ (§A.30) and returns to zero at $2L^*$.

Why it appears here. §3.4 tabulates it across leverage, §10.4 turns it into mental arithmetic, and §6.3 identifies it with the Itô correction.

Deeper. Any treatment of the geometric mean criterion; Willenbrock (2011) for the portfolio version.

A.26 The diversification return

Conceptually. A portfolio rebalanced to fixed weights compounds faster than the weighted average of its constituents' compound growth rates. The extra growth comes from the mechanical buying low and selling high that rebalancing performs. Equivalently, the portfolio's variance is lower than the average constituent variance, so the portfolio has less drag.

Formally. For a continuously rebalanced portfolio with fixed weights w,

$$g_p - \sum_i w_i g_i \;=\; \tfrac{1}{2}\Big(\sum_i w_i \sigma_i^2 - \sigma_p^2\Big) \;\ge\; 0, \qquad \sigma_p^2 = \mathbf{w}^\top \Sigma \mathbf{w}$$

Non-negativity is exactly the statement that a weighted average of variances is at least the variance of the weighted average. In discrete time, the per-period version is $\frac{1}{2}\,\mathrm{Var}_w(R)$, half the *cross-sectional* variance of returns across holdings.

Why it appears here. §2.3 identifies it as exactly the error that arises from averaging log returns across a portfolio. The error and the diversification return are the same quantity. §9.5 gives its magnitude on a real book.

Deeper. Booth & Fama (1992); Willenbrock (2011), who states that it is an artefact of rebalancing and not a source of extra profit.

A.27 The Sharpe ratio and annualisation

Conceptually. The Sharpe ratio is return per unit of risk. It is the excess return a strategy earned for each unit of volatility it took. It is the industry's default single number. It has two well-known weaknesses. It treats upside and downside volatility identically. And its estimate has far more error than most practitioners assume.

Formally. For excess returns $R^e = R - R_f$,

$$\mathrm{SR} \;=\; \frac{\mathbb{E}[R^e]}{\sqrt{\mathrm{Var}(R^e)}}, \qquad \mathrm{SR}_{\text{ann}} = \sqrt{A}\;\mathrm{SR}_{\text{per period}}$$

The $\sqrt{A}$ scaling assumes iid returns. It is as exact as the square-root-of-time rule: exact for log returns, approximate for simple ones. The standard error of a Sharpe ratio estimated over T periods is roughly $\sqrt{(1+\mathrm{SR}^2/2)/T}$. For a decade of daily data on a Sharpe-1 strategy, it is about 0.32. So a ten-year backtest cannot reliably distinguish a Sharpe ratio of 0.5 from one of 1.5.

Why it appears here. §8.3 shows that the convention changes the reported Sharpe ratio by $\sigma_{\text{ann}}/2$. The change is 0.10 for a long-only equity book and 0.30 for a levered one. So the Sharpe ratio is the one daily-frequency statistic whose convention the author must disclose.

Deeper. Lo, "The statistics of Sharpe ratios," *Financial Analysts Journal* 58(4), 2002, 36–52.

A.28 CAPM, factor models, alpha, beta, and idiosyncratic volatility

Conceptually. A factor model says that most of an asset's movement comes from the market or from a few other common drivers. Beta measures the sensitivity to a factor. Alpha is the average return left over after subtracting the return that the factor exposures explain. Idiosyncratic volatility is the volatility of that remaining return. Investors often claim alpha and rarely achieve it.

Formally. The CAPM regression on **simple excess returns** is

$$R_{i,t} - R_{f,t} \;=\; \alpha_i + \beta_i (R_{M,t} - R_{f,t}) + \varepsilon_{i,t}, \qquad \mathsf{Var}(\varepsilon_i) = \sigma^2_{\varepsilon,i}$$

Therefore $\sigma_i^2 = \beta_i^2 \sigma_M^2 + \sigma^2_{\varepsilon,i}$. This decomposes total variance into systematic and idiosyncratic parts. Multifactor versions add columns. **Fama–MacBeth** runs the cross-sectional regression period by period and averages the coefficients. Standard errors come from the time-series variation of the coefficients.

Why it appears here. §7.2 lists these models as simple-return domains, because the left-hand side must aggregate linearly. §9.3 shows the effect of running them in logs. The intercept contains a mechanical $-\frac{1}{2}\sigma^2_{\varepsilon,i}$ term. Between a stock with 50% idiosyncratic volatility and one with 15%, this term creates a spurious alpha spread of about 11 percentage points a year.

Deeper. Cochrane, *Asset Pricing*, revised ed., ch. 12; Fama & MacBeth (1973).

A.29 Mean-variance optimisation

Conceptually. Mean-variance optimisation chooses portfolio weights that balance expected return against variance. The portfolio's expected return is linear in the weights and its variance is quadratic. So the problem is a quadratic program. It is convex, fast to solve, and has a closed form in the unconstrained case. This tractability follows directly from using simple returns. A log-return objective removes it.

Formally.

$$\max_{\mathbf{w}} \;\; \mathbf{w}^\top \mu \;-\; \frac{\gamma}{2}\, \mathbf{w}^\top \Sigma \mathbf{w} \qquad \text{subject to } \mathbf{1}^\top \mathbf{w} = 1$$

Here γ is the risk-aversion coefficient. Without constraints, the solution is $\mathbf{w}^* \propto \Sigma^{-1}\mu$. The inputs are **arithmetic** expected simple returns. Feeding the optimiser compound growth rates silently subtracts an asset-specific $\sigma_i^2/2$. It tilts the solution toward low-volatility assets for no economic reason.

Why it appears here. §7.2 and §12.2 rely on it. The log-return version of the objective is a log-sum-exp (§A.6), not a quadratic program.

Deeper. Markowitz, "Portfolio selection," *Journal of Finance* 7(1), 1952, 77–91; Meucci, *Risk and Asset Allocation*, ch. 6.

A.30 The Kelly criterion

Conceptually. Suppose an investor makes the same kind of bet repeatedly and reinvests the proceeds each time. The bet size that maximises the growth rate of wealth is a specific fraction of capital, proportional to the edge and inversely proportional to the variance. Betting more than twice that fraction grows wealth more slowly than not betting at all. The criterion comes from maximising expected *log* wealth. That is why it belongs in this chapter.

Formally. Maximise $\mathbb{E}[\ln(1 + fR)]$ over the fraction f. Expanding gives $\mathbb{E}[\ln(1 + fR)] \approx f\mu - \frac{1}{2}f^2\sigma^2$, a downward parabola, with

$$f^* \;\approx\; \frac{\mu}{\sigma^2}, \qquad\qquad g^* \;=\; \frac{\mu^2}{2\sigma^2} \;=\; \frac{\mathsf{SR}^2}{2}$$

Here μ and σ are excess-return moments. Because $\ln W_T \;=\; \ln W_0 + \sum_t \ln(1 + f_t R_t)$ is additive, the multi-period problem separates into independent one-period problems. The investor is myopic, and no dynamic programming is needed. **Fractional Kelly** bets cf^* for $c < 1$. Growth is quadratic near its peak, so $c = \frac{1}{2}$ retains $1 - (1 - \frac{1}{2})^2 = 75\%$ of g^* at a quarter of the variance.

Why it appears here. §7.1 relies on it, including the contested question of its status. Maximising expected log wealth is optimal if and only if utility is logarithmic, so it may be a *normative* criterion or merely a good engineering heuristic.

Deeper. Kelly (1956); Samuelson (1979) for the objection; MacLean, Thorp & Ziemba, eds., *The Kelly Capital Growth Investment Criterion* (2011), which collects both sides.

A.31 Value at Risk and expected shortfall

Conceptually. Value at Risk and expected shortfall are two ways to summarise a loss distribution in one number. Value at Risk is the loss level that losses exceed only $1-\alpha$ of the time. Expected shortfall instead measures the average loss, given a loss in that bad tail, which is a better question. Expected shortfall is coherent in a technical sense that Value at Risk is not. It has largely replaced Value at Risk in regulation.

Formally. For a loss L over a fixed horizon,

$$\mathrm{VaR}_\alpha(L) = \inf\{\ell : \mathbb{P}(L \le \ell) \ge \alpha\}, \qquad \mathrm{ES}_\alpha(L) = \mathbb{E}\big[L \mid L \ge \mathrm{VaR}_\alpha(L)\big]$$

VaR is a quantile, and it fails subadditivity. A portfolio's VaR can exceed the sum of its parts' VaRs. ES is subadditive.

Why it appears here. §7.1 lists long-horizon VaR as a log-return application. A model in log space, mapped back through $e^r - 1 > -1$, respects the -100% floor. A Gaussian simple-return model at long horizons and high volatility puts real probability on losses beyond total ruin.

Deeper. McNeil, Frey & Embrechts, *Quantitative Risk Management*, revised ed., ch. 2 and 8; Artzner et al., "Coherent measures of risk," *Mathematical Finance* 9(3), 1999, 203–228.

A.32 Leveraged and inverse products

Conceptually. A $3\times$ ETF promises three times the index's return *each day*, not over any longer period. It resets daily, so its multi-period return depends on the path. Volatility drag reduces its growth nine times as much as the index's. Over a year it lags three times the index's compound growth by a predictable amount unrelated to fees.

Formally. A daily-reset $L\times$ fund has a simple return of exactly LR_t each day, before costs and financing. Its log drift is therefore $L\mu - \frac{1}{2}L^2\sigma^2$, against L times the index's compound growth, $L(\mu - \frac{1}{2}\sigma^2)$. The shortfall is

$$\tfrac{1}{2}(L^2 - L)\sigma^2$$

For $L = 3$ and $\sigma = 20\%$, this is 12 percentage points a year.

Why it appears here. §3.2 uses it as the clearest counterexample to the claim that logs are always better, because leverage is exactly linear in *simple* returns and not in logs. §10.4 turns the shortfall into a mental calculation.

Deeper. Cheng & Madhavan, "The dynamics of leveraged and inverse exchange-traded funds," *Journal of Investment Management* 7(4), 2009.

A.33 Market invariants: the three-stage pipeline

Conceptually. A better question than "should log returns be used?" is which transformation of this instrument's data is closest to independent and identically distributed across time. That transformed quantity is the invariant, and the analyst models it. The analyst then projects the invariant to the horizon and, only at the end, maps it back into the units in which investors count money. The answer differs by instrument: log returns for equities, yield *changes* for bonds, and implied-volatility changes for options.

Formally. Meucci's pipeline has three stages. **(1) Quest for invariance:** find X_t such that $\{X_t\}$ is iid. **(2) Projection:** estimate the distribution of $X_{t+1:t+h}$, which is easy precisely because the increments are iid and additive. **(3) Pricing:** map the projected invariant back to the P&L of the actual position. This stage reintroduces the non-linearity, and it must precede any aggregation across positions.

Why it appears here. §7.3 makes this pipeline the organising discipline of the chapter, and §12.1 notes that risk management has run it since RiskMetrics in 1996.

Deeper. Meucci, *Risk and Asset Allocation*, ch. 3; Meucci (2010).

A.34 Conditional means, medians, and retransformation bias

Conceptually. Exponentiating the prediction of a model fitted in log space gives the median outcome, not the average outcome. The gap between the two depends on how noisy the particular observation is. In finance the noise varies enormously across assets, so the gap varies too. The gap therefore reorders the predictions instead of shifting them all by a constant. This is the most damaging silent error in a finance ML pipeline.

Formally. For a model $\widehat{m}(x) = \widehat{\mathbb{E}}[r \mid x]$ with conditional residual variance $s^2(x)$, under conditional lognormality,

$$\mathbb{E}[R \mid x] \;=\; \exp\!\Big(\widehat{m}(x) + \tfrac{1}{2}s^2(x)\Big) - 1, \qquad \mathrm{Med}[R \mid x] = e^{\widehat{m}(x)} - 1$$

Duan's smearing estimator replaces the parametric correction with the empirical residual distribution, $\widehat{\mathbb{E}}[R \mid x] = \frac{1}{n}\sum_j \exp(\widehat{m}(x) + \widehat{\varepsilon}_j) - 1$. It assumes that the residuals are homoskedastic. Finance residuals are heteroskedastic.

Why it appears here. §9.7 and §11.5 rely on it. Two stocks with identical predicted log returns and daily volatilities of 1% and 5% differ by 12 bp per day in expected simple return. That is enough to look like a systematic short bias in high-volatility names.

Deeper. Duan (1983); Manning (1998); Goldberger (1968).

A.35 Gradient-boosted trees

Conceptually. Gradient boosting fits a shallow decision tree, examines its errors, fits another tree to those errors, and repeats this a thousand times. Each tree splits the data with yes-or-no questions about one feature at a time, such as "the 20-day return is above 3%". The model therefore responds only to the *order* of a feature's values, never to their spacing. So the log-versus-simple question does not arise for raw return features.

Formally. The ensemble is $F_M(x) = \sum_{k=1}^{M} \nu\, h_k(x)$. Each h_k is a regression tree fitted to the negative gradient of the loss at the current prediction (and, in second-order implementations, to the Hessian). The symbol ν is the learning rate. Each internal node is an **axis-aligned split** $\mathbb{1}\{x_j \le \theta\}$. Candidate thresholds come from the observed order statistics of x_j. The split criterion depends on a candidate only through the partition of rows it induces. The whole procedure is therefore invariant to any strictly increasing transform of any feature (§A.7).

Why it appears here. §11.2 states and proves that invariance and marks where it stops holding. It survives cumulative returns. It fails for averages, differences, ratios, and cross-sectional aggregates. §11.2 also lists the practical caveats. One caveat is that $R = -1$ leaves the domain of $\ln(1 + \cdot)$, which changes which rows are missing.

Deeper. Friedman, "Greedy function approximation: a gradient boosting machine," *Annals of Statistics* 29(5), 2001, 1189–1232; Chen & Guestrin (2016); Ke et al. (2017).

A.36 Feature scaling: robust, rank-Gauss, and volatility normalisation

Conceptually. These are three ways to make a feature comparable across assets and across time. Robust scaling recentres and rescales with order statistics, so outliers cannot dominate. Rank-Gauss discards the values entirely, keeps only their ordering, and reshapes the ordering into a normal distribution. Volatility normalisation divides by a recent volatility estimate. Only volatility normalisation adds information beyond improving conditioning.

Formally.

$$\underbrace{\frac{x - \mathrm{med}(x)}{\mathrm{IQR}(x)}}_{\text{robust}}, \qquad \underbrace{\Phi^{-1}\!\left(\frac{\mathrm{rank}(x) - \frac{1}{2}}{n}\right)}_{\text{rank-Gauss}}, \qquad \underbrace{\frac{r_t}{\widehat{\sigma}_{t-1}}}_{\text{volatility normalisation}}$$

Here Φ^{-1} is the inverse standard normal CDF, and the $-\frac{1}{2}$ is a plotting-position offset that keeps the argument inside $(0, 1)$. $\widehat{\sigma}_{t-1}$ is typically an EWMA, $\widehat{\sigma}_t^2 = \lambda\widehat{\sigma}_{t-1}^2 + (1-\lambda)r_t^2$, or a rolling standard deviation over 20–60 periods. The lag matters, because a $\widehat{\sigma}_t$ that uses r_t is lookahead.

Why it appears here. §5.4 argues that volatility normalisation matters roughly three orders of magnitude more than the convention choice. §11.3 notes that rank-Gauss is convention-invariant. A pipeline that ends in it therefore makes the convention question irrelevant.

Deeper. LeCun et al., "Efficient BackProp," in *Neural Networks: Tricks of the Trade* (1998), for why input conditioning matters to gradient descent.

A.37 Point-in-time data and purged cross-validation

Conceptually. A backtest can mislead in two ways. The first is using data that was not available at the time. Companies restate fundamentals, providers assign index membership retroactively, and today's database differs from what an investor would have seen then. The second is less obvious. If a label spans twelve months, an observation in the training set overlaps observations in the test set. The two sets are then dependent, and the overlap inflates the validation score.

Formally. Point-in-time data records, for each fact, both the date it refers to and the date it became known. A query can then reconstruct the information set $\mathcal{F}_t$ as of any past t. **Purging** removes from the training set every observation whose label window overlaps the test window. **Embargoing** also drops a buffer of observations immediately after the test window. This breaks serial dependence that purging alone leaves.

Why it appears here. §11.4 ranks these two practices first and second among the factors that affect out-of-sample performance. They rank above volatility normalisation, and far above the log-versus-simple choice, which ranks seventh.

Deeper. López de Prado (2018), ch. 7.

A.38 The information coefficient

Conceptually. The information coefficient measures how well a signal ranks what happens next. It is the correlation, across the cross-section, between today's prediction and the next period's realised return, averaged over days. Its values are small in absolute terms, and practitioners consistently misjudge what a good value is. A value of 0.03 supports a real business, and a sustained 0.10 usually indicates a bug.

Formally. Per period, $\text{IC}_t = \text{Corr}\left(\hat{y}_{i,t},\, y_{i,t+1}\right)$ across assets i. The **rank IC** uses Spearman correlation, the Pearson correlation of the ranks. It is therefore invariant to any strictly increasing transform of either argument (§A.7). The **fundamental law of active management** relates the IC to the achievable information ratio (IR), the Sharpe ratio of active returns: $\text{IR} \approx \text{IC}\sqrt{\text{breadth}}$.

Why it appears here. §8.2 uses it to show that the convention moves any smooth statistic far below its own sampling error. §11.6 recommends it as the evaluation metric, because the rank version is exactly convention-free.

Deeper. Grinold & Kahn, *Active Portfolio Management*, 2nd ed., ch. 6.

A.39 Robust loss functions

Conceptually. Squared error penalises a large error heavily. Four times the error gives sixteen times the penalty. So a few extreme observations can dominate a fit. Robust losses grow more slowly in the tails, which limits the influence of rare events on the model. Quantile loss estimates a chosen percentile instead of the mean, which is often what a return forecast should deliver.

Formally. Huber loss with threshold δ is quadratic near zero and linear beyond it:

$$L_\delta(e) = \begin{cases} \frac{1}{2}e^2, & |e| \le \delta \\ \delta(|e| - \frac{1}{2}\delta), & |e| > \delta \end{cases}$$

Quantile (pinball) loss for level τ is $L_\tau(e) = \max\left(\tau e,\, (\tau - 1)e\right)$, minimised in expectation at the τ-quantile. For $\tau = \frac{1}{2}$, it recovers absolute loss and estimates the median.

Why it appears here. §11.6 shows that squared error on log returns is crash-sensitive, while squared error on simple returns is rally-sensitive. It notes that Huber and quantile losses shrink the difference by down-weighting the tails in both conventions.

Deeper. Huber, *Robust Statistics*, 2nd ed.; Koenker, *Quantile Regression*.

Appendix B: Additional works cited

This appendix lists works cited in passing in the main text that do not appear in §13.

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